Mathematics

Class 3

Balochistan Textbook Board,

Quetta

Solved Exercises — Complete Book

This book is the Balochistan Textbook Board's Class 3 Mathematics book (English medium, based on the National Curriculum 2022-23). It has seven units: Whole Numbers, Number Operations, Fractions, Measurement of Length, Mass, Capacity and Temperature, Time, Geometry, and Data Handling and Probability, plus a Glossary at the end. In these solved notes every lesson is summarised and every exercise is solved step by step. Where the book gives only reading, counting practice or a practical activity, that work is stated clearly so the pupil can do it.

Unit 1: Whole Numbers (pages 5 to 35)

Students' Learning Outcomes (page 5): Count up to 9999 (4-digit numbers). Read and write up to 999 in numerals and in words. Recognise the place value of each digit in 4-digit numbers. Compare numbers using symbols and arrange numbers up to 9999 using appropriate language. Round numbers to the nearest tens, hundreds and thousands using

different concrete and pictorial representations. Read and write Roman numbers up to 20. Recognise and differentiate between even and odd numbers.

Counting up to 999 — Revision Exercise (page 6)

We have already learned numbers up to 999. Before we start, let's quickly recap.

Revision Exercise (page 6): Question 1 — Count forward and write numbers

Revision Exercise (page 6): Q1 (ii) — 489, _____, _____, _____, _____, _____

Missing word: 490, 491, 492, 493, 494

Revision Exercise (page 6): Q1 (iii) — 739, _____, _____, _____, _____, _____

Missing word: 740, 741, 742, 743, 744

Revision Exercise (page 6): Q1 (iv) — 965, _____, _____, _____, _____, _____

Missing word: 966, 967, 968, 969, 970

Revision Exercise (page 6): Q1 (i) is already completed in the book: 323, 324, 325, 326, 327, 328.

Revision Exercise (page 6): Question 2 — Count backward and write numbers

Revision Exercise (page 6): Q2 (ii) — 681, _____, _____, _____, _____, _____

Missing word: 680, 679, 678, 677, 676

Revision Exercise (page 6): Q2 (iii) — 952, _____, _____, _____, _____, _____

Missing word: 951, 950, 949, 948, 947

Revision Exercise (page 6): Q2 (iv) — 780, _____, _____, _____, _____, _____

Missing word: 779, 778, 777, 776, 775

Revision Exercise (page 6): Q2 (i) is already completed in the book: 457,

456, 455, 454, 453, 452.

Revision Exercise (page 6): Question 3 — Complete the missing numbers in sequence

Revision Exercise (page 6): Q3 (i) — 125, _____, _____, _____, _____, 130

Missing word: 126, 127, 128, 129

Revision Exercise (page 6): Q3 (ii) — 532, _____, _____, 535, _____, _____

Missing word: 533, 534, 536, 537

Revision Exercise (page 6): Q3 (iii) — _____, 846, _____, _____, _____, 850

Missing word: 845, 847, 848, 849

Counting Numbers up to 9999 (page 7)

Now we will learn 4-digit numbers up to 9999. Look at the 3-digit numbers below: 336 = Three hundred and thirty-six; 452 = Four hundred and fifty-two; 999 = Nine hundred and ninety-nine. Add one: 999 + 1 = 1000. In the above, 1 more block added which completed 1000. Since 999 is the greatest 3-digit number, when we add 1 to the 3-digit number that is 999 + 1, we get the smallest 4-digit number that is 1,000. We read it as 'one thousand'.

Counting Numbers up to 9999 (page 7): 999 + 1 = ?

Step 1: 999 is the greatest 3-digit number.

Step 2: Add 1 to 999.

Step 3: 999 + 1 = 1000.

Step 4: 1000 is the smallest 4-digit number.

Step 5: We read 1000 as 'one thousand'.

Answer: 1000

Count the numbers up to 9999 (page 8)

Page 8 gives a counting chart from 1000 to 9999 in rows of ten (1000-1009, 1010-1019, 2020-2029, 2030-2039, and so on up to 9990-9999). It is a reference chart for counting practice, not a fill-in exercise, so it is not reprinted row by row here — practise counting along the rows in the textbook.

Exercise: Count and write the numbers (pages 9 to 10)

Each row below is one counting run of ten numbers. The numbers printed in the book are shown first, then the completed row.

Exercise (page 9): Question 1 — Count and write the numbers from 1000 to 5999

Q: Exercise (page 9): Row starting 1000, 1001, 1002 — complete the row.

A: 1000, 1001, 1002, 1003, 1004, 1005, 1006, 1007, 1008, 1009.

Q: Exercise (page 9): Row containing 1014 — complete the row.

A: 1010, 1011, 1012, 1013, 1014, 1015, 1016, 1017, 1018, 1019.

Q: Exercise (page 9): Row 1500, 1503, 1509 — complete the row.

A: 1500, 1501, 1502, 1503, 1504, 1505, 1506, 1507, 1508, 1509.

Q: Exercise (page 9): Row 2000, 2005 — complete the row.

A: 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009.

Q: Exercise (page 9): Row 2661, 2666 — complete the row.

A: 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669.

Q: Exercise (page 9): Row 2800, 2804, 2808 — complete the row.

A: 2800, 2801, 2802, 2803, 2804, 2805, 2806, 2807, 2808, 2809.

Q: Exercise (page 9): Row 2992, 2997 — complete the row.

A: 2990, 2991, 2992, 2993, 2994, 2995, 2996, 2997, 2998, 2999.

Q: Exercise (page 9): Row 3000, 3009 — complete the row.

A: 3000, 3001, 3002, 3003, 3004, 3005, 3006, 3007, 3008, 3009.

Q: Exercise (page 9): Row 3452, 3456 — complete the row.

A: 3450, 3451, 3452, 3453, 3454, 3455, 3456, 3457, 3458, 3459.

Q: Exercise (page 9): Row 3671, 3678 — complete the row.

A: 3670, 3671, 3672, 3673, 3674, 3675, 3676, 3677, 3678, 3679.

Q: Exercise (page 9): Row starting 3950 — complete the row.

A: 3950, 3951, 3952, 3953, 3954, 3955, 3956, 3957, 3958, 3959.

Q: Exercise (page 9): Row 4000, 4004 — complete the row.

A: 4000, 4001, 4002, 4003, 4004, 4005, 4006, 4007, 4008, 4009.

Q: Exercise (page 9): Row 4321, 4327 — complete the row.

A: 4320, 4321, 4322, 4323, 4324, 4325, 4326, 4327, 4328, 4329.

Q: Exercise (page 9): Row 4662, 4668 — complete the row.

A: 4660, 4661, 4662, 4663, 4664, 4665, 4666, 4667, 4668, 4669.

Q: Exercise (page 9): Row 4900, 4905 — complete the row.

A: 4900, 4901, 4902, 4903, 4904, 4905, 4906, 4907, 4908, 4909.

Q: Exercise (page 9): Row 5000, 5009 — complete the row.

A: 5000, 5001, 5002, 5003, 5004, 5005, 5006, 5007, 5008, 5009.

Q: Exercise (page 9): Row 5351, 5357 — complete the row.

A: 5350, 5351, 5352, 5353, 5354, 5355, 5356, 5357, 5358, 5359.

Q: Exercise (page 9): Row 5723, 5728 — complete the row.

A: 5720, 5721, 5722, 5723, 5724, 5725, 5726, 5727, 5728, 5729.

Q: Exercise (page 9): Row 5891, 5897 — complete the row.

A: 5890, 5891, 5892, 5893, 5894, 5895, 5896, 5897, 5898, 5899.

Q: Exercise (page 9): Row 5990, 5994, 5999 — complete the row.

A: 5990, 5991, 5992, 5993, 5994, 5995, 5996, 5997, 5998, 5999.

Exercise (page 10): Question 2 — Count and write the numbers from 6000 to 9999

Q: Exercise (page 10): Row 6000, 6007 — complete the row.

A: 6000, 6001, 6002, 6003, 6004, 6005, 6006, 6007, 6008, 6009.

Q: Exercise (page 10): Row 6452, 6458 — complete the row.

A: 6450, 6451, 6452, 6453, 6454, 6455, 6456, 6457, 6458, 6459.

Q: Exercise (page 10): Row 6514, 6519 — complete the row.

A: 6510, 6511, 6512, 6513, 6514, 6515, 6516, 6517, 6518, 6519.

Q: Exercise (page 10): Row 6783, 6787 — complete the row.

A: 6780, 6781, 6782, 6783, 6784, 6785, 6786, 6787, 6788, 6789.

Q: Exercise (page 10): Row 6991, 6999 — complete the row.

A: 6990, 6991, 6992, 6993, 6994, 6995, 6996, 6997, 6998, 6999.

Q: Exercise (page 10): Row 7000, 7005 — complete the row.

A: 7000, 7001, 7002, 7003, 7004, 7005, 7006, 7007, 7008, 7009.

Q: Exercise (page 10): Row 7222, 7228 — complete the row.

A: 7220, 7221, 7222, 7223, 7224, 7225, 7226, 7227, 7228, 7229.

Q: Exercise (page 10): Row 7630, 7637 — complete the row.

A: 7630, 7631, 7632, 7633, 7634, 7635, 7636, 7637, 7638, 7639.

Q: Exercise (page 10): Row 7853, 7859 — complete the row.

A: 7850, 7851, 7852, 7853, 7854, 7855, 7856, 7857, 7858, 7859.

Q: Exercise (page 10): Row 7961, 7968 — complete the row.

A: 7960, 7961, 7962, 7963, 7964, 7965, 7966, 7967, 7968, 7969.

Q: Exercise (page 10): Row 8000, 8005 — complete the row.

A: 8000, 8001, 8002, 8003, 8004, 8005, 8006, 8007, 8008, 8009.

Q: Exercise (page 10): Row 8213, 8218 — complete the row.

A: 8210, 8211, 8212, 8213, 8214, 8215, 8216, 8217, 8218, 8219.

Q: Exercise (page 10): Row 8550, 8556 — complete the row.

A: 8550, 8551, 8552, 8553, 8554, 8555, 8556, 8557, 8558, 8559.

Q: Exercise (page 10): Row 8762, 8769 — complete the row.

A: 8760, 8761, 8762, 8763, 8764, 8765, 8766, 8767, 8768, 8769.

Q: Exercise (page 10): Row 8980, 8988 — complete the row.

A: 8980, 8981, 8982, 8983, 8984, 8985, 8986, 8987, 8988, 8989.

Q: Exercise (page 10): Row 9000, 9004 — complete the row.

A: 9000, 9001, 9002, 9003, 9004, 9005, 9006, 9007, 9008, 9009.

Q: Exercise (page 10): Row 9301, 9306 — complete the row.

A: 9300, 9301, 9302, 9303, 9304, 9305, 9306, 9307, 9308, 9309.

Q: Exercise (page 10): Row 9562, 9567 — complete the row.

A: 9560, 9561, 9562, 9563, 9564, 9565, 9566, 9567, 9568, 9569.

Q: Exercise (page 10): Row 9891, 9897 — complete the row.

A: 9890, 9891, 9892, 9893, 9894, 9895, 9896, 9897, 9898, 9899.

Q: Exercise (page 10): Row 9990, 9999 — complete the row.

A: 9990, 9991, 9992, 9993, 9994, 9995, 9996, 9997, 9998, 9999.

Exercise (page 11): Question 3 — Count and complete the missing numbers sequence

Q: Exercise (page 11): Q3 (i) — 1008, _____, _____, 1011, _____, 1013.

A: 1008, 1009, 1010, 1011, 1012, 1013.

Q: Exercise (page 11): Q3 (ii) — 2435, _____, 2437, _____, _____, 2440.

A: 2435, 2436, 2437, 2438, 2439, 2440.

Q: Exercise (page 11): Q3 (iii) — 4659, _____, 4661, _____, 4663, _____.

A: 4659, 4660, 4661, 4662, 4663, 4664.

Q: Exercise (page 11): Q3 (iv) — 6145, _____, _____, 6148, _____, _____.

A: 6145, 6146, 6147, 6148, 6149, 6150.

Q: Exercise (page 11): Q3 (v) — 5255, _____, 5257, _____, _____, 5260.

A: 5255, 5256, 5257, 5258, 5259, 5260.

Exercise (page 11): Question 4 — Count forward and write numbers

Exercise (page 11): Q4 (i) — 1236, 1237, _____, _____, _____, _____

Missing word: 1238, 1239, 1240, 1241

Exercise (page 11): Q4 (ii) — 3569, _____, _____, _____, _____

Missing word: 3570, 3571, 3572, 3573

Exercise (page 11): Q4 (iii) — 7143, _____, _____, _____, _____

Missing word: 7144, 7145, 7146, 7147

Exercise (page 11): Q4 (iv) — 5482, _____, _____, _____, _____

Missing word: 5483, 5484, 5485, 5486

Exercise (page 11): Q4 (v) — 6364, _____, _____, _____, _____

Missing word: 6365, 6366, 6367, 6368

Exercise (page 11): Question 5 — Count backward and write numbers

Exercise (page 11): Q5 (i) — 2576, 2575, _____, _____, _____, _____

Missing word: 2574, 2573, 2572, 2571

Exercise (page 11): Q5 (ii) — 7342, _____, _____, _____, _____

Missing word: 7341, 7340, 7339, 7338

Exercise (page 11): Q5 (iii) — 1859, _____, _____, _____, _____

Missing word: 1858, 1857, 1856, 1855

Exercise (page 11): Q5 (iv) — 5287, _____, _____, _____, _____

Missing word: 5286, 5285, 5284, 5283

Exercise (page 11): Q5 (v) — 8431, _____, _____, _____, _____

Missing word: 8430, 8429, 8428, 8427

Numbers up to 99 in numerals and in words (page 12)

Let us revise numbers in numerals and in words up to 99.

Revision Exercise (page 12): Question 1 — Read and write numbers in words

Revision Exercise (page 12): Q1 (b) — 99 in words

Missing word: Ninety-nine

Revision Exercise (page 12): Q1 (c) — 79 in words

Missing word: Seventy-nine

Revision Exercise (page 12): Q1 (d) — 77 in words

Missing word: Seventy-seven

Revision Exercise (page 12): Q1 (e) — 30 in words

Missing word: Thirty

Revision Exercise (page 12): Q1 (f) — 55 in words

Missing word: Fifty-five

Revision Exercise (page 12): Q1 (g) — 45 in words

Missing word: Forty-five

Revision Exercise (page 12): Q1 (h) — 87 in words

Missing word: Eighty-seven

Revision Exercise (page 12): Q1 (i) — 51 in words

Missing word: Fifty-one

Revision Exercise (page 12): Q1 (j) — 83 in words

Missing word: Eighty-three

Revision Exercise (page 12): Q1 (k) — 21 in words

Missing word: Twenty-one

Revision Exercise (page 12): Q1 (l) — 71 in words

Missing word: Seventy-one

Revision Exercise (page 12): Q1 (m) — 41 in words

Missing word: Forty-one

Revision Exercise (page 12): Q1 (n) — 97 in words

Missing word: Ninety-seven

Revision Exercise (page 12): Q1 (a) is already completed in the book: 65 = Sixty five.

Revision Exercise (page 12): Question 2 — Read and write numbers in numerals

Revision Exercise (page 12): Q2 (b) — Forty-eight in numerals

Missing word: 48

Revision Exercise (page 12): Q2 (c) — Forty-five in numerals

Missing word: 45

Revision Exercise (page 12): Q2 (d) — Thirteen in numerals

Missing word: 13

Revision Exercise (page 12): Q2 (e) — Sixty-seven in numerals

Missing word: 67

Revision Exercise (page 12): Q2 (f) — Fifty-five in numerals

Missing word: 55

Revision Exercise (page 12): Q2 (g) — Sixteen in numerals

Missing word: 16

Revision Exercise (page 12): Q2 (h) — Twelve in numerals

Missing word: 12

Revision Exercise (page 12): Q2 (i) — Ninety-six in numerals

Missing word: 96

Revision Exercise (page 12): Q2 (j) — Seventeen in numerals

Missing word: 17

Revision Exercise (page 12): Q2 (k) — Sixty six in numerals

Missing word: 66

Revision Exercise (page 12): Q2 (l) — Thirty-one in numerals

Missing word: 31

Revision Exercise (page 12): Q2 (m) — Twenty-one in numerals

Missing word: 21

Revision Exercise (page 12): Q2 (n) — Nineteen in numerals

Missing word: 19

Revision Exercise (page 12): Q2 (o) — Seventy-eight in numerals

Missing word: 78

Revision Exercise (page 12): Q2 (p) — Fifty-four in numerals

Missing word: 54

Revision Exercise (page 12): Q2 (a) is already completed in the book: Twelve = 12.

Numbers up to 999 in Numerals and in words

(pages 13 to 14)

Pages 13 to 14 give a reading table: 100 = One hundred; 112 = One hundred and twelve; 213 = Two hundred and thirteen; 214 = Two hundred and fourteen; 215 = Two hundred and fifteen; 216 = Two hundred and sixteen; 317 = Three hundred and seventeen; 318 = Three hundred and eighteen; 319 = Three hundred and nineteen; 320 = Three hundred and twenty; 421 = Four hundred and twenty one; 422 = Four hundred and twenty two; 423 = Four hundred and twenty three; 424 = Four hundred and twenty four; 525 = Five hundred and twenty five; 526 = Five hundred and twenty six; 527 = Five hundred and twenty seven; 528 = Five hundred and twenty eight; 629 = Six hundred and twenty nine; 630, 631, 632, 743, 744, 755, 766, 876, 887, 888, 989, 998, 999 with their words on page 14. This table is already completed in the book, so it is summarised here, not solved again. NOTE — the book's table itself has a slip on page 14: 630 is labelled 'Six hundred and thirty one' and 631 is labelled 'Six hundred and thirty'. The correct words are 630 = Six hundred and thirty and 631 = Six hundred and thirty one. The slip is in the book, not in these notes.

Exercise (page 14): Question 1 — Read and write in numerals (figures)

Exercise (page 14): Q1 (b) — One hundred and eighty-eight in numerals

Missing word: 188

Exercise (page 14): Q1 (c) — Four hundred and forty-five in numerals

Missing word: 445

Exercise (page 14): Q1 (d) — Three hundred and thirteen in numerals

Missing word: 313

Exercise (page 14): Q1 (e) — Three hundred and ninety-nine in numerals

Missing word: 399

Exercise (page 14): Q1 (f) — Two hundred and twelve in numerals

Missing word: 212

Exercise (page 14): Q1 (g) — Five hundred and ninety-six in numerals

Missing word: 596

Exercise (page 14): Q1 (h) — Four hundred and seventeen in numerals

Missing word: 417

Exercise (page 14): Q1 (a) is already completed in the book: One hundred and ten = 110.

Exercise (page 15): Question 1 (continued) — Read and write in numerals (figures)

Exercise (page 15): Q1 (i) — Six hundred and sixty-seven in numerals

Missing word: 667

Exercise (page 15): Q1 (j) — Six hundred and fifty-five in numerals

Missing word: 655

Exercise (page 15): Q1 (k) — Seven hundred and twenty-one in numerals

Missing word: 721

Exercise (page 15): Q1 (l) — Eight hundred and thirty-one in numerals

Missing word: 831

Exercise (page 15): Q1 (m) — Eight hundred and twenty-one in numerals

Missing word: 821

Exercise (page 15): Q1 (n) — Seven hundred and nineteen in numerals

Missing word: 719

Exercise (page 15): Q1 (o) — Nine hundred and ninety in numerals

Missing word: 990

Exercise (page 15): Q1 (p) — Nine hundred and fifty-four in numerals

Missing word: 954

Exercise (page 15): Question 2 — Read and write number in words

Exercise (page 15): Q2 (b) — 375 in words

Missing word: Three hundred seventy five

Exercise (page 15): Q2 (c) — 406 in words

Missing word: Four hundred six

Exercise (page 15): Q2 (d) — 554 in words

Missing word: Five hundred fifty four

Exercise (page 15): Q2 (e) — 612 in words

Missing word: Six hundred twelve

Exercise (page 15): Q2 (f) — 718 in words

Missing word: Seven hundred eighteen

Exercise (page 15): Q2 (j) — 815 in words

Missing word: Eight hundred fifteen

Exercise (page 15): Q2 (k) — 986 in words

Missing word: Nine hundred eighty six

Exercise (page 15): Q2 (a) is already completed in the book: 125 = One hundred twenty five. NOTE — the book's labels jump from (f) to (j) in this question; the labels are kept as printed.

Exercise (pages 15 to 16): Question 3 — Fill in the blanks

Exercise (page 15): Q3 — 115 in words

Missing word: One hundred fifteen

Exercise (page 15): Q3 — 523 in words

Missing word: Five hundred twenty three

Exercise (page 15): Q3 — Two hundred and thirty in numerals

Missing word: 230

Exercise (page 15): Q3 — Four hundred and fifty-two in numerals

Missing word: 452

Exercise (page 16): Q3 — 417 in words

Missing word: Four hundred seventeen

Exercise (page 16): Q3 — 691 in words

Missing word: Six hundred ninety one

Exercise (page 16): Q3 — Seven hundred and eleven in numerals

Missing word: 711

Exercise (page 16): Q3 — Eight hundred and seventy six in numerals

Missing word: 876

Exercise (page 16): Q3 — 225 in words

Missing word: Two hundred twenty five

Exercise (page 16): Q3 — 367 in words

Missing word: Three hundred sixty seven

Exercise (page 16): Q3 — Five hundred and eighty two in numerals

Missing word: 582

Exercise (page 16): Q3 — Three hundred and five in numerals

Missing word: 305

Exercise (page 16): Q3 — 732 in words

Missing word: Seven hundred thirty two

Exercise (page 16): Q3 — 999 in words

Missing word: Nine hundred ninety nine

Exercise (page 16): Q3 — Six hundred and ninety-one in numerals

Missing word: 691

Exercise (page 16): Q3 — Nine hundred and forty in numerals

Missing word: 940

Exercise (page 16): Question 4 — Match the numbers in words with the numbers in numbers

One hundred and ninety — 190

Six hundred and thirty four — 634

Four hundred and sixty-two — 462

Nine hundred and fifteen — 915

Seven hundred and eighty-six — 786

Three hundred and forty — 340

Eight hundred and forty eight — 848

Two hundred and seventy-three — 273

Five hundred and twenty five — 525

Exercise (page 16): Q4 — the book already joins 'One hundred and ninety' to 190 with an arrow; the remaining pairs are matched above.

Place value (pages 17 to 18)

We have already learnt about the place values of 3-digit numbers. We know that 99 is the greatest 2-digit number. 1 more than 99 is 100. 100 is the first 3-digit number. We can show it in place value as: 1 in the Hundreds place, 0 in the Tens place, 0 in the Ones place. We can show the number 111: 1 hundred, 1 ten and 1 one. We know that 999 is the

largest (greatest) 3-digit number. If we add 1 more in 999, it becomes 1000. It is the first (smallest) 4-digit number. We can show the place of each digit as: 1 in the Thousands place, 0 in the Hundreds place, 0 in the Tens place, 0 in the Ones place.

Example (page 18): Identify the place of each digit of 5432. Solution: Thousands = 5, Hundreds = 4, Tens = 3, Ones = 2. Example (page 18, already completed in the book): 232 = Hundreds 2, Tens 3, Ones 2; 6450 = Thousands 6, Hundreds 4, Tens 5, Ones 0; 5634 = Thousands 5, Hundreds 6, Tens 3, Ones 4; 7021 = Thousands 7, Hundreds 0, Tens 2, Ones 1. Example (page 18): Write the place value of 4532: 2 ones = 2; 3 tens = 30; 5 hundreds = 500; 4 thousands = 4000.

Exercise (page 19): Question 1 — Write the number and place of each digit in the blanks (count the blocks)

Exercise (page 19): Q1 — blocks show Thousands, Hundreds, Tens and Ones

Step 1: Count the thousands cubes: there are 2, so Thousands = 2.

Step 2: Count the hundreds flats: there are 3, so Hundreds = 3.

Step 3: Count the tens rods: there are 6, so Tens = 6.

Step 4: Count the ones cubes: there are 6, so Ones = 6.

Step 5: Put the digits together in order Thousands, Hundreds, Tens, Ones: 2366.

Answer: This number 2366 has 6 ones, 6 tens, 3 hundreds and 2 thousands.

Exercise (page 19): Question 2 — Write the place of each digit in the table for the given numbers

Exercise (page 19): Q2 (i) — 4325: Thousands, Hundreds, Tens, Ones

Missing word: Thousands 4, Hundreds 3, Tens 2, Ones 5

Exercise (page 19): Q2 (ii) — 6421: Thousands, Hundreds, Tens, Ones

Missing word: Thousands 6, Hundreds 4, Tens 2, Ones 1

Exercise (page 19): Q2 (iii) — 3402: Thousands, Hundreds, Tens, Ones

Missing word: Thousands 3, Hundreds 4, Tens 0, Ones 2

Exercise (page 19): Question 3 — Write in thousands, hundreds, tens and ones

Exercise (page 19): Q3 (ii) — 5225

Missing word: 5 thousands 2 hundreds, 2 tens and 5 ones

Exercise (page 19): Q3 (iii) — 4350

Missing word: 4 thousands 3 hundreds, 5 tens and 0 ones

Exercise (page 19): Q3 (iv) — 7057

Missing word: 7 thousands 0 hundreds, 5 tens and 7 ones

Exercise (page 19): Q3 (i) is already completed in the book: 1305 = 1 thousand 3 hundreds, 0 tens and 5 ones.

Exercise (page 20): Question 4 — Write the place value of digits of the given numbers

Exercise (page 20): Q4 (ii) — 6655: place values from ones to thousands

Missing word: 5 ones = 5; 5 tens = 50; 6 hundreds = 600; 6 thousands = 6000

Exercise (page 20): Q4 (iii) — 5721: place values from ones to thousands

Missing word: 1 one = 1; 2 tens = 20; 7 hundreds = 700; 5 thousands = 5000

Exercise (page 20): Q4 (iv) — 2115: place values from ones to thousands

Missing word: 5 ones = 5; 1 ten = 10; 1 hundred = 100; 2 thousands = 2000

Exercise (page 20): Q4 (i) is already completed in the book: 6543 = 3

ones, 40 tens, 500 hundreds, 6000 thousands.

Exercise (page 20): Question 4 (repeated number in the book) — Write the place value of circled digits of the given numbers

Exercise (page 20): Q4 (ii) — 5325, circled digit 2: place value

Missing word: 20 (2 tens)

Exercise (page 20): Q4 (iii) — 3452, circled digit 4: place value

Missing word: 400 (4 hundreds)

Exercise (page 20): Q4 (iv) — 4643, circled digit 4: place value

Missing word: 4000 (4 thousands)

Exercise (page 20): Q4 (v) — 5265, circled digit 2: place value

Missing word: 200 (2 hundreds)

Exercise (page 20): Q4 (vi) — 6522, circled digit 2: place value

Missing word: 2 (2 ones)

Exercise (page 20): Q4 (vii) — 7670, circled digit 7: place value

Missing word: 70 (7 tens)

Exercise (page 20): Q4 (viii) — 8476, circled digit 8: place value

Missing word: 8000 (8 thousands)

Exercise (page 20): Q4 (i) is already completed in the book: 244, circled digit 2 = 200. NOTE — the book numbers two different questions on page 20 both as '4'; both are solved here in order.

Comparing numbers up to 9999 using symbols

(pages 21 to 23)

The comparison of four digit numbers can be done in the same way as we

did for the 3-digit numbers. Key note: we use symbols to compare two numbers: < less than, > greater than, = equal to. Step I: If a number has more digits than other number, so it is always greater. Step II: If two numbers have the same number of digits, then the number with greater digit on the extreme left is greater.

Example (page 21): Compare 3745 and 4414

Step 1: Compare the thousands digits of 3745 and 4414.

Step 2: They are 3 and 4.

Step 3: As 3 < 4, 3745 is less than 4414.

Step 4: So we write 3745 < 4414.

Answer: 3745 < 4414

Example (pages 21 to 22): Compare 3364 and 3344

Step 1: Compare thousands digits: they are equal (3 = 3).

Step 2: Compare hundreds digits: they are equal (3 = 3).

Step 3: Compare tens digits: they are 6 and 4.

Step 4: As 6 > 4, 3364 is greater than 3344.

Step 5: So we write 3364 > 3344.

Answer: 3364 > 3344

Arranging numbers up to 9999 (page 22)

Now we shall learn to arrange the given 4-digit numbers in ascending or descending order.

Example (page 22): Arrange in ascending and descending order. (i) 2746, 2012, 3184, 4101, 3014: Ascending order = 2012, 2746, 3014, 3184, 4101. Descending order = 4101, 3184, 3014, 2746, 2012. (ii) 4523, 5201,

5100, 6152, 3717: Ascending order = 3717, 4523, 5100, 5201, 6152.

Descending order = 6152, 5201, 5100, 4523, 3717.

Exercise (pages 22 to 23): Question 1 — Compare the following numbers by using <, > or =

Exercise (page 22): Q1 (ii) — 2748 _____ 2741

Missing word: >

Exercise (page 22): Q1 (iii) — 5312 _____ 5390

Missing word: <

Exercise (page 22): Q1 (iv) — 1484 _____ 3456

Missing word: <

Exercise (page 22): Q1 (v) — 1432 _____ 4734

Missing word: <

Exercise (page 22): Q1 (vi) — 2861 _____ 2861

Missing word: =

Exercise (page 22): Q1 (i) is already completed in the book: 2340 > 1220.

Exercise (page 23): Q1 (vii) — 1172 _____ 6231

Missing word: <

Exercise (page 23): Q1 (viii) — 5634 _____ 5638

Missing word: <

Exercise (page 23): Q1 (ix) — 9999 _____ 9990

Missing word: >

Exercise (page 23): Q1 (x) — 6524 _____ 6821

Missing word: <

Exercise (page 23): Q1 (xi) — 8282 _____ 2826

Missing word: >

Exercise (page 23): Q1 (xii) — 4178 _____ 2184

Missing word: >

Exercise (page 23): Q1 (xiii) — 4181 _____ 4181

Missing word: =

Exercise (page 23): Q1 (xiv) — 6136 _____ 3125

Missing word: >

Exercise (page 23): Q1 (xv) — 7196 _____ 7284

Missing word: <

Exercise (page 23): Question 2 — Arrange the numbers given below in ascending order

Exercise (page 23): Q2 (i) — 2106, 1234, 4513, 2019, 3451, 1015 in ascending order

Missing word: 1015, 1234, 2019, 2106, 3451, 4513

Exercise (page 23): Q2 (ii) — 1005, 5238, 3621, 4918, 2203, 3415 in ascending order

Missing word: 1005, 2203, 3415, 3621, 4918, 5238

Exercise (page 23): Q2 (iii) — 6351, 4518, 3761, 6014, 4802, 5834 in ascending order

Missing word: 3761, 4518, 4802, 5834, 6014, 6351

Exercise (page 23): Q2 (iv) — 7541, 5315, 8011, 6643, 7019, 6408 in ascending order

Missing word: 5315, 6408, 6643, 7019, 7541, 8011

Exercise (page 23): Q2 (v) — 9201, 7318, 8425, 9523, 6703, 8136 in ascending order

Missing word: 6703, 7318, 8136, 8425, 9201, 9523

Exercise (page 23): Question 3 — Arrange the numbers given below in descending order

Exercise (page 23): Q3 (i) — 1525, 2342, 1439, 3657, 4764, 3983 in descending order

Missing word: 4764, 3983, 3657, 2342, 1525, 1439

Exercise (page 23): Q3 (ii) — 3246, 5694, 3172, 4381, 6539, 5425 in descending order

Missing word: 6539, 5694, 5425, 4381, 3246, 3172

Exercise (page 23): Q3 (iii) — 5918, 7165, 6341, 7018, 8212, 6111 in descending order

Missing word: 8212, 7165, 7018, 6341, 6111, 5918

Exercise (page 23): Q3 (iv) — 2006, 8739, 5615, 4891, 6010, 7213 in descending order

Missing word: 8739, 7213, 6010, 5615, 4891, 2006

Exercise (page 23): Q3 (v) — 8342, 5718, 9015, 8528, 7581, 6201 in descending order

Missing word: 9015, 8528, 8342, 7581, 6201, 5718

Rounding Numbers (pages 24 to 26)

Rounding means giving a number a nearby value. People sometimes round the numbers to make numbers that are easier to use or easier to remember. General Rounding Rules: Finding the place where you are going to do rounding, then look at the digit on the right. If the digit is 5 or more, round up. If the digit is 4 or less, round down.

Rounding numbers to the nearest 10 (page 24): The round to the nearest 10 gives a number the value of tens it is closest to. When the ones digit is 5 or more, round up. When it is 4 or less, round down. We can use a

number line to round off a number. Let's round off 17 to the nearest 10. It shows that 17 is nearest to 20 than to 10, so 17 is rounded up to 20. Rounding numbers to the nearest 100 (pages 24 to 25): Similarly, we can round off to the nearest 100. Let's round off 342 to the nearest 100. It shows that 342 is nearest to 300 than to 400, so 342 is rounded down to 300.

Another Method of Rounding Off Numbers — Rounding Numbers to the Nearest 1000 (page 25): We will use two steps to round off to the nearest 1000. If the digit in the hundreds place is less than 5, then the ones, tens and hundreds place digits are replaced by '0'. If the digit in the hundreds place is 5 or greater than 5, then the ones, tens and hundreds place digits are replaced by '0' and the thousands place digit is increased by '1'. Let's round off 5453 to the nearest 1000. In 5453 the digit in the hundreds place is less than 5. Here, we replace 4, 5 and 3 by '0'. So, 5453 rounded off to the nearest 1000 is 5000. Let's round off 5744 to the nearest 1000. In 5744 the digit in the hundreds place is greater than 5. Here, we increase the thousands place digit by '1' and replace the tens, ones and hundreds place digits by '0'. So, 5744 rounded off to the nearest 1000 is 6000.

Exercise (page 26): Question 1 — Round off to the nearest 10

Exercise (page 26): Q1 (i) — Round off 26 to the nearest 10

Step 1: Look at the ones digit of 26: it is 6.

Step 2: 6 is 5 or more, so round up.

Step 3: 26 rounds up to the next ten.

Answer: 30

Exercise (page 26): Q1 (ii) — Round off 38 to the nearest 10

Step 1: Look at the ones digit of 38: it is 8.

Step 2: 8 is 5 or more, so round up.

Answer: 40

Exercise (page 26): Q1 (iii) — Round off 47 to the nearest 10

Step 1: Look at the ones digit of 47: it is 7.

Step 2: 7 is 5 or more, so round up.

Answer: 50

Exercise (page 26): Q1 (iv) — Round off 66 to the nearest 10

Step 1: Look at the ones digit of 66: it is 6.

Step 2: 6 is 5 or more, so round up.

Answer: 70

Exercise (page 26): Q1 (v) — Round off 53 to the nearest 10

Step 1: Look at the ones digit of 53: it is 3.

Step 2: 3 is 4 or less, so round down.

Answer: 50

Exercise (page 26): Q1 (vi) — Round off 74 to the nearest 10

Step 1: Look at the ones digit of 74: it is 4.

Step 2: 4 is 4 or less, so round down.

Answer: 70

Exercise (page 26): Q1 (vii) — Round off 72 to the nearest 10

Step 1: Look at the ones digit of 72: it is 2.

Step 2: 2 is 4 or less, so round down.

Answer: 70

Exercise (page 26): Q1 (viii) — Round off 85 to the nearest 10

Step 1: Look at the ones digit of 85: it is 5.

Step 2: 5 means round up.

Answer: 90

Exercise (page 26): Question 2 — Round off to the nearest 100

Exercise (page 26): Q2 (i) — Round off 324 to the nearest 100

Step 1: Look at the tens digit of 324: it is 2.

Step 2: 2 is 4 or less, so round down.

Answer: 300

Exercise (page 26): Q2 (ii) — Round off 269 to the nearest 100

Step 1: Look at the tens digit of 269: it is 6.

Step 2: 6 is 5 or more, so round up.

Answer: 300

Exercise (page 26): Q2 (iii) — Round off 478 to the nearest 100

Step 1: Look at the tens digit of 478: it is 7.

Step 2: 7 is 5 or more, so round up.

Answer: 500

Exercise (page 26): Q2 (iv) — Round off 579 to the nearest 100

Step 1: Look at the tens digit of 579: it is 7.

Step 2: 7 is 5 or more, so round up.

Answer: 600

Exercise (page 26): Q2 (v) — Round off 710 to the nearest 100

Step 1: Look at the tens digit of 710: it is 1.

Step 2: 1 is 4 or less, so round down.

Answer: 700

Exercise (page 26): Q2 (vi) — Round off 668 to the nearest 100

Step 1: Look at the tens digit of 668: it is 6.

Step 2: 6 is 5 or more, so round up.

Answer: 700

Exercise (page 26): Q2 (vii) — Round off 788 to the nearest 100

Step 1: Look at the tens digit of 788: it is 8.

Step 2: 8 is 5 or more, so round up.

Answer: 800

Exercise (page 26): Q2 (viii) — Round off 887 to the nearest 100

Step 1: Look at the tens digit of 887: it is 8.

Step 2: 8 is 5 or more, so round up.

Answer: 900

Exercise (page 26): Question 3 — Round off to the nearest 1000

Exercise (page 26): Q3 (i) — Round off 1220 to the nearest 1000

Step 1: Look at the hundreds digit of 1220: it is 2.

Step 2: 2 is less than 5, so replace ones, tens and hundreds by 0.

Answer: 1000

Exercise (page 26): Q3 (ii) — Round off 3400 to the nearest 1000

Step 1: Look at the hundreds digit of 3400: it is 4.

Step 2: 4 is less than 5, so replace ones, tens and hundreds by 0.

Answer: 3000

Exercise (page 26): Q3 (iii) — Round off 4800 to the nearest 1000

Step 1: Look at the hundreds digit of 4800: it is 8.

Step 2: 8 is 5 or more, so increase the thousands digit by 1 and replace the rest by 0.

Answer: 5000

Exercise (page 26): Q3 (iv) — Round off 6070 to the nearest 1000

Step 1: Look at the hundreds digit of 6070: it is 0.

Step 2: 0 is less than 5, so replace ones, tens and hundreds by 0.

Answer: 6000

Exercise (page 26): Q3 (v) — Round off 6850 to the nearest 1000

Step 1: Look at the hundreds digit of 6850: it is 8.

Step 2: 8 is 5 or more, so increase the thousands digit by 1 and replace the rest by 0.

Answer: 7000

Exercise (page 26): Q3 (vi) — Round off 5450 to the nearest 1000

Step 1: Look at the hundreds digit of 5450: it is 4.

Step 2: 4 is less than 5, so replace ones, tens and hundreds by 0.

Answer: 5000

Exercise (page 26): Q3 (vii) — Round off 7950 to the nearest 1000

Step 1: Look at the hundreds digit of 7950: it is 9.

Step 2: 9 is 5 or more, so increase the thousands digit by 1 and replace the rest by 0.

Answer: 8000

Exercise (page 26): Q3 (viii) — Round off 4290 to the nearest 1000

Step 1: Look at the hundreds digit of 4290: it is 2.

Step 2: 2 is less than 5, so replace ones, tens and hundreds by 0.

Answer: 4000

Reading and Writing Roman Numbers up to 20

(pages 27 to 28)

Roman numbers were used by ancient Romans in between 800 - 900 B.C. These are frequently used in daily life. We have seen the Roman numbers on the dials of clocks, watches and clock towers in big cities. Sometimes these numbers are also used in examination papers and books. Roman numbers from 1 to 20 are based on the symbols: I, V, X. Read Roman numbers given below: 1 = I (i), 2 = II (ii), 3 = III (iii), 4 = IV (iv), 5 = V (v), 6 = VI (vi), 7 = VII (vii), 8 = VIII (viii), 9 = IX (ix), 10 = X (x), 11 = XI (xi), 12 = XII (xii), 13 = XIII (xiii), 14 = XIV (xiv), 15 = XV (xv), 16 = XVI (xvi), 17 = XVII (xvii), 18 = XVIII (xviii), 19 = XIX (xix), 20 = XX (xx).

Some rules to write Roman numbers are: 1. I, X can be repeated up to three times, so formed as: 1 = I, 2 = I + I = II, 3 = I + I + I = III, 10 = X, 12 = 10 + 1 + 1 = XII, 20 = 10 + 10 = XX. 2. When a symbol of smaller value is placed on the left of a symbol of bigger value, we subtract their values to find the Roman numeral: 5 - 1 = 4 = V - I = IV, and 10 - 1 = 9 = X - I = IX. 3. When a symbol of smaller value is placed on the right we add their value to find the Roman numeral, so formed as: 5 + 1 = 6 = V + I = VI, 5 + 2 = 7 = V + I + I = VII. Key note: There is no symbol for '0' in Roman numeration system. The symbol I can be written to the left or right of symbol V and X. Symbol V cannot be repeated. NOTE — the book prints rule 2's second example as '9 - 1 = 10 = X - I = IX'; the correct subtraction is 10 - 1 = 9 = X - I = IX, and that is what is used here.

Exercise (page 28): Question 1 — Write the following Roman numerals in correct order

Exercise (page 28): Q1 (i) — II, V, III, IV, I in correct order

Missing word: I, II, III, IV, V

Exercise (page 28): Q1 (ii) — XII, XIV, XI, XIII, XV in correct order

Missing word: XI, XII, XIII, XIV, XV

Exercise (page 28): Q1 (iii) — VI, VIII, IX, VII, X in correct order

Missing word: VI, VII, VIII, IX, X

Exercise (page 28): Q1 (iv) — XVII, XIX, XVIII, XVI, XX in correct order

Missing word: XVI, XVII, XVIII, XIX, XX

Exercise (page 28): Q1 (v) — XII, XIV, XV, XIII, XVI in correct order

Missing word: XII, XIII, XIV, XV, XVI

Exercise (page 28): Question 2 — Draw the clock hands to show the time given in boxes

Exercise (page 28): Q2 is a drawing task on clock faces with Roman numerals. Draw the hands as follows: Half past 3 — the minute hand points at VI (6) and the hour hand is halfway between III (3) and IV (4). Quarter past 3 — the minute hand points at III (3) and the hour hand is just past III (3). Half past 6 — the minute hand points at VI (6) and the hour hand is halfway between VI (6) and VII (7). Quarter past 5 — the minute hand points at III (3) and the hour hand is just past V (5).

Exercise (page 29): Question 3 — Read and write the missing Roman numerals

Exercise (page 29): Q3 row 1 — I, II, _____, IV, V, _____, VII, _____, IX, X

Missing word: III, VI, VIII

Exercise (page 29): Q3 row 2 — _____, _____, _____, XIV, _____, XVII, _____, _____, XX

Missing word: XI, XII, XIII, XV, XVI, XVIII, XIX

Exercise (page 29): Question 4 — Write the Roman numerals for each of the following

Exercise (page 29): Q4 — 5 in Roman numerals

Missing word: V

Exercise (page 29): Q4 — 7 in Roman numerals

Missing word: VII

Exercise (page 29): Q4 — 4 in Roman numerals

Missing word: IV

Exercise (page 29): Q4 — 9 in Roman numerals (first)

Missing word: IX

Exercise (page 29): Q4 — 9 in Roman numerals (second)

Missing word: IX

Exercise (page 29): Q4 — 10 in Roman numerals

Missing word: X

Exercise (page 29): Q4 — 11 in Roman numerals

Missing word: XI

Exercise (page 29): Q4 — 17 in Roman numerals

Missing word: XVII

Exercise (page 29): Q4 — 13 in Roman numerals

Missing word: XIII

Exercise (page 29): Q4 — 18 in Roman numerals

Missing word: XVIII

Exercise (page 29): Q4 — 15 in Roman numerals

Missing word: XV

Exercise (page 29): Q4 — 20 in Roman numerals

Missing word: XX

Exercise (page 29): Q4 — the book prints 9 twice in the left column; both are IX, and both are answered.

Exercise (page 29): Question 5 — Match the Natural number with Roman number

viii — 8

ix — 9

xi — 11

xiv — 14

xx — 20

xviii — 18

Exercise (page 29): Q5 — the book already joins 'ix' to 9 with a line; the remaining pairs are matched above.

Even and Odd Numbers (pages 30 to 31)

Even Numbers (page 30): The objects or things that are even can be grouped in pairs. For example in the given figure, the objects show the even numbers: 2, 4, 6, 8, 10. So, 2, 4, 6, 8, 10 are even numbers. Key note: Even numbers end in 0, 2, 4, 6 or 8. The numbers in which the digit at ones place is 0, 2, 4, 6, 8 are called even numbers. Remember: The numbers which are divisible by 2 are called even numbers.

Odd Numbers (page 31): The objects or things that are odd cannot be grouped in pairs. For example, in the given figure the objects show the odd numbers: 1, 3, 5, 7, 9. So, 1, 3, 5, 7, 9 are odd numbers. Key note: Odd numbers end in 1, 3, 5, 7 or 9. Remember: The numbers which are not divisible by 2 are called odd numbers. NOTE — the paragraph on

page 31 lists the ones digits as '3, 5, 7 or 9', leaving out 1, but the key note on the same page correctly lists 1, 3, 5, 7 or 9; the key-note version is used here.

The chart will help to identify even and odd numbers (page 32): a 0 to 99 chart in which the numbers given in red are even numbers (0, 2, 4, 6, 8, 10, 12, 14, 16, ..., 98) and the numbers given in blue are odd numbers (1, 3, 5, 7, 9, 11, 13, 15, ..., 99). This chart is already completed in the book, so it is summarised here. Teacher point (page 32): The teacher will involve students in a variety of activities of even and odd numbers up to 9999 — teacher work only.

Examples (page 33, all solved in the book): Tick the even numbers in 62, 43, 74, 83, 14, 39, 68, 91, 87, 22 — the even numbers are 62, 74, 14, 68 and 22. Tick the odd numbers in 54, 66, 33, 46, 73, 21, 84, 97, 60, 17 — the odd numbers are 33, 73, 21, 97 and 17. Write the even numbers between 5 and 13 — they are 6, 8, 10 and 12. Write odd numbers between 24 and 32 — they are 25, 27, 29 and 31.

Exercise (page 34): Question 1 — Tick the even numbers in the given numbers

Exercise (page 34): Q1 — 74, 61, 24, 56, 47, 88, 35, 68, 93, 20: the even numbers

Missing word: 74, 24, 56, 88, 68, 20

Exercise (page 34): Question 2 — Encircle the odd numbers in the given numbers

Exercise (page 34): Q2 — 66, 53, 42, 27, 38, 74, 83, 91, 12, 49: the odd numbers

Missing word: 53, 27, 83, 91, 49

Exercise (page 34): Q2 — the book already circles 53; the remaining odd

numbers are 27, 83, 91 and 49.

Exercise (page 34): Question 3 — Separate even and odd numbers

Exercise (page 34): Q3 — 10, 15, 29, 32, 42, 37, 81, 86, 65, 79, 56, 92: Even numbers

Missing word: 10, 32, 42, 86, 56, 92

Exercise (page 34): Q3 — 10, 15, 29, 32, 42, 37, 81, 86, 65, 79, 56, 92: Odd numbers

Missing word: 15, 29, 37, 81, 65, 79

Exercise (page 34): Question 4 — Write 'E' for even number and 'O' for odd number

Exercise (page 34): Q4 — 233

Missing word: O (odd)

Exercise (page 34): Q4 — 347

Missing word: O (odd)

Exercise (page 34): Q4 — 366

Missing word: E (even)

Exercise (page 34): Q4 — 421

Missing word: O (odd)

Exercise (page 34): Q4 — 462

Missing word: E (even)

Exercise (page 34): Q4 — 559

Missing word: O (odd)

Exercise (page 34): Q4 — 534

Missing word: E (even)

Exercise (page 34): Q4 — 683

Missing word: O (odd)

Exercise (page 34): Q4 — 674

Missing word: E (even)

Exercise (page 34): Q4 — 791

Missing word: O (odd)

Exercise (page 34): Q4 — 786

Missing word: E (even)

Exercise (page 34): Q4 — 876

Missing word: E (even)

Exercise (page 34): Q4 — 865

Missing word: O (odd)

Exercise (page 34): Q4 — 984

Missing word: E (even)

Exercise (page 34): Q4 — 999

Missing word: O (odd)

Exercise (page 34): Q4 — 217 = O is already completed in the book.

Unit 1 Summary (page 35)

Summary (page 35): Counting and reading the numbers up to 4 digits. Writing the numbers in words up to 999. Identifying the place value of numbers up to 4 digits. Comparing of numbers using symbols <, > or =. Arranging numbers in order from smallest to greatest is called ascending order. Arranging numbers in order from greatest to smallest is called descending order. Rounding off numbers nearest to 10, 100 and 1000. Roman numerals 1 - 20 are based on three symbols: I, V, X. All even numbers end in 0, 2, 4, 6 or 8. All odd numbers end in 1, 3, 5, 7 or 9.

Rounding means giving a number a nearby value. Vocabulary: Count, Thousand, Arrange, Place Value, Roman, before, after, Rounding Off, between. This closes Unit 1 (pages 5 to 35).

Unit 2: Number Operations (pages 36 to 84)

Students' Learning Outcomes (page 36): Add and subtract numbers mentally and in written form (with and without regrouping) including 4-digit numbers with 1, 2, 3 and 4-digit numbers. Solve real-world word problems (including missing numbers and money) involving addition and subtraction. Estimate the answer to an addition and subtraction question (using various approaches). Count and write in multiple steps. Write out developed times tables for 6, 7, 8, and 9 and write multiplication sentences using concrete and pictorial representations. Reinforce through concrete and pictorial representations that the multiplication of any two numbers can be done in any order. Multiply mentally and in written form: 2-digit numbers by 1-digit numbers, 3-digit numbers by 1-digit numbers. Solve real-world word problems involving multiplication. Reinforce through concrete and pictorial representation that the division of any two numbers cannot be done in any order (Commutative). Divide mentally and in written form: 2-digit numbers by 1-digit number (with and without remainder), 3-digit numbers with 1-digit numbers (with and without remainder). Divide a number by 1 and itself. Solve real-world word problems involving division. Solve real-world word problems involving addition, subtraction, multiplication and division. Recognise and extend a given number pattern in increasing and decreasing order. Find the missing number or operation in a number sentence (e.g., 20 + W = 100).

Addition — Revision Exercise (page 37)

We have already learnt addition of numbers up to 3 digits in class 2.

Revision Exercise (page 37): Question 1 — Add the following without regrouping

Revision Exercise (page 37): Q1 (ii) — 576 + 101

Step 1: Add the ones: 6 + 1 = 7.

Step 2: Add the tens: 7 + 0 = 7.

Step 3: Add the hundreds: 5 + 1 = 6.

Answer: 677

Revision Exercise (page 37): Q1 (iii) — 628 + 261

Step 1: Add the ones: 8 + 1 = 9.

Step 2: Add the tens: 2 + 6 = 8.

Step 3: Add the hundreds: 6 + 2 = 8.

Answer: 889

Revision Exercise (page 37): Q1 (iv) — 175 + 123

Step 1: Add the ones: 5 + 3 = 8.

Step 2: Add the tens: 7 + 2 = 9.

Step 3: Add the hundreds: 1 + 1 = 2.

Answer: 298

Revision Exercise (page 37): Q1 (v) — 816 + 83

Step 1: Add the ones: 6 + 3 = 9.

Step 2: Add the tens: 1 + 8 = 9.

Step 3: Add the hundreds: 8 + 0 = 8.

Answer: 899

Revision Exercise (page 37): Q1 (vi) — 623 + 324

Step 1: Add the ones: 3 + 4 = 7.

Step 2: Add the tens: 2 + 2 = 4.

Step 3: Add the hundreds: 6 + 3 = 9.

Answer: 947

Revision Exercise (page 37): Q1 (i) is already completed in the book: 432 + 60 = 492.

Revision Exercise (page 37): Question 2 — Add the following with regrouping

Revision Exercise (page 37): Q2 (ii) — 816 + 99

Step 1: Add the ones: 6 + 9 = 15. Write 5, carry 1 to the tens.

Step 2: Add the tens: 1 + 9 + 1 (carried) = 11. Write 1, carry 1 to the hundreds.

Step 3: Add the hundreds: 8 + 0 + 1 (carried) = 9.

Answer: 915

Revision Exercise (page 37): Q2 (iii) — 576 + 65

Step 1: Add the ones: 6 + 5 = 11. Write 1, carry 1 to the tens.

Step 2: Add the tens: 7 + 6 + 1 (carried) = 14. Write 4, carry 1 to the hundreds.

Step 3: Add the hundreds: 5 + 0 + 1 (carried) = 6.

Answer: 641

Revision Exercise (page 37): Q2 (iv) — 647 + 254

Step 1: Add the ones: 7 + 4 = 11. Write 1, carry 1 to the tens.

Step 2: Add the tens: 4 + 5 + 1 (carried) = 10. Write 0, carry 1 to the

hundreds.

Step 3: Add the hundreds: 6 + 2 + 1 (carried) = 9.

Answer: 901

Revision Exercise (page 37): Q2 (v) — 729 + 324

Step 1: Add the ones: 9 + 4 = 13. Write 3, carry 1 to the tens.

Step 2: Add the tens: 2 + 2 + 1 (carried) = 5.

Step 3: Add the hundreds: 7 + 3 = 10. Write 0, carry 1 to the thousands.

Step 4: Thousands: 1.

Answer: 1053

Revision Exercise (page 37): Q2 (vi) — 847 + 347

Step 1: Add the ones: 7 + 7 = 14. Write 4, carry 1 to the tens.

Step 2: Add the tens: 4 + 4 + 1 (carried) = 9.

Step 3: Add the hundreds: 8 + 3 = 11. Write 1, carry 1 to the thousands.

Step 4: Thousands: 1.

Answer: 1194

Revision Exercise (page 37): Q2 (i) is already completed in the book: 395 + 27 = 422 (carries 1 and 1 shown in the book).

Addition of 4-digit numbers with 1-digit numbers

— Mental Strategy (page 38)

Mental Strategy: We can use splitting up numbers for easier mental calculation. Example (page 38): Add 2458 + 4 mentally. Solution: Splitting 4 = 2 + 2. So 2458 + 4 = 2458 + 2 + 2 = (2458 + 2) + 2 = 2460 + 2 = 2462.

Exercise (page 38): Question 1 — Add using mental strategies

Exercise (page 38): Q1 (i) — 1324 + 8

Step 1: Add the ones: 4 + 8 = 12. Write 2, carry 1 to the tens.

Step 2: Tens: 2 + 1 (carried) = 3. Hundreds and thousands stay 3 and 1.

Answer: 1332

Exercise (page 38): Q1 (ii) — 3466 + 8

Step 1: Add the ones: 6 + 8 = 14. Write 4, carry 1 to the tens.

Step 2: Tens: 6 + 1 (carried) = 7. Other digits stay the same.

Answer: 3474

Exercise (page 38): Q1 (iii) — 4237 + 5

Step 1: Add the ones: 7 + 5 = 12. Write 2, carry 1 to the tens.

Step 2: Tens: 3 + 1 (carried) = 4. Other digits stay the same.

Answer: 4242

Exercise (page 38): Q1 (iv) — 5168 + 4

Step 1: Add the ones: 8 + 4 = 12. Write 2, carry 1 to the tens.

Step 2: Tens: 6 + 1 (carried) = 7. Other digits stay the same.

Answer: 5172

Exercise (page 38): Q1 (v) — 2656 + 7

Step 1: Add the ones: 6 + 7 = 13. Write 3, carry 1 to the tens.

Step 2: Tens: 5 + 1 (carried) = 6. Other digits stay the same.

Answer: 2663

Exercise (page 38): Q1 (vi) — 7516 + 6

Step 1: Add the ones: 6 + 6 = 12. Write 2, carry 1 to the tens.

Step 2: Tens: 1 + 1 (carried) = 2. Other digits stay the same.

Answer: 7522

Exercise (page 38): Q1 (vii) — 6747 + 8

Step 1: Add the ones: 7 + 8 = 15. Write 5, carry 1 to the tens.

Step 2: Tens: 4 + 1 (carried) = 5. Other digits stay the same.

Answer: 6755

Exercise (page 38): Q1 (viii) — 5235 + 9

Step 1: Add the ones: 5 + 9 = 14. Write 4, carry 1 to the tens.

Step 2: Tens: 3 + 1 (carried) = 4. Other digits stay the same.

Answer: 5244

Exercise (page 38): Q1 (ix) — 3526 + 6

Step 1: Add the ones: 6 + 6 = 12. Write 2, carry 1 to the tens.

Step 2: Tens: 2 + 1 (carried) = 3. Other digits stay the same.

Answer: 3532

Exercise (page 38): Q1 (x) — 8359 + 7

Step 1: Add the ones: 9 + 7 = 16. Write 6, carry 1 to the tens.

Step 2: Tens: 5 + 1 (carried) = 6. Other digits stay the same.

Answer: 8366

Exercise (page 38): Q1 (xi) — 7185 + 8

Step 1: Add the ones: 5 + 8 = 13. Write 3, carry 1 to the tens.

Step 2: Tens: 8 + 1 (carried) = 9. Other digits stay the same.

Answer: 7193

Exercise (page 38): Q1 (xii) — 9467 + 5

Step 1: Add the ones: 7 + 5 = 12. Write 2, carry 1 to the tens.

Step 2: Tens: 6 + 1 (carried) = 7. Other digits stay the same.

Answer: 9472

Exercise (page 38): Q1 (xiii) — 9877 + 9

Step 1: Add the ones: 7 + 9 = 16. Write 6, carry 1 to the tens.

Step 2: Tens: 7 + 1 (carried) = 8. Other digits stay the same.

Answer: 9886

Exercise (page 38): Q1 (xiv) — 8984 + 7

Step 1: Add the ones: 4 + 7 = 11. Write 1, carry 1 to the tens.

Step 2: Tens: 8 + 1 (carried) = 9. Other digits stay the same.

Answer: 8991

Addition of 4-digit numbers and ones (without regrouping) (page 39)

Example (page 39): Add 1114 + 3. On the abacus, 1114 plus 3 ones gives 1117. So 1114 + 3 = 1117.

Exercise (page 39): Question 1 — Add the following

Exercise (page 39): Q1 (i) — 1236 + 2

Step 1: Add the ones: 6 + 2 = 8.

Step 2: Thousands, hundreds and tens stay 1, 2 and 3.

Answer: 1238

Exercise (page 39): Q1 (ii) — 3546 + 3

Step 1: Add the ones: 6 + 3 = 9.

Step 2: Other digits stay the same.

Answer: 3549

Exercise (page 39): Q1 (iii) — 5362 + 6

Step 1: Add the ones: 2 + 6 = 8.

Step 2: Other digits stay the same.

Answer: 5368

Exercise (page 39): Q1 (iv) — 2414 + 5

Step 1: Add the ones: 4 + 5 = 9.

Step 2: Other digits stay the same.

Answer: 2419

Exercise (page 39): Q1 (v) — 7853 + 6

Step 1: Add the ones: 3 + 6 = 9.

Step 2: Other digits stay the same.

Answer: 7859

Exercise (page 39): Q1 (vi) — 9784 + 4

Step 1: Add the ones: 4 + 4 = 8.

Step 2: Other digits stay the same.

Answer: 9788

Exercise (page 39): Question 2 — Solve the following

Exercise (page 39): Q2 (i) — 1516 + 2

Step 1: Add the ones: 6 + 2 = 8.

Step 2: Other digits stay the same.

Answer: 1518

Exercise (page 39): Q2 (ii) — 4624 + 4

Step 1: Add the ones: 4 + 4 = 8.

Step 2: Other digits stay the same.

Answer: 4628

Exercise (page 39): Q2 (iii) — 6172 + 7

Step 1: Add the ones: 2 + 7 = 9.

Step 2: Other digits stay the same.

Answer: 6179

Exercise (page 39): Q2 (iv) — 2432 + 5

Step 1: Add the ones: 2 + 5 = 7.

Step 2: Other digits stay the same.

Answer: 2437

Exercise (page 39): Q2 (v) — 5363 + 5

Step 1: Add the ones: 3 + 5 = 8.

Step 2: Other digits stay the same.

Answer: 5368

Exercise (page 39): Q2 (vi) — 3984 + 5

Step 1: Add the ones: 4 + 5 = 9.

Step 2: Other digits stay the same.

Answer: 3989

Exercise (page 39): Q2 (vii) — 8251 + 8

Step 1: Add the ones: 1 + 8 = 9.

Step 2: Other digits stay the same.

Answer: 8259

Exercise (page 39): Q2 (viii) — 7824 + 3

Step 1: Add the ones: 4 + 3 = 7.

Step 2: Other digits stay the same.

Answer: 7827

Addition of 4-digit numbers and 2-digit numbers (without regrouping) (page 40)

Example (page 40): Add 2244 + 22. On the abacus, 2244 plus 22 gives 2266. So 2244 + 22 = 2266.

Exercise (page 40): Question 1 — Add the following

Exercise (page 40): Q1 (i) — 1462 + 17

Step 1: Add the ones: 2 + 7 = 9.

Step 2: Add the tens: 6 + 1 = 7.

Step 3: Thousands and hundreds stay 1 and 4.

Answer: 1479

Exercise (page 40): Q1 (ii) — 3854 + 23

Step 1: Add the ones: 4 + 3 = 7.

Step 2: Add the tens: 5 + 2 = 7.

Step 3: Thousands and hundreds stay 3 and 8.

Answer: 3877

Exercise (page 40): Q1 (iii) — 5273 + 25

Step 1: Add the ones: 3 + 5 = 8.

Step 2: Add the tens: 7 + 2 = 9.

Step 3: Thousands and hundreds stay 5 and 2.

Answer: 5298

Exercise (page 40): Q1 (iv) — 6942 + 37

Step 1: Add the ones: 2 + 7 = 9.

Step 2: Add the tens: 4 + 3 = 7.

Step 3: Thousands and hundreds stay 6 and 9.

Answer: 6979

Exercise (page 40): Q1 (v) — 8342 + 46

Step 1: Add the ones: 2 + 6 = 8.

Step 2: Add the tens: 4 + 4 = 8.

Step 3: Thousands and hundreds stay 8 and 3.

Answer: 8388

Exercise (page 40): Q1 (vi) — 9525 + 63

Step 1: Add the ones: 5 + 3 = 8.

Step 2: Add the tens: 2 + 6 = 8.

Step 3: Thousands and hundreds stay 9 and 5.

Answer: 9588

Exercise (page 40): Question 2 — Solve the following

Exercise (page 40): Q2 (i) — 2456 + 23

Step 1: Add the ones: 6 + 3 = 9.

Step 2: Add the tens: 5 + 2 = 7.

Step 3: Thousands and hundreds stay 2 and 4.

Answer: 2479

Exercise (page 40): Q2 (ii) — 6234 + 34

Step 1: Add the ones: 4 + 4 = 8.

Step 2: Add the tens: 3 + 3 = 6.

Step 3: Thousands and hundreds stay 6 and 2.

Answer: 6268

Exercise (page 40): Q2 (iii) — 4926 + 52

Step 1: Add the ones: 6 + 2 = 8.

Step 2: Add the tens: 2 + 5 = 7.

Step 3: Thousands and hundreds stay 4 and 9.

Answer: 4978

Exercise (page 40): Q2 (iv) — 5642 + 36

Step 1: Add the ones: 2 + 6 = 8.

Step 2: Add the tens: 4 + 3 = 7.

Step 3: Thousands and hundreds stay 5 and 6.

Answer: 5678

Exercise (page 40): Q2 (v) — 8343 + 43

Step 1: Add the ones: 3 + 3 = 6.

Step 2: Add the tens: 4 + 4 = 8.

Step 3: Thousands and hundreds stay 8 and 3.

Answer: 8386

Exercise (page 40): Q2 (vi) — 6152 + 65

Step 1: Add the ones: 2 + 5 = 7.

Step 2: Add the tens: 5 + 6 = 11. Write 1, carry 1 to the hundreds.

Step 3: Hundreds: 1 + 1 (carried) = 2. Thousands stays 6.

Answer: 6217

Exercise (page 40): Q2 (vii) — 3717 + 72

Step 1: Add the ones: 7 + 2 = 9.

Step 2: Add the tens: 1 + 7 = 8.

Step 3: Thousands and hundreds stay 3 and 7.

Answer: 3789

Exercise (page 40): Q2 (viii) — 9275 + 23

Step 1: Add the ones: 5 + 3 = 8.

Step 2: Add the tens: 7 + 2 = 9.

Step 3: Thousands and hundreds stay 9 and 2.

Answer: 9298

NOTE on page 40: Exercise Q2 (vi), 6152 + 65, regroups in the tens (5 + 6 = 11) although the section heading says 'without regrouping'. It is solved above with the carry shown, as printed.

Addition of 4-digit numbers and 3-digit numbers (without regrouping) (page 41)

Example (page 41): Add 3214 + 232. On the abacus, 3214 plus 232 gives 3446. So 3214 + 232 = 3446.

Exercise (page 41): Question 1 — Add the following

Exercise (page 41): Q1 (i) — 1245 + 113

Step 1: Add the ones: 5 + 3 = 8.

Step 2: Add the tens: 4 + 1 = 5.

Step 3: Add the hundreds: 2 + 1 = 3.

Step 4: Thousands stays 1.

Answer: 1358

Exercise (page 41): Q1 (ii) — 4516 + 243

Step 1: Add the ones: 6 + 3 = 9.

Step 2: Add the tens: 1 + 4 = 5.

Step 3: Add the hundreds: 5 + 2 = 7.

Step 4: Thousands stays 4.

Answer: 4759

Exercise (page 41): Q1 (iii) — 2464 + 323

Step 1: Add the ones: 4 + 3 = 7.

Step 2: Add the tens: 6 + 2 = 8.

Step 3: Add the hundreds: 4 + 3 = 7.

Step 4: Thousands stays 2.

Answer: 2787

Exercise (page 41): Q1 (iv) — 8356 + 443

Step 1: Add the ones: 6 + 3 = 9.

Step 2: Add the tens: 5 + 4 = 9.

Step 3: Add the hundreds: 3 + 4 = 7.

Step 4: Thousands stays 8.

Answer: 8799

Exercise (page 41): Q1 (v) — 6423 + 462

Step 1: Add the ones: 3 + 2 = 5.

Step 2: Add the tens: 2 + 6 = 8.

Step 3: Add the hundreds: 4 + 4 = 8.

Step 4: Thousands stays 6.

Answer: 6885

Exercise (page 41): Q1 (vi) — 9637 + 252

Step 1: Add the ones: 7 + 2 = 9.

Step 2: Add the tens: 3 + 5 = 8.

Step 3: Add the hundreds: 6 + 2 = 8.

Step 4: Thousands stays 9.

Answer: 9889

Exercise (page 41): Question 2 — Solve the following

Exercise (page 41): Q2 (i) — 2342 + 123

Step 1: Add the ones: 2 + 3 = 5.

Step 2: Add the tens: 4 + 2 = 6.

Step 3: Add the hundreds: 3 + 1 = 4.

Step 4: Thousands stays 2.

Answer: 2465

Exercise (page 41): Q2 (ii) — 3456 + 431

Step 1: Add the ones: 6 + 1 = 7.

Step 2: Add the tens: 5 + 3 = 8.

Step 3: Add the hundreds: 4 + 4 = 8.

Step 4: Thousands stays 3.

Answer: 3887

Exercise (page 41): Q2 (iii) — 5163 + 524

Step 1: Add the ones: 3 + 4 = 7.

Step 2: Add the tens: 6 + 2 = 8.

Step 3: Add the hundreds: 1 + 5 = 6.

Step 4: Thousands stays 5.

Answer: 5687

Exercise (page 41): Q2 (iv) — 1574 + 225

Step 1: Add the ones: 4 + 5 = 9.

Step 2: Add the tens: 7 + 2 = 9.

Step 3: Add the hundreds: 5 + 2 = 7.

Step 4: Thousands stays 1.

Answer: 1799

Exercise (page 41): Q2 (v) — 4123 + 645

Step 1: Add the ones: 3 + 5 = 8.

Step 2: Add the tens: 2 + 4 = 6.

Step 3: Add the hundreds: 1 + 6 = 7.

Step 4: Thousands stays 4.

Answer: 4768

Exercise (page 41): Q2 (vi) — 6321 + 447

Step 1: Add the ones: 1 + 7 = 8.

Step 2: Add the tens: 2 + 4 = 6.

Step 3: Add the hundreds: 3 + 4 = 7.

Step 4: Thousands stays 6.

Answer: 6768

Exercise (page 41): Q2 (vii) — 8462 + 126

Step 1: Add the ones: 2 + 6 = 8.

Step 2: Add the tens: 6 + 2 = 8.

Step 3: Add the hundreds: 4 + 1 = 5.

Step 4: Thousands stays 8.

Answer: 8588

Exercise (page 41): Q2 (viii) — 9325 + 634

Step 1: Add the ones: 5 + 4 = 9.

Step 2: Add the tens: 2 + 3 = 5.

Step 3: Add the hundreds: 3 + 6 = 9.

Step 4: Thousands stays 9.

Answer: 9959

Exercise (page 41): Q2 (ix) — 7213 + 274

Step 1: Add the ones: 3 + 4 = 7.

Step 2: Add the tens: 1 + 7 = 8.

Step 3: Add the hundreds: 2 + 2 = 4.

Step 4: Thousands stays 7.

Answer: 7487

Addition of two 4-digit numbers (without regrouping) (page 42)

Example (page 42): Add 4352 and 3343. On the abacus, 4352 plus 3343 gives 7695. So 4352 + 3343 = 7695.

Exercise (page 42): Question 1 — Add the following

Exercise (page 42): Q1 (i) — 1734 + 1234

Step 1: Add the ones: 4 + 4 = 8.

Step 2: Add the tens: 3 + 3 = 6.

Step 3: Add the hundreds: 7 + 2 = 9.

Step 4: Add the thousands: 1 + 1 = 2.

Answer: 2968

Exercise (page 42): Q1 (ii) — 2079 + 4320

Step 1: Add the ones: 9 + 0 = 9.

Step 2: Add the tens: 7 + 2 = 9.

Step 3: Add the hundreds: 0 + 3 = 3.

Step 4: Add the thousands: 2 + 4 = 6.

Answer: 6399

Exercise (page 42): Q1 (iii) — 4070 + 1700

Step 1: Add the ones: 0 + 0 = 0.

Step 2: Add the tens: 7 + 0 = 7.

Step 3: Add the hundreds: 0 + 7 = 7.

Step 4: Add the thousands: 4 + 1 = 5.

Answer: 5770

Exercise (page 42): Q1 (iv) — 2451 + 3327

Step 1: Add the ones: 1 + 7 = 8.

Step 2: Add the tens: 5 + 2 = 7.

Step 3: Add the hundreds: 4 + 3 = 7.

Step 4: Add the thousands: 2 + 3 = 5.

Answer: 5778

Exercise (page 42): Q1 (v) — 2737 + 4141

Step 1: Add the ones: 7 + 1 = 8.

Step 2: Add the tens: 3 + 4 = 7.

Step 3: Add the hundreds: 7 + 1 = 8.

Step 4: Add the thousands: 2 + 4 = 6.

Answer: 6878

Exercise (page 42): Q1 (vi) — 3251 + 4328

Step 1: Add the ones: 1 + 8 = 9.

Step 2: Add the tens: 5 + 2 = 7.

Step 3: Add the hundreds: 2 + 3 = 5.

Step 4: Add the thousands: 3 + 4 = 7.

Answer: 7579

Exercise (page 42): Q1 (vii) — 1273 + 2411

Step 1: Add the ones: 3 + 1 = 4.

Step 2: Add the tens: 7 + 1 = 8.

Step 3: Add the hundreds: 2 + 4 = 6.

Step 4: Add the thousands: 1 + 2 = 3.

Answer: 3684

Exercise (page 42): Q1 (viii) — 6450 + 2334

Step 1: Add the ones: 0 + 4 = 4.

Step 2: Add the tens: 5 + 3 = 8.

Step 3: Add the hundreds: 4 + 3 = 7.

Step 4: Add the thousands: 6 + 2 = 8.

Answer: 8784

Exercise (page 42): Q1 (ix) — 7346 + 1242

Step 1: Add the ones: 6 + 2 = 8.

Step 2: Add the tens: 4 + 4 = 8.

Step 3: Add the hundreds: 3 + 2 = 5.

Step 4: Add the thousands: 7 + 1 = 8.

Answer: 8588

Addition of 4-digit numbers and 1-digit numbers (with regrouping) (page 43)

Example (page 43): Add 1564 + 7. Step 1: Add ones: 4 ones + 7 ones = 11 ones. Regroup 11 ones = 1 ten + 1 one. Write 1 in the ones place, carry 1 ten to tens. Step 2: Add tens: 1 ten + 6 tens = 7 tens. Write 7 in the tens place. Step 3: Write 5 in the hundreds place. Step 4: Write 1 in the thousands place. So 1564 + 7 = 1571.

Exercise (page 43): Question 1 — Add the following

Exercise (page 43): Q1 (i) — 1368 + 3

Step 1: Add the ones: 8 + 3 = 11. Write 1, carry 1 to the tens.

Step 2: Tens: 6 + 1 (carried) = 7.

Step 3: Hundreds and thousands stay 3 and 1.

Answer: 1371

Exercise (page 43): Q1 (ii) — 3697 + 5

Step 1: Add the ones: 7 + 5 = 12. Write 2, carry 1 to the tens.

Step 2: Tens: 9 + 1 (carried) = 10. Write 0, carry 1 to the hundreds.

Step 3: Hundreds: 6 + 1 (carried) = 7. Thousands stays 3.

Answer: 3702

Exercise (page 43): Q1 (iii) — 2586 + 8

Step 1: Add the ones: 6 + 8 = 14. Write 4, carry 1 to the tens.

Step 2: Tens: 8 + 1 (carried) = 9.

Step 3: Hundreds and thousands stay 5 and 2.

Answer: 2594

Exercise (page 43): Q1 (iv) — 5294 + 9

Step 1: Add the ones: 4 + 9 = 13. Write 3, carry 1 to the tens.

Step 2: Tens: 9 + 1 (carried) = 10. Write 0, carry 1 to the hundreds.

Step 3: Hundreds: 2 + 1 (carried) = 3. Thousands stays 5.

Answer: 5303

Exercise (page 43): Q1 (v) — 6459 + 6

Step 1: Add the ones: 9 + 6 = 15. Write 5, carry 1 to the tens.

Step 2: Tens: 5 + 1 (carried) = 6.

Step 3: Hundreds and thousands stay 4 and 6.

Answer: 6465

Exercise (page 43): Q1 (vi) — 4379 + 2

Step 1: Add the ones: 9 + 2 = 11. Write 1, carry 1 to the tens.

Step 2: Tens: 7 + 1 (carried) = 8.

Step 3: Hundreds and thousands stay 3 and 4.

Answer: 4381

Exercise (page 43): Q1 (vii) — 6179 + 7

Step 1: Add the ones: 9 + 7 = 16. Write 6, carry 1 to the tens.

Step 2: Tens: 7 + 1 (carried) = 8.

Step 3: Hundreds and thousands stay 1 and 6.

Answer: 6186

Exercise (page 43): Q1 (viii) — 5498 + 9

Step 1: Add the ones: 8 + 9 = 17. Write 7, carry 1 to the tens.

Step 2: Tens: 9 + 1 (carried) = 10. Write 0, carry 1 to the hundreds.

Step 3: Hundreds: 4 + 1 (carried) = 5. Thousands stays 5.

Answer: 5507

Exercise (page 43): Q1 (ix) — 7859 + 7

Step 1: Add the ones: 9 + 7 = 16. Write 6, carry 1 to the tens.

Step 2: Tens: 5 + 1 (carried) = 6.

Step 3: Hundreds and thousands stay 8 and 7.

Answer: 7866

Exercise (page 43): Q1 (x) — 9476 + 7

Step 1: Add the ones: 6 + 7 = 13. Write 3, carry 1 to the tens.

Step 2: Tens: 7 + 1 (carried) = 8.

Step 3: Hundreds and thousands stay 4 and 9.

Answer: 9483

Exercise (page 43): Q1 (xi) — 7266 + 8

Step 1: Add the ones: 6 + 8 = 14. Write 4, carry 1 to the tens.

Step 2: Tens: 6 + 1 (carried) = 7.

Step 3: Hundreds and thousands stay 2 and 7.

Answer: 7274

Exercise (page 43): Q1 (xii) — 8188 + 6

Step 1: Add the ones: 8 + 6 = 14. Write 4, carry 1 to the tens.

Step 2: Tens: 8 + 1 (carried) = 9.

Step 3: Hundreds and thousands stay 1 and 8.

Answer: 8194

Addition of 4-digit numbers and 2-digit numbers

(with regrouping) (page 44)

Example (page 44): Add 1568 + 53. Step 1: Add ones: 8 ones + 3 ones = 11 ones. Regroup 11 ones = 1 ten + 1 one. Write 1 in the ones place, carry 1 ten to tens. Step 2: Add tens: 1 ten + 6 tens + 5 tens = 12 tens. Regrouping 12 tens = 1 hundred + 2 tens. Write 2 in the tens place. Carry 1 to hundred place. Step 3: Add hundreds: 1 hundred + 5 hundreds = 6 hundreds. Write 6 hundreds in the hundreds place. Step 4: Write 1 in the thousands place. So 1568 + 53 = 1621.

Exercise (page 44): Question 1 — Add the following

Exercise (page 44): Q1 (i) — 1279 + 43

Step 1: Add the ones: 9 + 3 = 12. Write 2, carry 1 to the tens.

Step 2: Add the tens: 7 + 4 + 1 (carried) = 12. Write 2, carry 1 to the hundreds.

Step 3: Hundreds: 2 + 1 (carried) = 3. Thousands stays 1.

Answer: 1322

Exercise (page 44): Q1 (ii) — 2468 + 53

Step 1: Add the ones: 8 + 3 = 11. Write 1, carry 1 to the tens.

Step 2: Add the tens: 6 + 5 + 1 (carried) = 12. Write 2, carry 1 to the hundreds.

Step 3: Hundreds: 4 + 1 (carried) = 5. Thousands stays 2.

Answer: 2521

Exercise (page 44): Q1 (iii) — 4957 + 74

Step 1: Add the ones: 7 + 4 = 11. Write 1, carry 1 to the tens.

Step 2: Add the tens: 5 + 7 + 1 (carried) = 13. Write 3, carry 1 to the

hundreds.

Step 3: Hundreds: 9 + 1 (carried) = 10. Write 0, carry 1 to the thousands.

Step 4: Thousands: 4 + 1 (carried) = 5.

Answer: 5031

Exercise (page 44): Q1 (iv) — 5386 + 26

Step 1: Add the ones: 6 + 6 = 12. Write 2, carry 1 to the tens.

Step 2: Add the tens: 8 + 2 + 1 (carried) = 11. Write 1, carry 1 to the hundreds.

Step 3: Hundreds: 3 + 1 (carried) = 4. Thousands stays 5.

Answer: 5412

Exercise (page 44): Q1 (v) — 6145 + 78

Step 1: Add the ones: 5 + 8 = 13. Write 3, carry 1 to the tens.

Step 2: Add the tens: 4 + 7 + 1 (carried) = 12. Write 2, carry 1 to the hundreds.

Step 3: Hundreds: 1 + 1 (carried) = 2. Thousands stays 6.

Answer: 6223

Exercise (page 44): Q1 (vi) — 8534 + 98

Step 1: Add the ones: 4 + 8 = 12. Write 2, carry 1 to the tens.

Step 2: Add the tens: 3 + 9 + 1 (carried) = 13. Write 3, carry 1 to the hundreds.

Step 3: Hundreds: 5 + 1 (carried) = 6. Thousands stays 8.

Answer: 8632

Exercise (page 44): Q1 (vii) — 3594 + 38

Step 1: Add the ones: 4 + 8 = 12. Write 2, carry 1 to the tens.

Step 2: Add the tens: 9 + 3 + 1 (carried) = 13. Write 3, carry 1 to the hundreds.

Step 3: Hundreds: 5 + 1 (carried) = 6. Thousands stays 3.

Answer: 3632

Exercise (page 44): Q1 (viii) — 7678 + 45

Step 1: Add the ones: 8 + 5 = 13. Write 3, carry 1 to the tens.

Step 2: Add the tens: 7 + 4 + 1 (carried) = 12. Write 2, carry 1 to the hundreds.

Step 3: Hundreds: 6 + 1 (carried) = 7. Thousands stays 7.

Answer: 7723

Exercise (page 44): Q1 (ix) — 9789 + 48

Step 1: Add the ones: 9 + 8 = 17. Write 7, carry 1 to the tens.

Step 2: Add the tens: 8 + 4 + 1 (carried) = 13. Write 3, carry 1 to the hundreds.

Step 3: Hundreds: 7 + 1 (carried) = 8. Thousands stays 9.

Answer: 9837

Exercise (page 44): Q1 (x) — 6298 + 37

Step 1: Add the ones: 8 + 7 = 15. Write 5, carry 1 to the tens.

Step 2: Add the tens: 9 + 3 + 1 (carried) = 13. Write 3, carry 1 to the hundreds.

Step 3: Hundreds: 2 + 1 (carried) = 3. Thousands stays 6.

Answer: 6335

Exercise (page 44): Q1 (xi) — 9877 + 65

Step 1: Add the ones: 7 + 5 = 12. Write 2, carry 1 to the tens.

Step 2: Add the tens: 7 + 6 + 1 (carried) = 14. Write 4, carry 1 to the

hundreds.

Step 3: Hundreds: 8 + 1 (carried) = 9. Thousands stays 9.

Answer: 9942

Exercise (page 44): Q1 (xii) — 8459 + 73

Step 1: Add the ones: 9 + 3 = 12. Write 2, carry 1 to the tens.

Step 2: Add the tens: 5 + 7 + 1 (carried) = 13. Write 3, carry 1 to the hundreds.

Step 3: Hundreds: 4 + 1 (carried) = 5. Thousands stays 8.

Answer: 8532

Addition of 4-digit numbers and 3-digit numbers (with regrouping) (page 45)

Example (page 45): Add 1876 + 235. Step 1: Add ones: 6 ones + 5 ones = 11 ones. Regroup 11 ones = 1 ten + 1 one. Write 1 in the ones place, carry 1 ten to tens. Step 2: Add tens: 1 ten + 7 tens + 3 tens = 11 tens. Regrouping 11 tens = 1 hundred + 1 ten. Write 1 in the tens place. Carried 1 to hundreds place. Step 3: Add hundreds: 1 hundred + 8 hundreds + 2 hundreds = 11 hundreds. Write 1 hundred in the hundreds place and carry 1 thousand to thousands place. Step 4: Add: 1 + 1 = 2 thousands. Write it in thousand place. So 1876 + 235 = 2111.

Exercise (page 45): Question 1 — Add the following

Exercise (page 45): Q1 (i) — 2157 + 964

Step 1: Add the ones: 7 + 4 = 11. Write 1, carry 1 to the tens.

Step 2: Add the tens: 5 + 6 + 1 (carried) = 12. Write 2, carry 1 to the hundreds.

Step 3: Add the hundreds: 1 + 9 + 1 (carried) = 11. Write 1, carry 1 to the thousands.

Step 4: Thousands: 2 + 1 (carried) = 3.

Answer: 3121

Exercise (page 45): Q1 (ii) — 4625 + 487

Step 1: Add the ones: 5 + 7 = 12. Write 2, carry 1 to the tens.

Step 2: Add the tens: 2 + 8 + 1 (carried) = 11. Write 1, carry 1 to the hundreds.

Step 3: Add the hundreds: 6 + 4 + 1 (carried) = 11. Write 1, carry 1 to the thousands.

Step 4: Thousands: 4 + 1 (carried) = 5.

Answer: 5112

Exercise (page 45): Q1 (iii) — 7245 + 865

Step 1: Add the ones: 5 + 5 = 10. Write 0, carry 1 to the tens.

Step 2: Add the tens: 4 + 6 + 1 (carried) = 11. Write 1, carry 1 to the hundreds.

Step 3: Add the hundreds: 2 + 8 + 1 (carried) = 11. Write 1, carry 1 to the thousands.

Step 4: Thousands: 7 + 1 (carried) = 8.

Answer: 8110

Exercise (page 45): Q1 (iv) — 5364 + 739

Step 1: Add the ones: 4 + 9 = 13. Write 3, carry 1 to the tens.

Step 2: Add the tens: 6 + 3 + 1 (carried) = 10. Write 0, carry 1 to the hundreds.

Step 3: Add the hundreds: 3 + 7 + 1 (carried) = 11. Write 1, carry 1 to

the thousands.

Step 4: Thousands: 5 + 1 (carried) = 6.

Answer: 6103

Exercise (page 45): Q1 (v) — 2593 + 628

Step 1: Add the ones: 3 + 8 = 11. Write 1, carry 1 to the tens.

Step 2: Add the tens: 9 + 2 + 1 (carried) = 12. Write 2, carry 1 to the hundreds.

Step 3: Add the hundreds: 5 + 6 + 1 (carried) = 12. Write 2, carry 1 to the thousands.

Step 4: Thousands: 2 + 1 (carried) = 3.

Answer: 3221

Exercise (page 45): Q1 (vi) — 5819 + 483

Step 1: Add the ones: 9 + 3 = 12. Write 2, carry 1 to the tens.

Step 2: Add the tens: 1 + 8 + 1 (carried) = 10. Write 0, carry 1 to the hundreds.

Step 3: Add the hundreds: 8 + 4 + 1 (carried) = 13. Write 3, carry 1 to the thousands.

Step 4: Thousands: 5 + 1 (carried) = 6.

Answer: 6302

Exercise (page 45): Q1 (vii) — 4278 + 945

Step 1: Add the ones: 8 + 5 = 13. Write 3, carry 1 to the tens.

Step 2: Add the tens: 7 + 4 + 1 (carried) = 12. Write 2, carry 1 to the hundreds.

Step 3: Add the hundreds: 2 + 9 + 1 (carried) = 12. Write 2, carry 1 to the thousands.

Step 4: Thousands: 4 + 1 (carried) = 5.

Answer: 5223

Exercise (page 45): Q1 (viii) — 9784 + 429

Step 1: Add the ones: 4 + 9 = 13. Write 3, carry 1 to the tens.

Step 2: Add the tens: 8 + 2 + 1 (carried) = 11. Write 1, carry 1 to the hundreds.

Step 3: Add the hundreds: 7 + 4 + 1 (carried) = 12. Write 2, carry 1 to the thousands.

Step 4: Thousands: 9 + 1 (carried) = 10.

Answer: 10213

Exercise (page 45): Q1 (ix) — 6472 + 659

Step 1: Add the ones: 2 + 9 = 11. Write 1, carry 1 to the tens.

Step 2: Add the tens: 7 + 5 + 1 (carried) = 13. Write 3, carry 1 to the hundreds.

Step 3: Add the hundreds: 4 + 6 + 1 (carried) = 11. Write 1, carry 1 to the thousands.

Step 4: Thousands: 6 + 1 (carried) = 7.

Answer: 7131

Exercise (page 45): Q1 (x) — 8938 + 778

Step 1: Add the ones: 8 + 8 = 16. Write 6, carry 1 to the tens.

Step 2: Add the tens: 3 + 7 + 1 (carried) = 11. Write 1, carry 1 to the hundreds.

Step 3: Add the hundreds: 9 + 7 + 1 (carried) = 17. Write 7, carry 1 to the thousands.

Step 4: Thousands: 8 + 1 (carried) = 9.

Answer: 9716

Exercise (page 45): Q1 (xi) — 5697 + 457

Step 1: Add the ones: 7 + 7 = 14. Write 4, carry 1 to the tens.

Step 2: Add the tens: 9 + 5 + 1 (carried) = 15. Write 5, carry 1 to the hundreds.

Step 3: Add the hundreds: 6 + 4 + 1 (carried) = 11. Write 1, carry 1 to the thousands.

Step 4: Thousands: 5 + 1 (carried) = 6.

Answer: 6154

Exercise (page 45): Q1 (xii) — 9465 + 678

Step 1: Add the ones: 5 + 8 = 13. Write 3, carry 1 to the tens.

Step 2: Add the tens: 6 + 7 + 1 (carried) = 14. Write 4, carry 1 to the hundreds.

Step 3: Add the hundreds: 4 + 6 + 1 (carried) = 11. Write 1, carry 1 to the thousands.

Step 4: Thousands: 9 + 1 (carried) = 10.

Answer: 10143

Addition of two 4-digit numbers (with regrouping)

(page 46)

Example (page 46): Add 5937 and 3289. 7 ones + 9 ones = 16 ones = 1 ten 6 ones. Write 6 in the ones place and carry 1 ten. 3 tens + 8 tens + 1 ten = 12 tens = 1 hundred 2 tens. Write 2 in the tens place and carry 1. 9 hundreds + 2 hundreds + 1 hundred = 12 hundreds = 1 thousand 2

hundreds. Write 2 in the hundred place and carry 1 thousand. 5 thousands + 3 thousands + 1 thousand = 9 thousands. Write 9 in the thousand place. So, 9226 answer.

Exercise (page 46): Question 1 — Add the following

Exercise (page 46): Q1 (i) — 1425 + 3366

Step 1: Add the ones: 5 + 6 = 11. Write 1, carry 1 to the tens.

Step 2: Add the tens: 2 + 6 + 1 (carried) = 9.

Step 3: Add the hundreds: 4 + 3 = 7.

Step 4: Add the thousands: 1 + 3 = 4.

Answer: 4791

Exercise (page 46): Q1 (ii) — 3435 + 1237

Step 1: Add the ones: 5 + 7 = 12. Write 2, carry 1 to the tens.

Step 2: Add the tens: 3 + 3 + 1 (carried) = 7.

Step 3: Add the hundreds: 4 + 2 = 6.

Step 4: Add the thousands: 3 + 1 = 4.

Answer: 4672

Exercise (page 46): Q1 (iii) — 4079 + 2695

Step 1: Add the ones: 9 + 5 = 14. Write 4, carry 1 to the tens.

Step 2: Add the tens: 7 + 9 + 1 (carried) = 17. Write 7, carry 1 to the hundreds.

Step 3: Add the hundreds: 0 + 6 + 1 (carried) = 7.

Step 4: Add the thousands: 4 + 2 = 6.

Answer: 6774

Exercise (page 46): Q1 (iv) — 4478 + 4588

Step 1: Add the ones: 8 + 8 = 16. Write 6, carry 1 to the tens.

Step 2: Add the tens: 7 + 8 + 1 (carried) = 16. Write 6, carry 1 to the hundreds.

Step 3: Add the hundreds: 4 + 5 + 1 (carried) = 10. Write 0, carry 1 to the thousands.

Step 4: Add the thousands: 4 + 4 + 1 (carried) = 9.

Answer: 9066

Exercise (page 46): Q1 (v) — 5002 + 4039

Step 1: Add the ones: 2 + 9 = 11. Write 1, carry 1 to the tens.

Step 2: Add the tens: 0 + 3 + 1 (carried) = 4.

Step 3: Add the hundreds: 0 + 0 = 0.

Step 4: Add the thousands: 5 + 4 = 9.

Answer: 9041

Exercise (page 46): Q1 (vi) — 6655 + 2555

Step 1: Add the ones: 5 + 5 = 10. Write 0, carry 1 to the tens.

Step 2: Add the tens: 5 + 5 + 1 (carried) = 11. Write 1, carry 1 to the hundreds.

Step 3: Add the hundreds: 6 + 5 + 1 (carried) = 12. Write 2, carry 1 to the thousands.

Step 4: Add the thousands: 6 + 2 + 1 (carried) = 9.

Answer: 9210

Exercise (page 46): Q1 (vii) — 8343 + 1447

Step 1: Add the ones: 3 + 7 = 10. Write 0, carry 1 to the tens.

Step 2: Add the tens: 4 + 4 + 1 (carried) = 9.

Step 3: Add the hundreds: 3 + 4 = 7.

Step 4: Add the thousands: 8 + 1 = 9.

Answer: 9790

Exercise (page 46): Q1 (viii) — 6554 + 2408

Step 1: Add the ones: 4 + 8 = 12. Write 2, carry 1 to the tens.

Step 2: Add the tens: 5 + 0 + 1 (carried) = 6.

Step 3: Add the hundreds: 5 + 4 = 9.

Step 4: Add the thousands: 6 + 2 = 8.

Answer: 8962

Exercise (page 46): Q1 (ix) — 5999 + 2555

Step 1: Add the ones: 9 + 5 = 14. Write 4, carry 1 to the tens.

Step 2: Add the tens: 9 + 5 + 1 (carried) = 15. Write 5, carry 1 to the hundreds.

Step 3: Add the hundreds: 9 + 5 + 1 (carried) = 15. Write 5, carry 1 to the thousands.

Step 4: Add the thousands: 5 + 2 + 1 (carried) = 8.

Answer: 8554

Subtraction of ones from 4-digit numbers (without regrouping) (page 47)

Example (page 47): Solve 6417 - 3. On the abacus, 6417 minus 3 ones gives 6414. So 6417 - 3 = 6414.

Exercise (page 47): Question 1 — Subtract the following

Exercise (page 47): Q1 (i) — 2615 - 3

Step 1: Subtract the ones: 5 - 3 = 2.

Step 2: Other digits stay the same.

Answer: 2612

Exercise (page 47): Q1 (ii) — 7146 - 2

Step 1: Subtract the ones: 6 - 2 = 4.

Step 2: Other digits stay the same.

Answer: 7144

Exercise (page 47): Q1 (iii) — 5377 - 4

Step 1: Subtract the ones: 7 - 4 = 3.

Step 2: Other digits stay the same.

Answer: 5373

Exercise (page 47): Q1 (iv) — 4639 - 5

Step 1: Subtract the ones: 9 - 5 = 4.

Step 2: Other digits stay the same.

Answer: 4634

Exercise (page 47): Q1 (v) — 9878 - 6

Step 1: Subtract the ones: 8 - 6 = 2.

Step 2: Other digits stay the same.

Answer: 9872

Exercise (page 47): Q1 (vi) — 8259 - 8

Step 1: Subtract the ones: 9 - 8 = 1.

Step 2: Other digits stay the same.

Answer: 8251

Exercise (page 47): Question 2 — Solve the following

Exercise (page 47): Q2 (i) — 1736 - 4

Step 1: Subtract the ones: 6 - 4 = 2.

Step 2: Other digits stay the same.

Answer: 1732

Exercise (page 47): Q2 (ii) — 4368 - 6

Step 1: Subtract the ones: 8 - 6 = 2.

Step 2: Other digits stay the same.

Answer: 4362

Exercise (page 47): Q2 (iii) — 7249 - 7

Step 1: Subtract the ones: 9 - 7 = 2.

Step 2: Other digits stay the same.

Answer: 7242

Exercise (page 47): Q2 (iv) — 5275 - 3

Step 1: Subtract the ones: 5 - 3 = 2.

Step 2: Other digits stay the same.

Answer: 5272

Exercise (page 47): Q2 (v) — 6157 - 5

Step 1: Subtract the ones: 7 - 5 = 2.

Step 2: Other digits stay the same.

Answer: 6152

Exercise (page 47): Q2 (vi) — 8349 - 6

Step 1: Subtract the ones: 9 - 6 = 3.

Step 2: Other digits stay the same.

Answer: 8343

Exercise (page 47): Q2 (vii) — 3849 - 5

Step 1: Subtract the ones: 9 - 5 = 4.

Step 2: Other digits stay the same.

Answer: 3844

Exercise (page 47): Q2 (viii) — 9258 - 6

Step 1: Subtract the ones: 8 - 6 = 2.

Step 2: Other digits stay the same.

Answer: 9252

Exercise (page 47): Q2 (ix) — 5947 - 4

Step 1: Subtract the ones: 7 - 4 = 3.

Step 2: Other digits stay the same.

Answer: 5943

Subtraction of 2-digit numbers from 4-digit numbers (without regrouping) (page 48)

Example (page 48): Solve 8563 - 31. On the abacus, 8563 minus 31 gives 8532. So 8563 - 31 = 8532.

Exercise (page 48): Question 1 — Subtract the following

Exercise (page 48): Q1 (i) — 3454 - 32

Step 1: Subtract the ones: 4 - 2 = 2.

Step 2: Subtract the tens: 5 - 3 = 2.

Step 3: Thousands and hundreds stay 3 and 4.

Answer: 3422

Exercise (page 48): Q1 (ii) — 5078 - 56

Step 1: Subtract the ones: 8 - 6 = 2.

Step 2: Subtract the tens: 7 - 5 = 2.

Step 3: Thousands and hundreds stay 5 and 0.

Answer: 5022

Exercise (page 48): Q1 (iii) — 2135 - 24

Step 1: Subtract the ones: 5 - 4 = 1.

Step 2: Subtract the tens: 3 - 2 = 1.

Step 3: Thousands and hundreds stay 2 and 1.

Answer: 2111

Exercise (page 48): Q1 (iv) — 8479 - 45

Step 1: Subtract the ones: 9 - 5 = 4.

Step 2: Subtract the tens: 7 - 4 = 3.

Step 3: Thousands and hundreds stay 8 and 4.

Answer: 8434

Exercise (page 48): Q1 (v) — 6268 - 27

Step 1: Subtract the ones: 8 - 7 = 1.

Step 2: Subtract the tens: 6 - 2 = 4.

Step 3: Thousands and hundreds stay 6 and 2.

Answer: 6241

Exercise (page 48): Q1 (vi) — 9547 - 34

Step 1: Subtract the ones: 7 - 4 = 3.

Step 2: Subtract the tens: 4 - 3 = 1.

Step 3: Thousands and hundreds stay 9 and 5.

Answer: 9513

Exercise (page 48): Question 2 — Solve the following

Exercise (page 48): Q2 (i) — 1647 - 35

Step 1: Subtract the ones: 7 - 5 = 2.

Step 2: Subtract the tens: 4 - 3 = 1.

Step 3: Thousands and hundreds stay 1 and 6.

Answer: 1612

Exercise (page 48): Q2 (ii) — 6055 - 24

Step 1: Subtract the ones: 5 - 4 = 1.

Step 2: Subtract the tens: 5 - 2 = 3.

Step 3: Thousands and hundreds stay 6 and 0.

Answer: 6031

Exercise (page 48): Q2 (iii) — 3278 - 46

Step 1: Subtract the ones: 8 - 6 = 2.

Step 2: Subtract the tens: 7 - 4 = 3.

Step 3: Thousands and hundreds stay 3 and 2.

Answer: 3232

Exercise (page 48): Q2 (iv) — 8175 - 44

Step 1: Subtract the ones: 5 - 4 = 1.

Step 2: Subtract the tens: 7 - 4 = 3.

Step 3: Thousands and hundreds stay 8 and 1.

Answer: 8131

Exercise (page 48): Q2 (v) — 5748 - 36

Step 1: Subtract the ones: 8 - 6 = 2.

Step 2: Subtract the tens: 4 - 3 = 1.

Step 3: Thousands and hundreds stay 5 and 7.

Answer: 5712

Exercise (page 48): Q2 (vi) — 9356 - 33

Step 1: Subtract the ones: 6 - 3 = 3.

Step 2: Subtract the tens: 5 - 3 = 2.

Step 3: Thousands and hundreds stay 9 and 3.

Answer: 9323

Exercise (page 48): Q2 (vii) — 4399 - 75

Step 1: Subtract the ones: 9 - 5 = 4.

Step 2: Subtract the tens: 9 - 7 = 2.

Step 3: Thousands and hundreds stay 4 and 3.

Answer: 4324

Exercise (page 48): Q2 (viii) — 3567 - 65

Step 1: Subtract the ones: 7 - 5 = 2.

Step 2: Subtract the tens: 6 - 6 = 0.

Step 3: Thousands and hundreds stay 3 and 5.

Answer: 3502

Exercise (page 48): Q2 (ix) — 7185 - 54

Step 1: Subtract the ones: 5 - 4 = 1.

Step 2: Subtract the tens: 8 - 5 = 3.

Step 3: Thousands and hundreds stay 7 and 1.

Answer: 7131

Subtraction of 3-digit numbers from 4-digit

numbers (without regrouping) (page 49)

Example (page 49): Solve 1532 - 321. On the abacus, 1532 minus 321 gives 1211. So 1532 - 321 = 1211.

Exercise (page 49): Question 1 — Subtract the following

Exercise (page 49): Q1 (i) — 2564 - 342

Step 1: Subtract the ones: 4 - 2 = 2.

Step 2: Subtract the tens: 6 - 4 = 2.

Step 3: Subtract the hundreds: 5 - 3 = 2.

Step 4: Thousands stays 2.

Answer: 2222

Exercise (page 49): Q1 (ii) — 6385 - 374

Step 1: Subtract the ones: 5 - 4 = 1.

Step 2: Subtract the tens: 8 - 7 = 1.

Step 3: Subtract the hundreds: 3 - 3 = 0.

Step 4: Thousands stays 6.

Answer: 6011

Exercise (page 49): Q1 (iii) — 1569 - 438

Step 1: Subtract the ones: 9 - 8 = 1.

Step 2: Subtract the tens: 6 - 3 = 3.

Step 3: Subtract the hundreds: 5 - 4 = 1.

Step 4: Thousands stays 1.

Answer: 1131

Exercise (page 49): Q1 (iv) — 4843 - 732

Step 1: Subtract the ones: 3 - 2 = 1.

Step 2: Subtract the tens: 4 - 3 = 1.

Step 3: Subtract the hundreds: 8 - 7 = 1.

Step 4: Thousands stays 4.

Answer: 4111

Exercise (page 49): Q1 (v) — 3695 - 261

Step 1: Subtract the ones: 5 - 1 = 4.

Step 2: Subtract the tens: 9 - 6 = 3.

Step 3: Subtract the hundreds: 6 - 2 = 4.

Step 4: Thousands stays 3.

Answer: 3434

Exercise (page 49): Q1 (vi) — 7482 - 161

Step 1: Subtract the ones: 2 - 1 = 1.

Step 2: Subtract the tens: 8 - 6 = 2.

Step 3: Subtract the hundreds: 4 - 1 = 3.

Step 4: Thousands stays 7.

Answer: 7321

Exercise (page 49): Question 2 — Solve the following

Exercise (page 49): Q2 (i) — 1532 - 421

Step 1: Subtract the ones: 2 - 1 = 1.

Step 2: Subtract the tens: 3 - 2 = 1.

Step 3: Subtract the hundreds: 5 - 4 = 1.

Step 4: Thousands stays 1.

Answer: 1111

Exercise (page 49): Q2 (ii) — 3274 - 152

Step 1: Subtract the ones: 4 - 2 = 2.

Step 2: Subtract the tens: 7 - 5 = 2.

Step 3: Subtract the hundreds: 2 - 1 = 1.

Step 4: Thousands stays 3.

Answer: 3122

Exercise (page 49): Q2 (iii) — 5786 - 453

Step 1: Subtract the ones: 6 - 3 = 3.

Step 2: Subtract the tens: 8 - 5 = 3.

Step 3: Subtract the hundreds: 7 - 4 = 3.

Step 4: Thousands stays 5.

Answer: 5333

Exercise (page 49): Q2 (iv) — 2644 - 532

Step 1: Subtract the ones: 4 - 2 = 2.

Step 2: Subtract the tens: 4 - 3 = 1.

Step 3: Subtract the hundreds: 6 - 5 = 1.

Step 4: Thousands stays 2.

Answer: 2112

Exercise (page 49): Q2 (v) — 4358 - 137

Step 1: Subtract the ones: 8 - 7 = 1.

Step 2: Subtract the tens: 5 - 3 = 2.

Step 3: Subtract the hundreds: 3 - 1 = 2.

Step 4: Thousands stays 4.

Answer: 4221

Exercise (page 49): Q2 (vi) — 6789 - 463

Step 1: Subtract the ones: 9 - 3 = 6.

Step 2: Subtract the tens: 8 - 6 = 2.

Step 3: Subtract the hundreds: 7 - 4 = 3.

Step 4: Thousands stays 6.

Answer: 6326

Exercise (page 49): Q2 (vii) — 7528 - 214

Step 1: Subtract the ones: 8 - 4 = 4.

Step 2: Subtract the tens: 2 - 1 = 1.

Step 3: Subtract the hundreds: 5 - 2 = 3.

Step 4: Thousands stays 7.

Answer: 7314

Exercise (page 49): Q2 (viii) — 8643 - 411

Step 1: Subtract the ones: 3 - 1 = 2.

Step 2: Subtract the tens: 4 - 1 = 3.

Step 3: Subtract the hundreds: 6 - 4 = 2.

Step 4: Thousands stays 8.

Answer: 8232

Exercise (page 49): Q2 (ix) — 9786 - 644

Step 1: Subtract the ones: 6 - 4 = 2.

Step 2: Subtract the tens: 8 - 4 = 4.

Step 3: Subtract the hundreds: 7 - 6 = 1.

Step 4: Thousands stays 9.

Answer: 9142

Subtraction of two 4-digit numbers (without regrouping) (page 50)

Worked example (page 50): Subtract 6842 - 4541. Step 1: Subtract 2 ones - 1 one = 1 one. Write 1 in the ones place. Step 2: Subtract 4 tens - 4 tens = 0. Write 0 in the tens place. Step 3: Subtract 8 hundreds - 5 hundreds = 3 hundreds. Write 3 in the hundred place. Step 4: Subtract 6 thousands - 4 thousands = 2 thousands. Write 2 in the thousands place. So 6842 - 4541 = 2301. NOTE — the book's Step 1 text says 'Write 1 in tens place'; the ones answer is written in the ones place, as solved here.

Exercise (page 50): Question 1 — Subtract the following

Exercise (page 50): Q1 (i) — 1340 - 1230

Step 1: Subtract the ones: 0 - 0 = 0.

Step 2: Subtract the tens: 4 - 3 = 1.

Step 3: Subtract the hundreds: 3 - 2 = 1.

Step 4: Subtract the thousands: 1 - 1 = 0.

Answer: 110

Exercise (page 50): Q1 (ii) — 1752 - 1311

Step 1: Subtract the ones: 2 - 1 = 1.

Step 2: Subtract the tens: 5 - 1 = 4.

Step 3: Subtract the hundreds: 7 - 3 = 4.

Step 4: Subtract the thousands: 1 - 1 = 0.

Answer: 441

Exercise (page 50): Q1 (iii) — 2474 - 1363

Step 1: Subtract the ones: 4 - 3 = 1.

Step 2: Subtract the tens: 7 - 6 = 1.

Step 3: Subtract the hundreds: 4 - 3 = 1.

Step 4: Subtract the thousands: 2 - 1 = 1.

Answer: 1111

Exercise (page 50): Q1 (iv) — 3766 - 2233

Step 1: Subtract the ones: 6 - 3 = 3.

Step 2: Subtract the tens: 6 - 3 = 3.

Step 3: Subtract the hundreds: 7 - 2 = 5.

Step 4: Subtract the thousands: 3 - 2 = 1.

Answer: 1533

Exercise (page 50): Q1 (v) — 3954 - 3543

Step 1: Subtract the ones: 4 - 3 = 1.

Step 2: Subtract the tens: 5 - 4 = 1.

Step 3: Subtract the hundreds: 9 - 5 = 4.

Step 4: Subtract the thousands: 3 - 3 = 0.

Answer: 411

Exercise (page 50): Q1 (vi) — 4426 - 3600

Step 1: Subtract the ones: 6 - 0 = 6.

Step 2: Subtract the tens: 2 - 0 = 2.

Step 3: Subtract the hundreds: 4 - 6 cannot be done, so borrow 1 thousand: 14 - 6 = 8.

Step 4: Subtract the thousands: 4 - 1 (borrowed) - 3 = 0.

Answer: 826

NOTE on page 50: Exercise Q1 (vi), 4426 - 3600, needs borrowing in the hundreds (4 - 6) although the section heading says 'without regrouping'. It is solved above with the borrow shown.

Exercise (page 50): Q1 (vii) — 4756 - 4040

Step 1: Subtract the ones: 6 - 0 = 6.

Step 2: Subtract the tens: 5 - 4 = 1.

Step 3: Subtract the hundreds: 7 - 0 = 7.

Step 4: Subtract the thousands: 4 - 4 = 0.

Answer: 716

Exercise (page 50): Q1 (viii) — 5922 - 4621

Step 1: Subtract the ones: 2 - 1 = 1.

Step 2: Subtract the tens: 2 - 2 = 0.

Step 3: Subtract the hundreds: 9 - 6 = 3.

Step 4: Subtract the thousands: 5 - 4 = 1.

Answer: 1301

Exercise (page 50): Q1 (ix) — 6462 - 5252

Step 1: Subtract the ones: 2 - 2 = 0.

Step 2: Subtract the tens: 6 - 5 = 1.

Step 3: Subtract the hundreds: 4 - 2 = 2.

Step 4: Subtract the thousands: 6 - 5 = 1.

Answer: 1210

Exercise (page 50): Question 2 — Solve the following

Exercise (page 50): Q2 (i) — 8888 - 5324

Step 1: Subtract the ones: 8 - 4 = 4.

Step 2: Subtract the tens: 8 - 2 = 6.

Step 3: Subtract the hundreds: 8 - 3 = 5.

Step 4: Subtract the thousands: 8 - 5 = 3.

Answer: 3564

Exercise (page 50): Q2 (ii) — 6850 - 4540

Step 1: Subtract the ones: 0 - 0 = 0.

Step 2: Subtract the tens: 5 - 4 = 1.

Step 3: Subtract the hundreds: 8 - 5 = 3.

Step 4: Subtract the thousands: 6 - 4 = 2.

Answer: 2310

Exercise (page 50): Q2 (iii) — 4891 - 1520

Step 1: Subtract the ones: 1 - 0 = 1.

Step 2: Subtract the tens: 9 - 2 = 7.

Step 3: Subtract the hundreds: 8 - 5 = 3.

Step 4: Subtract the thousands: 4 - 1 = 3.

Answer: 3371

Exercise (page 50): Q2 (iv) — 8967 - 4351

Step 1: Subtract the ones: 7 - 1 = 6.

Step 2: Subtract the tens: 6 - 5 = 1.

Step 3: Subtract the hundreds: 9 - 3 = 6.

Step 4: Subtract the thousands: 8 - 4 = 4.

Answer: 4616

Exercise (page 50): Q2 (v) — 8756 - 3542

Step 1: Subtract the ones: 6 - 2 = 4.

Step 2: Subtract the tens: 5 - 4 = 1.

Step 3: Subtract the hundreds: 7 - 5 = 2.

Step 4: Subtract the thousands: 8 - 3 = 5.

Answer: 5214

Exercise (page 50): Q2 (vi) — 5794 - 3683

Step 1: Subtract the ones: 4 - 3 = 1.

Step 2: Subtract the tens: 9 - 8 = 1.

Step 3: Subtract the hundreds: 7 - 6 = 1.

Step 4: Subtract the thousands: 5 - 3 = 2.

Answer: 2111

Subtraction of ones from 4-digit numbers (with regrouping) (page 51)

Example (page 51): Solve 2351 - 3. Step 1: Regroup tens and ones. Regroup 1 ten from 5 tens. We have 11 ones. Now subtract: 11 - 3 = 8. Write 8 ones in the ones place. Step 2: Write 4 tens in the tens place. Step 3: Write 3 hundreds in the hundred place. Step 4: Write 2 thousands in the thousand place. So 2351 - 3 = 2348.

Exercise (page 51): Question 1 — Subtract the following

Exercise (page 51): Q1 (i) — 1362 - 5

Step 1: Ones: 2 - 5 cannot be done, so regroup 1 ten: 12 - 5 = 7.

Step 2: Tens: 6 - 1 (regrouped) = 5. Hundreds and thousands stay 3 and 1.

Answer: 1357

Exercise (page 51): Q1 (ii) — 3541 - 2

Step 1: Ones: 1 - 2 cannot be done, so regroup 1 ten: 11 - 2 = 9.

Step 2: Tens: 4 - 1 (regrouped) = 3. Hundreds and thousands stay 5 and 3.

Answer: 3539

Exercise (page 51): Q1 (iii) — 5785 - 7

Step 1: Ones: 5 - 7 cannot be done, so regroup 1 ten: 15 - 7 = 8.

Step 2: Tens: 8 - 1 (regrouped) = 7. Hundreds and thousands stay 7 and 5.

Answer: 5778

Exercise (page 51): Q1 (iv) — 2673 - 6

Step 1: Ones: 3 - 6 cannot be done, so regroup 1 ten: 13 - 6 = 7.

Step 2: Tens: 7 - 1 (regrouped) = 6. Hundreds and thousands stay 6 and 2.

Answer: 2667

Exercise (page 51): Q1 (v) — 6131 - 4

Step 1: Ones: 1 - 4 cannot be done, so regroup 1 ten: 11 - 4 = 7.

Step 2: Tens: 3 - 1 (regrouped) = 2. Hundreds and thousands stay 1 and 6.

Answer: 6127

Exercise (page 51): Q1 (vi) — 7484 - 9

Step 1: Ones: 4 - 9 cannot be done, so regroup 1 ten: 14 - 9 = 5.

Step 2: Tens: 8 - 1 (regrouped) = 7. Hundreds and thousands stay 4 and 7.

Answer: 7475

Exercise (page 51): Q1 (vii) — 4364 - 5

Step 1: Ones: 4 - 5 cannot be done, so regroup 1 ten: 14 - 5 = 9.

Step 2: Tens: 6 - 1 (regrouped) = 5. Hundreds and thousands stay 3 and 4.

Answer: 4359

Exercise (page 51): Q1 (viii) — 8576 - 9

Step 1: Ones: 6 - 9 cannot be done, so regroup 1 ten: 16 - 9 = 7.

Step 2: Tens: 7 - 1 (regrouped) = 6. Hundreds and thousands stay 5 and 8.

Answer: 8567

Exercise (page 51): Q1 (ix) — 6341 - 3

Step 1: Ones: 1 - 3 cannot be done, so regroup 1 ten: 11 - 3 = 8.

Step 2: Tens: 4 - 1 (regrouped) = 3. Hundreds and thousands stay 3 and 6.

Answer: 6338

Exercise (page 51): Q1 (x) — 9685 - 7

Step 1: Ones: 5 - 7 cannot be done, so regroup 1 ten: 15 - 7 = 8.

Step 2: Tens: 8 - 1 (regrouped) = 7. Hundreds and thousands stay 6 and 9.

Answer: 9678

Exercise (page 51): Q1 (xi) — 5163 - 6

Step 1: Ones: 3 - 6 cannot be done, so regroup 1 ten: 13 - 6 = 7.

Step 2: Tens: 6 - 1 (regrouped) = 5. Hundreds and thousands stay 1 and 5.

Answer: 5157

Exercise (page 51): Q1 (xii) — 9254 - 8

Step 1: Ones: 4 - 8 cannot be done, so regroup 1 ten: 14 - 8 = 6.

Step 2: Tens: 5 - 1 (regrouped) = 4. Hundreds and thousands stay 2 and 9.

Answer: 9246

Subtraction of 2-digit numbers from 4-digit numbers (with regrouping) (page 52)

Example (page 52): Solve 3263 - 76. Step 1: Regroup tens and ones. Take 1 ten from 6 tens. We have 13 ones. Now subtract (13 - 6 = 7). Write 7 ones in the ones place. Step 2: Take 1 hundred from 2 hundreds. We have 15 tens. Now subtract: 15 - 7 = 8. Write 8 in the tens place. Step 3: Write 1 hundred in the hundred place. Step 4: Write 3 thousands in the thousand place. So 3263 - 76 = 3187.

Exercise (page 52): Question 1 — Subtract the following

Exercise (page 52): Q1 (i) — 1246 - 68

Step 1: Ones: 6 - 8 cannot be done, so regroup 1 ten: 16 - 8 = 8.

Step 2: Tens: 4 - 1 (regrouped) = 3; 3 - 6 cannot be done, so regroup 1 hundred: 13 - 6 = 7.

Step 3: Hundreds: 2 - 1 (regrouped) = 1. Thousands stays 1.

Answer: 1178

Exercise (page 52): Q1 (ii) — 4685 - 76

Step 1: Ones: 5 - 6 cannot be done, so regroup 1 ten: 15 - 6 = 9.

Step 2: Tens: 8 - 1 (regrouped) - 7 = 0.

Step 3: Hundreds and thousands stay 6 and 4.

Answer: 4609

Exercise (page 52): Q1 (iii) — 6342 - 84

Step 1: Ones: 2 - 4 cannot be done, so regroup 1 ten: 12 - 4 = 8.

Step 2: Tens: 4 - 1 (regrouped) = 3; 3 - 8 cannot be done, so regroup 1 hundred: 13 - 8 = 5.

Step 3: Hundreds: 3 - 1 (regrouped) = 2. Thousands stays 6.

Answer: 6258

Exercise (page 52): Q1 (iv) — 1286 - 59

Step 1: Ones: 6 - 9 cannot be done, so regroup 1 ten: 16 - 9 = 7.

Step 2: Tens: 8 - 1 (regrouped) - 5 = 2.

Step 3: Hundreds and thousands stay 2 and 1.

Answer: 1227

Exercise (page 52): Q1 (v) — 7458 - 79

Step 1: Ones: 8 - 9 cannot be done, so regroup 1 ten: 18 - 9 = 9.

Step 2: Tens: 5 - 1 (regrouped) = 4; 4 - 7 cannot be done, so regroup 1 hundred: 14 - 7 = 7.

Step 3: Hundreds: 4 - 1 (regrouped) = 3. Thousands stays 7.

Answer: 7379

Exercise (page 52): Q1 (vi) — 5263 - 77

Step 1: Ones: 3 - 7 cannot be done, so regroup 1 ten: 13 - 7 = 6.

Step 2: Tens: 6 - 1 (regrouped) = 5; 5 - 7 cannot be done, so regroup 1 hundred: 15 - 7 = 8.

Step 3: Hundreds: 2 - 1 (regrouped) = 1. Thousands stays 5.

Answer: 5186

Exercise (page 52): Q1 (vii) — 8452 - 64

Step 1: Ones: 2 - 4 cannot be done, so regroup 1 ten: 12 - 4 = 8.

Step 2: Tens: 5 - 1 (regrouped) = 4; 4 - 6 cannot be done, so regroup 1 hundred: 14 - 6 = 8.

Step 3: Hundreds: 4 - 1 (regrouped) = 3. Thousands stays 8.

Answer: 8388

Exercise (page 52): Q1 (viii) — 3133 - 55

Step 1: Ones: 3 - 5 cannot be done, so regroup 1 ten: 13 - 5 = 8.

Step 2: Tens: 3 - 1 (regrouped) = 2; 2 - 5 cannot be done, so regroup 1 hundred: 12 - 5 = 7.

Step 3: Hundreds: 1 - 1 (regrouped) = 0. Thousands stays 3.

Answer: 3078

Exercise (page 52): Q1 (ix) — 9452 - 54

Step 1: Ones: 2 - 4 cannot be done, so regroup 1 ten: 12 - 4 = 8.

Step 2: Tens: 5 - 1 (regrouped) = 4; 4 - 5 cannot be done, so regroup 1 hundred: 14 - 5 = 9.

Step 3: Hundreds: 4 - 1 (regrouped) = 3. Thousands stays 9.

Answer: 9398

Exercise (page 52): Q1 (x) — 6274 - 75

Step 1: Ones: 4 - 5 cannot be done, so regroup 1 ten: 14 - 5 = 9.

Step 2: Tens: 7 - 1 (regrouped) = 6; 6 - 7 cannot be done, so regroup 1 hundred: 16 - 7 = 9.

Step 3: Hundreds: 2 - 1 (regrouped) = 1. Thousands stays 6.

Answer: 6199

Exercise (page 52): Q1 (xi) — 8395 - 78

Step 1: Ones: 5 - 8 cannot be done, so regroup 1 ten: 15 - 8 = 7.

Step 2: Tens: 9 - 1 (regrouped) - 7 = 1.

Step 3: Hundreds and thousands stay 3 and 8.

Answer: 8317

Exercise (page 52): Q1 (xii) — 9174 - 95

Step 1: Ones: 4 - 5 cannot be done, so regroup 1 ten: 14 - 5 = 9.

Step 2: Tens: 7 - 1 (regrouped) = 6; 6 - 9 cannot be done, so regroup 1 hundred: 16 - 9 = 7.

Step 3: Hundreds: 1 - 1 (regrouped) = 0. Thousands stays 9.

Answer: 9079

Subtraction of 3-digit numbers from 4-digit numbers (with regrouping) (page 53)

Example (page 53): Solve 5943 - 754. Step 1: Regrouping tens and ones. Take 1 ten from 4 tens. We have 13 ones. Now subtract: 13 - 4 = 9. Write 9 ones in the ones place. Step 2: Take 1 hundred from 9 hundreds. We have 13 tens. Now we will subtract: 13 - 5 = 8. Write 8 tens in the tens place. Step 3: Subtract hundreds: 8 - 7 = 1. Write 1 hundred in the hundred place. Step 4: Write 5 thousands as it is in the thousand place. So 5943 - 754 = 5189.

Exercise (page 53): Question 1 — Subtract the following

Exercise (page 53): Q1 (i) — 1354 - 265

Step 1: Ones: 4 - 5 cannot be done, so regroup 1 ten: 14 - 5 = 9.

Step 2: Tens: 5 - 1 (regrouped) = 4; 4 - 6 cannot be done, so regroup 1 hundred: 14 - 6 = 8.

Step 3: Hundreds: 3 - 1 (regrouped) - 2 = 0. Thousands stays 1.

Answer: 1089

Exercise (page 53): Q1 (ii) — 3462 - 563

Step 1: Ones: 2 - 3 cannot be done, so regroup 1 ten: 12 - 3 = 9.

Step 2: Tens: 6 - 1 (regrouped) = 5; 5 - 6 cannot be done, so regroup 1 hundred: 15 - 6 = 9.

Step 3: Hundreds: 4 - 1 (regrouped) = 3; 3 - 5 cannot be done, so regroup 1 thousand: 13 - 5 = 8.

Step 4: Thousands: 3 - 1 (regrouped) = 2.

Answer: 2899

Exercise (page 53): Q1 (iii) — 4271 - 263

Step 1: Ones: 1 - 3 cannot be done, so regroup 1 ten: 11 - 3 = 8.

Step 2: Tens: 7 - 1 (regrouped) - 6 = 0.

Step 3: Hundreds: 2 - 2 = 0. Thousands stays 4.

Answer: 4008

Exercise (page 53): Q1 (iv) — 2675 - 686

Step 1: Ones: 5 - 6 cannot be done, so regroup 1 ten: 15 - 6 = 9.

Step 2: Tens: 7 - 1 (regrouped) = 6; 6 - 8 cannot be done, so regroup 1 hundred: 16 - 8 = 8.

Step 3: Hundreds: 6 - 1 (regrouped) = 5; 5 - 6 cannot be done, so regroup 1 thousand: 15 - 6 = 9.

Step 4: Thousands: 2 - 1 (regrouped) = 1.

Answer: 1989

Exercise (page 53): Q1 (v) — 3495 - 398

Step 1: Ones: 5 - 8 cannot be done, so regroup 1 ten: 15 - 8 = 7.

Step 2: Tens: 9 - 1 (regrouped) = 8; 8 - 9 cannot be done, so regroup 1 hundred: 18 - 9 = 9.

Step 3: Hundreds: 4 - 1 (regrouped) - 3 = 0. Thousands stays 3.

Answer: 3097

Exercise (page 53): Q1 (vi) — 5455 - 267

Step 1: Ones: 5 - 7 cannot be done, so regroup 1 ten: 15 - 7 = 8.

Step 2: Tens: 5 - 1 (regrouped) = 4; 4 - 6 cannot be done, so regroup 1 hundred: 14 - 6 = 8.

Step 3: Hundreds: 4 - 1 (regrouped) - 2 = 1. Thousands stays 5.

Answer: 5188

Exercise (page 53): Q1 (vii) — 6844 - 455

Step 1: Ones: 4 - 5 cannot be done, so regroup 1 ten: 14 - 5 = 9.

Step 2: Tens: 4 - 1 (regrouped) = 3; 3 - 5 cannot be done, so regroup 1 hundred: 13 - 5 = 8.

Step 3: Hundreds: 8 - 1 (regrouped) - 4 = 3. Thousands stays 6.

Answer: 6389

Exercise (page 53): Q1 (viii) — 7527 - 538

Step 1: Ones: 7 - 8 cannot be done, so regroup 1 ten: 17 - 8 = 9.

Step 2: Tens: 2 - 1 (regrouped) = 1; 1 - 3 cannot be done, so regroup 1 hundred: 11 - 3 = 8.

Step 3: Hundreds: 5 - 1 (regrouped) = 4; 4 - 5 cannot be done, so regroup 1 thousand: 14 - 5 = 9.

Step 4: Thousands: 7 - 1 (regrouped) = 6.

Answer: 6989

Exercise (page 53): Q1 (ix) — 6458 - 639

Step 1: Ones: 8 - 9 cannot be done, so regroup 1 ten: 18 - 9 = 9.

Step 2: Tens: 5 - 1 (regrouped) - 3 = 1.

Step 3: Hundreds: 4 - 6 cannot be done, so regroup 1 thousand: 14 - 6 = 8.

Step 4: Thousands: 6 - 1 (regrouped) = 5.

Answer: 5819

Exercise (page 53): Q1 (x) — 5431 - 532

Step 1: Ones: 1 - 2 cannot be done, so regroup 1 ten: 11 - 2 = 9.

Step 2: Tens: 3 - 1 (regrouped) = 2; 2 - 3 cannot be done, so regroup 1 hundred: 12 - 3 = 9.

Step 3: Hundreds: 4 - 1 (regrouped) = 3; 3 - 5 cannot be done, so regroup 1 thousand: 13 - 5 = 8.

Step 4: Thousands: 5 - 1 (regrouped) = 4.

Answer: 4899

Exercise (page 53): Q1 (xi) — 8642 - 453

Step 1: Ones: 2 - 3 cannot be done, so regroup 1 ten: 12 - 3 = 9.

Step 2: Tens: 4 - 1 (regrouped) = 3; 3 - 5 cannot be done, so regroup 1 hundred: 13 - 5 = 8.

Step 3: Hundreds: 6 - 1 (regrouped) - 4 = 1. Thousands stays 8.

Answer: 8189

Exercise (page 53): Q1 (xii) — 9786 - 897

Step 1: Ones: 6 - 7 cannot be done, so regroup 1 ten: 16 - 7 = 9.

Step 2: Tens: 8 - 1 (regrouped) = 7; 7 - 9 cannot be done, so regroup 1

hundred: 17 - 9 = 8.

Step 3: Hundreds: 7 - 1 (regrouped) = 6; 6 - 8 cannot be done, so regroup 1 thousand: 16 - 8 = 8.

Step 4: Thousands: 9 - 1 (regrouped) = 8.

Answer: 8889

Subtraction of two 4-digit numbers (with regrouping) (page 54)

Worked example (page 54): Subtract 5243 - 2787. Step 1: Take 1 ten from 4 tens. We have 13 ones. Subtract: 13 - 7 = 6. Write 6 in the ones place. Step 2: Take 1 hundred from 2 hundreds. We have 13 tens. Subtract: 13 - 8 = 5. Write 5 in the tens place. Step 3: Take 1 thousand from 5 thousands. We will have 11 hundreds. Subtract: 11 - 7 = 4. Write 4 hundreds in the hundred place. Step 4: Subtract: 4 - 2 = 2 thousands. Write 2 thousands in the thousand place. So 5243 - 2787 = 2456.

Exercise (page 54): Question 1 — Subtract the following

Exercise (page 54): Q1 (i) — 2325 - 1228

Step 1: Ones: 5 - 8 cannot be done, so regroup 1 ten: 15 - 8 = 7.

Step 2: Tens: 2 - 1 (regrouped) = 1; 1 - 2 cannot be done, so regroup 1 hundred: 11 - 2 = 9.

Step 3: Hundreds: 3 - 1 (regrouped) - 2 = 0.

Step 4: Thousands: 2 - 1 = 1.

Answer: 1097

Exercise (page 54): Q1 (ii) — 2166 - 2037

Step 1: Ones: 6 - 7 cannot be done, so regroup 1 ten: 16 - 7 = 9.

Step 2: Tens: 6 - 1 (regrouped) - 3 = 2.

Step 3: Hundreds: 1 - 0 = 1.

Step 4: Thousands: 2 - 2 = 0.

Answer: 129

Exercise (page 54): Q1 (iii) — 3834 - 3446

Step 1: Ones: 4 - 6 cannot be done, so regroup 1 ten: 14 - 6 = 8.

Step 2: Tens: 3 - 1 (regrouped) = 2; 2 - 4 cannot be done, so regroup 1 hundred: 12 - 4 = 8.

Step 3: Hundreds: 8 - 1 (regrouped) - 4 = 3.

Step 4: Thousands: 3 - 3 = 0.

Answer: 388

Exercise (page 54): Q1 (iv) — 4962 - 4058

Step 1: Ones: 2 - 8 cannot be done, so regroup 1 ten: 12 - 8 = 4.

Step 2: Tens: 6 - 1 (regrouped) - 5 = 0.

Step 3: Hundreds: 9 - 0 = 9.

Step 4: Thousands: 4 - 4 = 0.

Answer: 904

Exercise (page 54): Q1 (v) — 4435 - 2259

Step 1: Ones: 5 - 9 cannot be done, so regroup 1 ten: 15 - 9 = 6.

Step 2: Tens: 3 - 1 (regrouped) = 2; 2 - 5 cannot be done, so regroup 1 hundred: 12 - 5 = 7.

Step 3: Hundreds: 4 - 1 (regrouped) - 2 = 1.

Step 4: Thousands: 4 - 2 = 2.

Answer: 2176

Exercise (page 54): Q1 (vi) — 5576 - 3357

Step 1: Ones: 6 - 7 cannot be done, so regroup 1 ten: 16 - 7 = 9.

Step 2: Tens: 7 - 1 (regrouped) - 5 = 1.

Step 3: Hundreds: 5 - 3 = 2.

Step 4: Thousands: 5 - 3 = 2.

Answer: 2219

Exercise (page 54): Q1 (vii) — 5837 - 6639 is FLAGGED, not solved: the number to be subtracted (6639) is larger than 5837, so this subtraction cannot be done by the book's Grade 3 method (the difference would be -802, which is outside Grade 3 work). Ask the teacher — the book may have misprinted one of the numbers. No corrected number has been invented here.

Exercise (page 54): Q1 (viii) — 6473 - 5258

Step 1: Ones: 3 - 8 cannot be done, so regroup 1 ten: 13 - 8 = 5.

Step 2: Tens: 7 - 1 (regrouped) - 5 = 1.

Step 3: Hundreds: 4 - 2 = 2.

Step 4: Thousands: 6 - 5 = 1.

Answer: 1215

Exercise (page 54): Q1 (ix) — 7933 - 7035

Step 1: Ones: 3 - 5 cannot be done, so regroup 1 ten: 13 - 5 = 8.

Step 2: Tens: 3 - 1 (regrouped) = 2; 2 - 3 cannot be done, so regroup 1 hundred: 12 - 3 = 9.

Step 3: Hundreds: 9 - 1 (regrouped) - 0 = 8.

Step 4: Thousands: 7 - 7 = 0.

Answer: 898

Exercise (page 54): Question 2 — Subtract the following

Exercise (page 54): Q2 (i) — 4897 - 1528

Step 1: Ones: 7 - 8 cannot be done, so regroup 1 ten: 17 - 8 = 9.

Step 2: Tens: 9 - 1 (regrouped) - 2 = 6.

Step 3: Hundreds: 8 - 5 = 3.

Step 4: Thousands: 4 - 1 = 3.

Answer: 3369

Exercise (page 54): Q2 (ii) — 7824 - 2304

Step 1: Subtract the ones: 4 - 4 = 0.

Step 2: Subtract the tens: 2 - 0 = 2.

Step 3: Subtract the hundreds: 8 - 3 = 5.

Step 4: Subtract the thousands: 7 - 2 = 5.

Answer: 5520

Exercise (page 54): Q2 (iii) — 6824 - 2307

Step 1: Ones: 4 - 7 cannot be done, so regroup 1 ten: 14 - 7 = 7.

Step 2: Tens: 2 - 1 (regrouped) - 0 = 1.

Step 3: Hundreds: 8 - 3 = 5.

Step 4: Thousands: 6 - 2 = 4.

Answer: 4517

Exercise (page 54): Q2 (iv) — 7877 - 5508

Step 1: Ones: 7 - 8 cannot be done, so regroup 1 ten: 17 - 8 = 9.

Step 2: Tens: 7 - 1 (regrouped) - 0 = 6.

Step 3: Hundreds: 8 - 5 = 3.

Step 4: Thousands: 7 - 5 = 2.

Answer: 2369

Exercise (page 54): Q2 (v) — 8050 - 3033

Step 1: Ones: 0 - 3 cannot be done, so regroup across the tens: 10 - 3 = 7 (tens becomes 4).

Step 2: Tens: 4 - 3 = 1.

Step 3: Hundreds: 0 - 0 = 0.

Step 4: Thousands: 8 - 3 = 5.

Answer: 5017

Exercise (page 54): Q2 (vi) — 9523 - 8214

Step 1: Ones: 3 - 4 cannot be done, so regroup 1 ten: 13 - 4 = 9.

Step 2: Tens: 2 - 1 (regrouped) - 1 = 0.

Step 3: Hundreds: 5 - 2 = 3.

Step 4: Thousands: 9 - 8 = 1.

Answer: 1309

Real-World Word Problems — Addition (page 55)

Example (page 55): A train travelled 2360 km on the first day and 1235 km on the second day. How many kilometres did it travel in two days?

Solution: 2360 + 1235 = 3595 km.

Exercise (page 55): Word problems

Exercise (page 55): (i) — In Malakand 1335 houses and in Waziristan 660 houses are damaged during the earthquake. How many houses are damaged in both cities?

Step 1: Add the houses: 1335 + 660.

Step 2: Ones: 5 + 0 = 5. Tens: 3 + 6 = 9. Hundreds: 3 + 6 = 9. Thousands: 1.

Answer: 1995 houses

Exercise (page 55): (ii) — A shopkeeper sold 2472 kilograms of rice in the first week and 2215 kilograms in the second week. How much rice did he sell in two weeks?

Step 1: Add the rice: 2472 + 2215.

Step 2: Ones: 2 + 5 = 7. Tens: 7 + 1 = 8. Hundreds: 4 + 2 = 6. Thousands: 2 + 2 = 4.

Answer: 4687 kilograms

Exercise (page 55): (iii) — There are 4256 motor bikes and 2639 bicycles in the city. How many motor bikes and bicycles are there altogether in the city?

Step 1: Add them: 4256 + 2639.

Step 2: Ones: 6 + 9 = 15. Write 5, carry 1.

Step 3: Tens: 5 + 3 + 1 (carried) = 9. Hundreds: 2 + 6 = 8. Thousands: 4 + 2 = 6.

Answer: 6895 altogether

Exercise (page 55): (iv) — There are 3799 apples, and 2405 mangoes in the shop. How many fruits are there in the shop altogether?

Step 1: Add them: 3799 + 2405.

Step 2: Ones: 9 + 5 = 14. Write 4, carry 1.

Step 3: Tens: 9 + 0 + 1 (carried) = 10. Write 0, carry 1.

Step 4: Hundreds: 7 + 4 + 1 (carried) = 12. Write 2, carry 1.

Step 5: Thousands: 3 + 2 + 1 (carried) = 6.

Answer: 6204 fruits

Exercise (page 55): (v) — Samad has 3492 goats and 509 cows. How many cattle does Samad have altogether?

Step 1: Add them: 3492 + 509.

Step 2: Ones: 2 + 9 = 11. Write 1, carry 1.

Step 3: Tens: 9 + 0 + 1 (carried) = 10. Write 0, carry 1.

Step 4: Hundreds: 4 + 5 + 1 (carried) = 10. Write 0, carry 1.

Step 5: Thousands: 3 + 1 (carried) = 4.

Answer: 4001 cattle

Real-World Word Problems — Subtraction (page 56)

Example (page 56): Total number of books in a bookshop is 6666, 3498 books are sold. How many books are left? Solution: 6666 - 3498 = 3168 books left.

Exercise (page 56): Word problems

Exercise (page 56): (i) — Faryal had Rs. 9656. She gave Rs. 8325 to her brother. How much money does she have left?

Step 1: Subtract: 9656 - 8325.

Step 2: Ones: 6 - 5 = 1. Tens: 5 - 2 = 3. Hundreds: 6 - 3 = 3. Thousands: 9 - 8 = 1.

Answer: Rs. 1331 left

Exercise (page 56): (ii) — A shopkeeper had 4834 packets of sweets. He sold 2423 packets of sweets. How many packets are left?

Step 1: Subtract: 4834 - 2423.

Step 2: Ones: 4 - 3 = 1. Tens: 3 - 2 = 1. Hundreds: 8 - 4 = 4. Thousands: 4 - 2 = 2.

Answer: 2411 packets left

Exercise (page 56): (iii) — In the exam 6891 students appeared. The 4371 students passed the exam. How many students failed in the exam?

Step 1: Subtract: 6891 - 4371.

Step 2: Ones: 1 - 1 = 0. Tens: 9 - 7 = 2. Hundreds: 8 - 3 = 5. Thousands: 6 - 4 = 2.

Answer: 2520 students failed

Exercise (page 56): (iv) — Ahmed has Rs. 9180 in his bank account. He took out Rs. 3597. How much money is left?

Step 1: Subtract: 9180 - 3597.

Step 2: Ones: 0 - 7 cannot be done, so regroup 1 ten: 10 - 7 = 3.

Step 3: Tens: 8 - 1 (regrouped) = 7; 7 - 9 cannot be done, so regroup 1 hundred: 17 - 9 = 8.

Step 4: Hundreds: 1 - 1 (regrouped) = 0; 0 - 5 cannot be done, so regroup 1 thousand: 10 - 5 = 5.

Step 5: Thousands: 9 - 1 (regrouped) - 3 = 5.

Answer: Rs. 5583 left

Exercise (page 56): (v) — Sadia has Rs. 8000. She gave Rs. 4255

for Zakat. Find the amount left.

Step 1: Subtract: 8000 - 4255.

Step 2: Regroup across the zeros: 8000 = 7 thousands, 9 hundreds, 9 tens, 10 ones.

Step 3: Ones: 10 - 5 = 5. Tens: 9 - 5 = 4. Hundreds: 9 - 2 = 7. Thousands: 7 - 4 = 3.

Answer: Rs. 3745 left

Estimating Sums (pages 57 to 58)

Estimation is helpful when we are adding, to decide whether the answer is reasonable or not. Example (page 57): Ahmed bought 476 kg apples and Anwar bought 320 kg apples. How many kg of apples did they buy altogether? Solution: 476 + 320 = ? We have exact sum 476 + 320 = 796. We find estimate of the sum: 476 rounds to 500, 320 rounds to 300. Then, add the rounded numbers: 500 + 300 = 800. So, the estimated mass of apples is 800 kg. Example (page 57): Estimate the sum: 3897 + 2216. Solution: 3897 + 2216 = ? We have exact sum 3897 + 2216 = 6113. We find estimated sum: 3897 rounds to 4000, 2216 rounds to 2000. Then, add the rounded numbers: 4000 + 2000 = 6000. So, the estimated sum is 6000.

Exercise (page 58): Question 1 — Find the estimated sum (round to nearest 100)

Exercise (page 58): Q1 (ii) — 275 + 125

Step 1: Round each number to the nearest 100: 275 rounds to 300, 125 rounds to 100.

Step 2: Add the rounded numbers: 300 + 100.

Answer: 400 (estimated sum; exact sum is 400)

Exercise (page 58): Q1 (iii) — 382 + 219

Step 1: Round each number to the nearest 100: 382 rounds to 400, 219 rounds to 200.

Step 2: Add the rounded numbers: 400 + 200.

Answer: 600 (estimated sum; exact sum is 601)

Exercise (page 58): Q1 (iv) — 492 + 234

Step 1: Round each number to the nearest 100: 492 rounds to 500, 234 rounds to 200.

Step 2: Add the rounded numbers: 500 + 200.

Answer: 700 (estimated sum; exact sum is 726)

Exercise (page 58): Q1 (v) — 589 + 325

Step 1: Round each number to the nearest 100: 589 rounds to 600, 325 rounds to 300.

Step 2: Add the rounded numbers: 600 + 300.

Answer: 900 (estimated sum; exact sum is 914)

Exercise (page 58): Q1 (second iv) — 678 + 323

Step 1: Round each number to the nearest 100: 678 rounds to 700, 323 rounds to 300.

Step 2: Add the rounded numbers: 700 + 300.

Answer: 1000 (estimated sum; exact sum is 1001)

Exercise (page 58): Q1 (i) is already completed in the book: 269 + 114 = 383, and 300 + 100 = 400. NOTE — the book numbers two different items on page 58 as '(iv)' in Question 1; both are solved in order and the

repeat is flagged here.

Exercise (page 58): Question 2 — Find the estimated sum (round to nearest thousand)

Exercise (page 58): Q2 (ii) — 3782 + 2218

Step 1: Round each number to the nearest thousand: 3782 rounds to 4000, 2218 rounds to 2000.

Step 2: Add the rounded numbers: 4000 + 2000.

Answer: 6000 (estimated sum; exact sum is 6000)

Exercise (page 58): Q2 (iii) — 4592 + 2138

Step 1: Round each number to the nearest thousand: 4592 rounds to 5000, 2138 rounds to 2000.

Step 2: Add the rounded numbers: 5000 + 2000.

Answer: 7000 (estimated sum; exact sum is 6730)

Exercise (page 58): Q2 (iv) — 5826 + 3335

Step 1: Round each number to the nearest thousand: 5826 rounds to 6000, 3335 rounds to 3000.

Step 2: Add the rounded numbers: 6000 + 3000.

Answer: 9000 (estimated sum; exact sum is 9161)

Exercise (page 58): Q2 (i) is already completed in the book: 2696 + 1112 = 3808, and 3000 + 1000 = 4000.

Estimate Differences (pages 59 to 60)

Estimation is helpful when we are subtracting, to decide whether the answer is reasonable or not. Example (page 59): Arshad has 484 marbles. Qamar has 314 marbles. How many more marbles does Arshad have than

Qamar? Solution: 484 - 314 = ? We have exact difference 484 - 314 = 170. We estimate the difference: 484 rounds to 500, 314 rounds to 300. Then subtract the rounded numbers: 500 - 300 = 200. So, Arshad has estimated 200 more marbles than Qamar. Example (page 59): Estimate the difference: 2892 - 2324. Solution: 2892 - 2324 = ? We have exact difference 2892 - 2324 = 568. We find the estimated difference: 2892 rounds to 3000, 2324 rounds to 2000. Then, subtract the rounded numbers: 3000 - 2000 = 1000. So, the estimated difference is 1000.

Exercise (page 60): Question 1 — Find estimated difference (round to nearest hundred)

Exercise (page 60): Q1 (ii) — 385 - 221

Step 1: Round each number to the nearest hundred: 385 rounds to 400, 221 rounds to 200.

Step 2: Subtract the rounded numbers: 400 - 200.

Answer: 200 (estimated difference; exact difference is 164)

Exercise (page 60): Q1 (iii) — 468 - 315

Step 1: Round each number to the nearest hundred: 468 rounds to 500, 315 rounds to 300.

Step 2: Subtract the rounded numbers: 500 - 300.

Answer: 200 (estimated difference; exact difference is 153)

Exercise (page 60): Q1 (iv) — 693 - 325

Step 1: Round each number to the nearest hundred: 693 rounds to 700, 325 rounds to 300.

Step 2: Subtract the rounded numbers: 700 - 300.

Answer: 400 (estimated difference; exact difference is 368)

Exercise (page 60): Q1 (v) — 786 - 424

Step 1: Round each number to the nearest hundred: 786 rounds to 800, 424 rounds to 400.

Step 2: Subtract the rounded numbers: 800 - 400.

Answer: 400 (estimated difference; exact difference is 362)

Exercise (page 60): Q1 (second iv) — 876 - 534

Step 1: Round each number to the nearest hundred: 876 rounds to 900, 534 rounds to 500.

Step 2: Subtract the rounded numbers: 900 - 500.

Answer: 400 (estimated difference; exact difference is 342)

Exercise (page 60): Q1 (i) is already completed in the book: 589 - 422 = 167, and 600 - 400 = 200. NOTE — the book numbers two different items on page 60 as '(iv)' in Question 1; both are solved in order and the repeat is flagged here.

Exercise (page 60): Question 2 — Find estimated difference (round to nearest thousand)

Exercise (page 60): Q2 (ii) — 4978 - 2436

Step 1: Round each number to the nearest thousand: 4978 rounds to 5000, 2436 rounds to 2000.

Step 2: Subtract the rounded numbers: 5000 - 2000.

Answer: 3000 (estimated difference; exact difference is 2542)

Exercise (page 60): Q2 (iii) — 4695 - 2472

Step 1: Round each number to the nearest thousand: 4695 rounds to 5000, 2472 rounds to 2000.

Step 2: Subtract the rounded numbers: 5000 - 2000.

Answer: 3000 (estimated difference; exact difference is 2223)

Exercise (page 60): Q2 (iv) — 5875 - 3641

Step 1: Round each number to the nearest thousand: 5875 rounds to 6000, 3641 rounds to 4000.

Step 2: Subtract the rounded numbers: 6000 - 4000.

Answer: 2000 (estimated difference; exact difference is 2234)

Exercise (page 60): Q2 (i) is already completed in the book: 2788 - 2222 = 566, and 3000 - 2000 = 1000.

Multiplication — Counting and writing in multiple steps (page 61)

There are 3 pairs of shoes. How many shoes are there altogether? Number of pairs of shoes = 3. Number of shoes = 2 + 2 + 2 = 6. We can write it as: 3 times 2 = 6, or 3 x 2 = 6, or the product of 3 and 2 is 6. So, we can use the term product for multiplication of two numbers. Few more examples are given below: Stars: 3 + 3 + 3 = 9; 3 times 3 = 9; 3 x 3 = 9. Dots: 2 + 2 + 2 + 2 + 2 + 2 + 2 + 2 = 16; 8 times 2 = 16; 8 x 2 = 16. NOTE on the hearts example (page 61): the book's addition line prints five 5s (5 + 5 + 5 + 5 + 5) but gives the answer 20 and writes '5 times 4 = 20' and '5 x 4 = 20'. Five 5s would make 25, so the correct addition line is four 5s: 5 + 5 + 5 + 5 = 20, matching 5 times 4 = 20 and 5 x 4 = 20. The corrected line is used here.

Multiplication Tables (pages 62 to 65)

Here we shall learn to develop the multiplication tables (time tables) of 6,

7, 8 and 9. Multiplication Table of 6 (page 62): 6 ones are 6 (6 x 1 = 6); 6 twos are 12 (6 x 2 = 12); 6 threes are 18 (6 x 3 = 18); 6 fours are 24 (6 x 4 = 24); 6 fives are 30 (6 x 5 = 30); 6 sixes are 36 (6 x 6 = 36); 6 sevens are 42 (6 x 7 = 42); 6 eights are 48 (6 x 8 = 48); 6 nines are 54 (6 x 9 = 54); 6 tens are 60 (6 x 10 = 60). The table is built by repeated addition: 6 + 6 + 6 + 6 + 6 + 6 + 6 + 6 + 6 + 6, giving 6, 12, 18, 24, 30, 36, 42, 48, 54, 60.

Multiplication Table of 7 (page 63): 7 ones are 7 (7 x 1 = 7); 7 twos are 14 (7 x 2 = 14); 7 threes are 21 (7 x 3 = 21); 7 fours are 28 (7 x 4 = 28); 7 fives are 35 (7 x 5 = 35); 7 sixes are 42 (7 x 6 = 42); 7 sevens are 49 (7 x 7 = 49); 7 eights are 56 (7 x 8 = 56); 7 nines are 63 (7 x 9 = 63); 7 tens are 70 (7 x 10 = 70). The table is built by repeated addition: 7 + 7 + 7 + 7 + 7 + 7 + 7 + 7 + 7 + 7, giving 7, 14, 21, 28, 35, 42, 49, 56, 63, 70.

Multiplication Table of 8 (page 64): 8 ones are 8 (8 x 1 = 8); 8 twos are 16 (8 x 2 = 16); 8 threes are 24 (8 x 3 = 24); 8 fours are 32 (8 x 4 = 32); 8 fives are 40 (8 x 5 = 40); 8 sixes are 48 (8 x 6 = 48); 8 sevens are 56 (8 x 7 = 56); 8 eights are 64 (8 x 8 = 64); 8 nines are 72 (8 x 9 = 72); 8 tens are 80 (8 x 10 = 80). The table is built by repeated addition: 8 + 8 + 8 + 8 + 8 + 8 + 8 + 8 + 8 + 8, giving 8, 16, 24, 32, 40, 48, 56, 64, 72, 80.

Multiplication Table of 9 (page 65): 9 ones are 9 (9 x 1 = 9); 9 twos are 18 (9 x 2 = 18); 9 threes are 27 (9 x 3 = 27); 9 fours are 36 (9 x 4 = 36); 9 fives are 45 (9 x 5 = 45); 9 sixes are 54 (9 x 6 = 54); 9 sevens are 63 (9 x 7 = 63); 9 eights are 72 (9 x 8 = 72); 9 nines are 81 (9 x 9 = 81); 9 tens are 90 (9 x 10 = 90). The table is built by repeated addition: 9 + 9 + 9 + 9 + 9 + 9 + 9 + 9 + 9 + 9, giving 9, 18, 27, 36, 45, 54, 63, 72, 81, 90.

Multiplication Sentence (page 66)

Six groups of 2 stars: 2 + 2 + 2 + 2 + 2 + 2 = 12. 6 times 2 equal to 12. 6 x 2 = 12 (Multiplication sentence). In the sentence, 6 is the Factor, 2 is the Factor, and 12 is the Product. Or in words: Six multiplied by 2 is twelve. Or in vertical form: 6 x 2 = 12. So, the product of 6 and 2 is 12.

Page 66: Write the multiplication sentence

Page 66 (i) — 8 groups of 3 dots: 3 + 3 + 3 + 3 + 3 + 3 + 3 + 3 = ?

Missing word: 24, and the multiplication sentence is 8 x 3 = 24

Page 66 (ii) — 9 groups of 4 hearts: 4 + 4 + 4 + 4 + 4 + 4 + 4 + 4 + 4 = ?

Missing word: 36, and the multiplication sentence is 9 x 4 = 36

Changing Order in Multiplication (page 67)

Example (page 67): Multiply 7 and 2 (change order). Solution: 2 x 7 = 14 and 7 x 2 = 14. So, 2 x 7 = 7 x 2. Example (page 67): Multiply 6 and 2 (change order). Solution: 2 x 6 = 12 and 6 x 2 = 12. So, if we change the order in multiplication the product remains the same.

Exercise (page 67): Question 1 — Write the product in the blanks

Exercise (page 67): Q1 (i) — 6 x 3 and 3 x 6

Missing word: 18 and 18

Exercise (page 67): Q1 (ii) — 7 x 5 and 5 x 7

Missing word: 35 and 35

Exercise (page 67): Q1 (iii) — 8 x 4 and 4 x 8

Missing word: 32 and 32

Exercise (page 67): Q1 (iv) — 9 x 2 and 2 x 9

Missing word: 18 and 18

Exercise (page 67): Q1 (v) — 6 x 5 and 5 x 6

Missing word: 30 and 30

Exercise (page 67): Q1 (vi) — 7 x 6 and 6 x 7

Missing word: 42 and 42

Exercise (page 67): Q1 (vii) — 8 x 6 and 6 x 8

Missing word: 48 and 48

Exercise (page 67): Q1 (viii) — 6 x 9 and 9 x 6

Missing word: 54 and 54

Exercise (page 67): Q1 (ix) — 9 x 7 and 7 x 9

Missing word: 63 and 63

Multiplication of 2-digit numbers by 1-digit numbers (page 68)

Example (page 68): Multiply 62 by 7. Step 1: Multiply digit at ones place: 2 x 7 = 14 ones. Write 4 in ones place and regroup 1 ten in tens place. Step 2: Multiply the digit at tens place: 6 x 7 = 42 tens. So, 42 tens and 1 ten = 43. Then write 3 in tens and 4 in hundreds place. So, 434 answer.

Exercise (page 68): Question 1 — Solve the following

Exercise (page 68): Q1 (1) — 11 x 3

Step 1: Multiply the ones: 1 x 3 = 3.

Step 2: Multiply the tens: 1 x 3 = 3.

Answer: 33

Exercise (page 68): Q1 (2) — 24 x 2

Step 1: Multiply the ones: 4 x 2 = 8.

Step 2: Multiply the tens: 2 x 2 = 4.

Answer: 48

Exercise (page 68): Q1 (3) — 34 x 6

Step 1: Multiply the ones: 4 x 6 = 24. Write 4, regroup 2 tens.

Step 2: Multiply the tens: 3 x 6 = 18 tens, plus 2 tens = 20 tens. Write 0 in tens and 2 in hundreds.

Answer: 204

Exercise (page 68): Q1 (4) — 47 x 8

Step 1: Multiply the ones: 7 x 8 = 56. Write 6, regroup 5 tens.

Step 2: Multiply the tens: 4 x 8 = 32 tens, plus 5 tens = 37 tens. Write 7 in tens and 3 in hundreds.

Answer: 376

Exercise (page 68): Question 2 — Multiply

Exercise (page 68): Q2 (1) — 12 x 6

Step 1: Multiply the ones: 2 x 6 = 12. Write 2, regroup 1 ten.

Step 2: Multiply the tens: 1 x 6 = 6 tens, plus 1 ten = 7 tens.

Answer: 72

Exercise (page 68): Q2 (2) — 24 x 7

Step 1: Multiply the ones: 4 x 7 = 28. Write 8, regroup 2 tens.

Step 2: Multiply the tens: 2 x 7 = 14 tens, plus 2 tens = 16 tens. Write 6 in tens and 1 in hundreds.

Answer: 168

Exercise (page 68): Q2 (3) — 30 x 8

Step 1: Multiply the ones: 0 x 8 = 0.

Step 2: Multiply the tens: 3 x 8 = 24 tens. Write 4 in tens and 2 in

hundreds.

Answer: 240

Exercise (page 68): Q2 (4) — 45 x 9

Step 1: Multiply the ones: 5 x 9 = 45. Write 5, regroup 4 tens.

Step 2: Multiply the tens: 4 x 9 = 36 tens, plus 4 tens = 40 tens. Write 0 in tens and 4 in hundreds.

Answer: 405

Exercise (page 68): Q2 (5) — 73 x 8

Step 1: Multiply the ones: 3 x 8 = 24. Write 4, regroup 2 tens.

Step 2: Multiply the tens: 7 x 8 = 56 tens, plus 2 tens = 58 tens. Write 8 in tens and 5 in hundreds.

Answer: 584

Exercise (page 68): Q2 (6) — 88 x 6

Step 1: Multiply the ones: 8 x 6 = 48. Write 8, regroup 4 tens.

Step 2: Multiply the tens: 8 x 6 = 48 tens, plus 4 tens = 52 tens. Write 2 in tens and 5 in hundreds.

Answer: 528

Multiplication of 3-digit numbers by 1-digit numbers (pages 69 to 70)

Example (page 69): Solve 432 x 6. Step 1: Multiply ones first: 2 x 6 = 12 ones. Write 2 in ones place and regroup 1 ten in tens place. Step 2: Multiply the digit at tens place: 3 x 6 = 18 tens. So, 18 tens and 1 ten = 19 tens. Write 9 in tens place and regroup 1 hundred in hundreds place.

Step 3: Multiply the digit at hundreds place: 4 x 6 hundreds = 24 hundreds. So, 24 hundreds and 1 hundred = 25 hundreds. Write 5 in hundreds place and 2 in thousands place. So, 2592 answer.

Exercise (page 70): Question 1 — Solve the following

Exercise (page 70): Q1 (1) — 214 x 2

Step 1: Multiply the ones: 4 x 2 = 8.

Step 2: Multiply the tens: 1 x 2 = 2.

Step 3: Multiply the hundreds: 2 x 2 = 4.

Answer: 428

Exercise (page 70): Q1 (2) — 165 x 3

Step 1: Multiply the ones: 5 x 3 = 15. Write 5, regroup 1 ten.

Step 2: Multiply the tens: 6 x 3 = 18 tens, plus 1 ten = 19 tens. Write 9, regroup 1 hundred.

Step 3: Multiply the hundreds: 1 x 3 = 3 hundreds, plus 1 hundred = 4 hundreds.

Answer: 495

Exercise (page 70): Q1 (3) — 472 x 8

Step 1: Multiply the ones: 2 x 8 = 16. Write 6, regroup 1 ten.

Step 2: Multiply the tens: 7 x 8 = 56 tens, plus 1 ten = 57 tens. Write 7, regroup 5 hundreds.

Step 3: Multiply the hundreds: 4 x 8 = 32 hundreds, plus 5 hundreds = 37 hundreds. Write 7 in hundreds and 3 in thousands.

Answer: 3776

Exercise (page 70): Q1 (4) — 351 x 5

Step 1: Multiply the ones: 1 x 5 = 5.

Step 2: Multiply the tens: 5 x 5 = 25 tens. Write 5, regroup 2 hundreds.

Step 3: Multiply the hundreds: 3 x 5 = 15 hundreds, plus 2 hundreds = 17 hundreds. Write 7 in hundreds and 1 in thousands.

Answer: 1755

Exercise (page 70): Q1 (5) — 543 x 6

Step 1: Multiply the ones: 3 x 6 = 18. Write 8, regroup 1 ten.

Step 2: Multiply the tens: 4 x 6 = 24 tens, plus 1 ten = 25 tens. Write 5, regroup 2 hundreds.

Step 3: Multiply the hundreds: 5 x 6 = 30 hundreds, plus 2 hundreds = 32 hundreds. Write 2 in hundreds and 3 in thousands.

Answer: 3258

Exercise (page 70): Q1 (6) — 762 x 7

Step 1: Multiply the ones: 2 x 7 = 14. Write 4, regroup 1 ten.

Step 2: Multiply the tens: 6 x 7 = 42 tens, plus 1 ten = 43 tens. Write 3, regroup 4 hundreds.

Step 3: Multiply the hundreds: 7 x 7 = 49 hundreds, plus 4 hundreds = 53 hundreds. Write 3 in hundreds and 5 in thousands.

Answer: 5334

Exercise (page 70): Q1 (7) — 654 x 8

Step 1: Multiply the ones: 4 x 8 = 32. Write 2, regroup 3 tens.

Step 2: Multiply the tens: 5 x 8 = 40 tens, plus 3 tens = 43 tens. Write 3, regroup 4 hundreds.

Step 3: Multiply the hundreds: 6 x 8 = 48 hundreds, plus 4 hundreds = 52 hundreds. Write 2 in hundreds and 5 in thousands.

Answer: 5232

Exercise (page 70): Q1 (8) — 841 x 9

Step 1: Multiply the ones: 1 x 9 = 9.

Step 2: Multiply the tens: 4 x 9 = 36 tens. Write 6, regroup 3 hundreds.

Step 3: Multiply the hundreds: 8 x 9 = 72 hundreds, plus 3 hundreds = 75 hundreds. Write 5 in hundreds and 7 in thousands.

Answer: 7569

Exercise (page 70): Q1 (9) — 975 x 8

Step 1: Multiply the ones: 5 x 8 = 40. Write 0, regroup 4 tens.

Step 2: Multiply the tens: 7 x 8 = 56 tens, plus 4 tens = 60 tens. Write 0, regroup 6 hundreds.

Step 3: Multiply the hundreds: 9 x 8 = 72 hundreds, plus 6 hundreds = 78 hundreds. Write 8 in hundreds and 7 in thousands.

Answer: 7800

Exercise (page 70): Q1 (10) — 526 x 7

Step 1: Multiply the ones: 6 x 7 = 42. Write 2, regroup 4 tens.

Step 2: Multiply the tens: 2 x 7 = 14 tens, plus 4 tens = 18 tens. Write 8, regroup 1 hundred.

Step 3: Multiply the hundreds: 5 x 7 = 35 hundreds, plus 1 hundred = 36 hundreds. Write 6 in hundreds and 3 in thousands.

Answer: 3682

Exercise (page 70): Q1 (11) — 731 x 6

Step 1: Multiply the ones: 1 x 6 = 6.

Step 2: Multiply the tens: 3 x 6 = 18 tens. Write 8, regroup 1 hundred.

Step 3: Multiply the hundreds: 7 x 6 = 42 hundreds, plus 1 hundred = 43 hundreds. Write 3 in hundreds and 4 in thousands.

Answer: 4386

Exercise (page 70): Q1 (12) — 953 x 9

Step 1: Multiply the ones: 3 x 9 = 27. Write 7, regroup 2 tens.

Step 2: Multiply the tens: 5 x 9 = 45 tens, plus 2 tens = 47 tens. Write 7, regroup 4 hundreds.

Step 3: Multiply the hundreds: 9 x 9 = 81 hundreds, plus 4 hundreds = 85 hundreds. Write 5 in hundreds and 8 in thousands.

Answer: 8577

Exercise (page 70): Question 2 — Multiply

Exercise (page 70): Q2 (1) — 276 x 9

Step 1: Multiply the ones: 6 x 9 = 54. Write 4, regroup 5 tens.

Step 2: Multiply the tens: 7 x 9 = 63 tens, plus 5 tens = 68 tens. Write 8, regroup 6 hundreds.

Step 3: Multiply the hundreds: 2 x 9 = 18 hundreds, plus 6 hundreds = 24 hundreds. Write 4 in hundreds and 2 in thousands.

Answer: 2484

Exercise (page 70): Q2 (2) — 452 x 7

Step 1: Multiply the ones: 2 x 7 = 14. Write 4, regroup 1 ten.

Step 2: Multiply the tens: 5 x 7 = 35 tens, plus 1 ten = 36 tens. Write 6, regroup 3 hundreds.

Step 3: Multiply the hundreds: 4 x 7 = 28 hundreds, plus 3 hundreds = 31 hundreds. Write 1 in hundreds and 3 in thousands.

Answer: 3164

Exercise (page 70): Q2 (3) — 631 x 8

Step 1: Multiply the ones: 1 x 8 = 8.

Step 2: Multiply the tens: 3 x 8 = 24 tens. Write 4, regroup 2 hundreds.

Step 3: Multiply the hundreds: 6 x 8 = 48 hundreds, plus 2 hundreds = 50 hundreds. Write 0 in hundreds and 5 in thousands.

Answer: 5048

Exercise (page 70): Q2 (4) — 785 x 6

Step 1: Multiply the ones: 5 x 6 = 30. Write 0, regroup 3 tens.

Step 2: Multiply the tens: 8 x 6 = 48 tens, plus 3 tens = 51 tens. Write 1, regroup 5 hundreds.

Step 3: Multiply the hundreds: 7 x 6 = 42 hundreds, plus 5 hundreds = 47 hundreds. Write 7 in hundreds and 4 in thousands.

Answer: 4710

Exercise (page 70): Q2 (5) — 354 x 5

Step 1: Multiply the ones: 4 x 5 = 20. Write 0, regroup 2 tens.

Step 2: Multiply the tens: 5 x 5 = 25 tens, plus 2 tens = 27 tens. Write 7, regroup 2 hundreds.

Step 3: Multiply the hundreds: 3 x 5 = 15 hundreds, plus 2 hundreds = 17 hundreds. Write 7 in hundreds and 1 in thousands.

Answer: 1770

Exercise (page 70): Q2 (6) — 546 x 3

Step 1: Multiply the ones: 6 x 3 = 18. Write 8, regroup 1 ten.

Step 2: Multiply the tens: 4 x 3 = 12 tens, plus 1 ten = 13 tens. Write 3, regroup 1 hundred.

Step 3: Multiply the hundreds: 5 x 3 = 15 hundreds, plus 1 hundred = 16 hundreds. Write 6 in hundreds and 1 in thousands.

Answer: 1638

Multiplication of 2-digit numbers by 1-digit numbers mentally (page 71)

i. Let us multiply 26 and 5 using mental calculation method by splitting up or distribution numbers. 26 x 5 = (20 + 6) x 5 (Splitting 26 = 20 + 6) = (20 x 5) + (6 x 5) = 100 + 30 = 130. ii. Now we multiply 48 x 3 using mental calculation method by rounding off one of them to nearest 10. 48 can be rounded off to the nearest 10, which is 50. We multiply 50 x 3 = 150. Now subtract the amount we rounded off. This was 2 x 3 = 6. Here, 150 - 6 = 144. We have 48 x 3 = 144.

Exercise (page 71): Question 1 — Multiply the following

Exercise (page 71): Q1 (i) — 16 x 3

Step 1: Split 16 = 10 + 6.

Step 2: (10 x 3) + (6 x 3) = 30 + 18.

Answer: 48

Exercise (page 71): Q1 (ii) — 17 x 6

Step 1: Multiply the ones: 7 x 6 = 42. Write 2, regroup 4 tens.

Step 2: Multiply the tens: 1 x 6 = 6 tens, plus 4 tens = 10 tens. Write 0 in tens and 1 in hundreds.

Answer: 102

Exercise (page 71): Q1 (iii) — 18 x 2

Step 1: Multiply the ones: 8 x 2 = 16. Write 6, regroup 1 ten.

Step 2: Multiply the tens: 1 x 2 = 2 tens, plus 1 ten = 3 tens.

Answer: 36

Exercise (page 71): Q1 (iv) — 45 x 7

Step 1: Multiply the ones: 5 x 7 = 35. Write 5, regroup 3 tens.

Step 2: Multiply the tens: 4 x 7 = 28 tens, plus 3 tens = 31 tens. Write 1 in tens and 3 in hundreds.

Answer: 315

Exercise (page 71): Q1 (v) — 52 x 4

Step 1: Multiply the ones: 2 x 4 = 8.

Step 2: Multiply the tens: 5 x 4 = 20 tens. Write 0 in tens and 2 in hundreds.

Answer: 208

Exercise (page 71): Q1 (vi) — 57 x 7

Step 1: Multiply the ones: 7 x 7 = 49. Write 9, regroup 4 tens.

Step 2: Multiply the tens: 5 x 7 = 35 tens, plus 4 tens = 39 tens. Write 9 in tens and 3 in hundreds.

Answer: 399

Exercise (page 71): Q1 (vii) — 68 x 8

Step 1: Multiply the ones: 8 x 8 = 64. Write 4, regroup 6 tens.

Step 2: Multiply the tens: 6 x 8 = 48 tens, plus 6 tens = 54 tens. Write 4 in tens and 5 in hundreds.

Answer: 544

Exercise (page 71): Q1 (viii) — 57 x 9

Step 1: Multiply the ones: 7 x 9 = 63. Write 3, regroup 6 tens.

Step 2: Multiply the tens: 5 x 9 = 45 tens, plus 6 tens = 51 tens. Write 1 in tens and 5 in hundreds.

Answer: 513

Exercise (page 71): Q1 (ix) — 74 x 5

Step 1: Multiply the ones: 4 x 5 = 20. Write 0, regroup 2 tens.

Step 2: Multiply the tens: 7 x 5 = 35 tens, plus 2 tens = 37 tens. Write 7 in tens and 3 in hundreds.

Answer: 370

Exercise (page 71): Q1 (x) — 55 x 9

Step 1: Multiply the ones: 5 x 9 = 45. Write 5, regroup 4 tens.

Step 2: Multiply the tens: 5 x 9 = 45 tens, plus 4 tens = 49 tens. Write 9 in tens and 4 in hundreds.

Answer: 495

Exercise (page 71): Q1 (xi) — 68 x 9

Step 1: Multiply the ones: 8 x 9 = 72. Write 2, regroup 7 tens.

Step 2: Multiply the tens: 6 x 9 = 54 tens, plus 7 tens = 61 tens. Write 1 in tens and 6 in hundreds.

Answer: 612

Exercise (page 71): Q1 (xii) — 66 x 8

Step 1: Multiply the ones: 6 x 8 = 48. Write 8, regroup 4 tens.

Step 2: Multiply the tens: 6 x 8 = 48 tens, plus 4 tens = 52 tens. Write 2 in tens and 5 in hundreds.

Answer: 528

Real-World Word Problems — Multiplication (page 72)

Example (page 72): There are 165 toys in a box. How many toys will be

packed in 6 such boxes? Toys in one box = 165. Number of boxes = 6. Total toys in 6 boxes = 165 x 6. Step 1: Multiply the ones: 5 x 6 = 30. Write 0, regroup 3 tens. Step 2: Multiply the tens: 6 x 6 = 36 tens, plus 3 tens = 39 tens. Write 9, regroup 3 hundreds. Step 3: Multiply the hundreds: 1 x 6 = 6 hundreds, plus 3 hundreds = 9 hundreds. So the total is 990 toys. NOTE — the book prints this answer as 991; the correct product of 165 x 6 is 990, and 990 is used here.

Exercise (page 72): Word problems

Exercise (page 72): (1) — There are 6 friends. Each one has 23 balls. How many balls do they have altogether?

Step 1: Multiply: 6 x 23.

Step 2: Ones: 3 x 6 = 18. Write 8, regroup 1 ten.

Step 3: Tens: 2 x 6 = 12 tens, plus 1 ten = 13 tens. Write 3 in tens and 1 in hundreds.

Answer: 138 balls

Exercise (page 72): (2) — Usman has 8 packets of colour pencils. Each packet contains 31 pencils. How many colour pencils does he have in all?

Step 1: Multiply: 8 x 31.

Step 2: Ones: 1 x 8 = 8.

Step 3: Tens: 3 x 8 = 24 tens. Write 4 in tens and 2 in hundreds.

Answer: 248 colour pencils

Exercise (page 72): (3) — There are 40 apples in one box. How many apples are there in such 8 boxes?

Step 1: Multiply: 40 x 8.

Step 2: Ones: 0 x 8 = 0.

Step 3: Tens: 4 x 8 = 32 tens. Write 2 in tens and 3 in hundreds.

Answer: 320 apples

Exercise (page 72): (4) — A bus carries 54 people. How many people are there in 7 such buses?

Step 1: Multiply: 54 x 7.

Step 2: Ones: 4 x 7 = 28. Write 8, regroup 2 tens.

Step 3: Tens: 5 x 7 = 35 tens, plus 2 tens = 37 tens. Write 7 in tens and 3 in hundreds.

Answer: 378 people

Exercise (page 72): (5) — A person earns Rs. 275 in a day. How many rupees does he earn in 6 days?

Step 1: Multiply: 275 x 6.

Step 2: Ones: 5 x 6 = 30. Write 0, regroup 3 tens.

Step 3: Tens: 7 x 6 = 42 tens, plus 3 tens = 45 tens. Write 5, regroup 4 hundreds.

Step 4: Hundreds: 2 x 6 = 12 hundreds, plus 4 hundreds = 16 hundreds. Write 6 in hundreds and 1 in thousands.

Answer: Rs. 1650

Exercise (page 72): (6) — There are 320 pages in a book. Find total number of pages in 9 such books.

Step 1: Multiply: 320 x 9.

Step 2: Ones: 0 x 9 = 0.

Step 3: Tens: 2 x 9 = 18 tens. Write 8, regroup 1 hundred.

Step 4: Hundreds: 3 x 9 = 27 hundreds, plus 1 hundred = 28 hundreds. Write 8 in hundreds and 2 in thousands.

Answer: 2880 pages

Division (page 73)

Division of any two numbers cannot be done in any order. In division we cannot change the order. 14 / 7 and 7 / 14 are not the same. Scenario 1: Divide 14 by 7. Group the 14 balls into 2 equal groups. Each group will have 7 balls. So, 14 / 7 = 2. Scenario 2: Divide 7 / 14. Try to divide 7 balls into 14 equal groups. Each group will have less than 1/2 a ball each, which may be hard to represent with physical objects. So, 7 / 14 = 1/2, which is not the same as 14 / 7 = 2. So, in division we cannot change the order: 14 / 7 is not equal to 7 / 14.

Division of 2-digit numbers by 1-digit numbers (with and without remainder) (pages 74 to 75)

Example (page 74): Divide 69 by 3. Divide a 2-digit number with a 1-digit number by long division. Divide the tens first: 6 / 3 = 2. Write down 2 as quotient in tens place. Then bring down 9 ones from dividend. Divide the number in ones place: 9 / 3 = 3. Write 3 as quotient in ones place. So, 69 / 3 = 23. Example (page 74): (i) 72 / 6. Long division: 6 into 7 goes 1 time, remainder 1; bring down 2 to make 12; 12 / 6 = 2. So 72 / 6 = 12. Key note (page 74): In the Division Process, the number being divided is the Dividend, the number we divide by is the Divisor, the answer is the Quotient, and what is left over is the Remainder (example in the book: 11 / 5 = 2 remainder 1).

Exercise (page 75): Question 1 — Solve the following

Exercise (page 75): Q1 (i) — 32 divided by 4

Step 1: Divide the tens: 3 tens cannot be divided by 4, so divide 32 by 4.

Step 2: 4 x 8 = 32, so the quotient is 8 with no remainder.

Answer: 8

Exercise (page 75): Q1 (ii) — 45 divided by 5

Step 1: Divide the tens: 4 tens cannot be divided by 5, so divide 45 by 5.

Step 2: 5 x 9 = 45, so the quotient is 9 with no remainder.

Answer: 9

Exercise (page 75): Q1 (iii) — 93 divided by 7

Step 1: Divide the tens: 9 / 7 = 1, remainder 2 tens.

Step 2: Bring down 3 ones to make 23. 23 / 7 = 3, remainder 2 (7 x 3 = 21).

Answer: 13 remainder 2

Exercise (page 75): Q1 (iv) — 80 divided by 6

Step 1: Divide the tens: 8 / 6 = 1, remainder 2 tens.

Step 2: Bring down 0 ones to make 20. 20 / 6 = 3, remainder 2 (6 x 3 = 18).

Answer: 13 remainder 2

Exercise (page 75): Question 2 — Solve the following

Exercise (page 75): Q2 (i) — 24 / 6

Missing word: 4

Exercise (page 75): Q2 (ii) — 33 / 5

Missing word: 6 remainder 3

Exercise (page 75): Q2 (iii) — 48 / 8

Missing word: 6

Exercise (page 75): Q2 (iv) — 24 / 4

Missing word: 6

Exercise (page 75): Q2 (v) — 14 / 7

Missing word: 2

Exercise (page 75): Q2 (vi) — 85 / 5

Missing word: 17

Exercise (page 75): Q2 (vii) — 65 / 4

Missing word: 16 remainder 1

Exercise (page 75): Q2 (viii) — 46 / 2

Missing word: 23

Exercise (page 75): Q2 (ix) — 58 / 3

Missing word: 19 remainder 1

Exercise (page 75): Q2 (x) — 28 / 2

Missing word: 14

Exercise (page 75): Q2 (xi) — 95 / 3

Missing word: 31 remainder 2

Exercise (page 75): Q2 (xii) — 55 / 5

Missing word: 11

Exercise (page 75): Question 3 — Divide the following

Exercise (page 75): Q3 (i) — 84 by 7

Missing word: 12

Exercise (page 75): Q3 (ii) — 66 by 6

Missing word: 11

Exercise (page 75): Q3 (iii) — 89 by 4

Missing word: 22 remainder 1

Exercise (page 75): Q3 (iv) — 47 by 5

Missing word: 9 remainder 2

Exercise (page 75): Q3 (v) — 75 by 5

Missing word: 15

Exercise (page 75): Q3 (vi) — 92 by 9

Missing word: 10 remainder 2

Division of 3-digit numbers by 1-digit numbers

(with and without remainder) (page 76)

Example (page 76): 684 / 6. Write 684 / 6 by using long division symbol. Divide the hundreds: 6 / 6 = 1. Write down 1 as quotient in hundreds. Divide the tens: 8 / 6 = 1. 8 tens - 6 tens = 2 tens are left. Write it down in tens. Then bring down 4 ones from dividend. It becomes 24. Divide the ones: 24 / 6 = 4. Write down 4 as quotient in ones. So, 114 answer.

Exercise (page 77): Question 1 — Solve the following

Exercise (page 77): Q1 (i) — 455 divided by 5

Step 1: Divide the hundreds: 4 hundreds cannot be divided by 5, so divide 45 tens by 5: 45 / 5 = 9.

Step 2: Bring down 5 ones: 5 / 5 = 1.

Answer: 91

Exercise (page 77): Q1 (ii) — 643 divided by 6

Step 1: Divide the hundreds: 6 / 6 = 1.

Step 2: Divide the tens: 4 tens cannot be divided by 6, so bring down 3 ones to make 43. 43 / 6 = 7, remainder 1 (6 x 7 = 42).

Answer: 107 remainder 1

Exercise (page 77): Q1 (iii) — 848 divided by 4

Step 1: Divide the hundreds: 8 / 4 = 2.

Step 2: Divide the tens: 4 / 4 = 1.

Step 3: Divide the ones: 8 / 4 = 2.

Answer: 212

Exercise (page 77): Q1 (iv) — 784 divided by 7

Step 1: Divide the hundreds: 7 / 7 = 1.

Step 2: Divide the tens: 8 / 7 = 1, remainder 1 ten.

Step 3: Bring down 4 ones to make 14. 14 / 7 = 2.

Answer: 112

Exercise (page 77): Question 2 — Solve the following

Exercise (page 77): Q2 (i) — 188 / 6

Missing word: 31 remainder 2

Exercise (page 77): Q2 (ii) — 968 / 8

Missing word: 121

Exercise (page 77): Q2 (iii) — 453 / 3

Missing word: 151

Exercise (page 77): Q2 (iv) — 726 / 6

Missing word: 121

Exercise (page 77): Q2 (v) — 932 / 7

Missing word: 133 remainder 1

Exercise (page 77): Q2 (vi) — 675 / 5

Missing word: 135

Exercise (page 77): Q2 (vii) — 826 / 2

Missing word: 413

Exercise (page 77): Q2 (viii) — 421 / 9

Missing word: 46 remainder 7

Exercise (page 77): Q2 (ix) — 449 / 4

Missing word: 112 remainder 1

Exercise (page 77): Q2 (x) — 178 / 8

Missing word: 22 remainder 2

Exercise (page 77): Q2 (xi) — 847 / 7

Missing word: 121

Exercise (page 77): Q2 (xii) — 927 / 9

Missing word: 103

Exercise (page 77): Question 3 — Divide the following

Exercise (page 77): Q3 (i) — 835 / 7

Missing word: 119 remainder 2

Exercise (page 77): Q3 (ii) — 639 / 3

Missing word: 213

Exercise (page 77): Q3 (iii) — 355 / 5

Missing word: 71

Exercise (page 77): Q3 (iv) — 414 / 6

Missing word: 69

Exercise (page 77): Q3 (v) — 209 / 8

Missing word: 26 remainder 1

Exercise (page 77): Q3 (vi) — 999 / 9

Missing word: 111

Divide Mentally (page 78)

Divide by splitting numbers. Example (page 78): Divide 35 / 5. Solution: = 35 / 5 (Splitting 35 = 30 + 5) = 30 / 5 + 5 / 5 (Divide each number by 5) = 6 + 1 = 7. So, 35 / 5 = 7.

Exercise (page 78): Question 1 — Recall the table and divide mentally

Exercise (page 78): Q1 (i) — 48 / 8

Missing word: 6

Exercise (page 78): Q1 (ii) — 36 / 6

Missing word: 6

Exercise (page 78): Q1 (iii) — 63 / 7

Missing word: 9

Exercise (page 78): Q1 (iv) — 72 / 9

Missing word: 8

Exercise (page 78): Q1 (v) — 42 / 6

Missing word: 7

Exercise (page 78): Q1 (vi) — 35 / 7

Missing word: 5

Exercise (page 78): Q1 (vii) — 25 / 5

Missing word: 5

Exercise (page 78): Q1 (viii) — 64 / 8

Missing word: 8

Exercise (page 78): Q1 (ix) — 690 / 6

Missing word: 115

Exercise (page 78): Q1 (x) — 496 / 4

Missing word: 124

Exercise (page 78): Q1 (xi) — 992 / 8

Missing word: 124

Exercise (page 78): Q1 (xii) — 917 / 7

Missing word: 131

Exercise (page 78): Q1 (xiii) — 785 / 5

Missing word: 157

Exercise (page 78): Q1 (xiv) — 972 / 6

Missing word: 162

Exercise (page 78): Q1 (xv) — 896 / 8

Missing word: 112

Exercise (page 78): Q1 (xvi) — 999 / 9

Missing word: 111

Dividing a number by 1 and itself (page 79)

Any number divided by 1 is the number itself: 5 / 1 = 5 and 8 / 1 = 8. Any number divided by itself is 1: 5 / 5 = 1 and 8 / 8 = 1.

Exercise (page 79): Question 1 — Divide the following

Exercise (page 79): Q1 (2) — 2 / 1

Missing word: 2

Exercise (page 79): Q1 (3) — 3 / 1

Missing word: 3

Exercise (page 79): Q1 (4) — 4 / 1

Missing word: 4

Exercise (page 79): Q1 (5) — 5 / 1

Missing word: 5

Exercise (page 79): Q1 (6) — 8 / 1

Missing word: 8

Exercise (page 79): Q1 (1) is already completed in the book: 1 / 1 = 1.

Exercise (page 79): Question 2 — Divide the following

Exercise (page 79): Q2 (2) — 8 / 1

Missing word: 8

Exercise (page 79): Q2 (3) — 7 / 1

Missing word: 7

Exercise (page 79): Q2 (4) — 0 / 1

Missing word: 0

Exercise (page 79): Q2 (5) — 9 / 1

Missing word: 9

Exercise (page 79): Q2 (6) — 1 / 1

Missing word: 1

Exercise (page 79): Q2 (1) is already completed in the book: 6 / 1 = 6.

Exercise (page 79): Question 3 — Complete the missing numbers

Exercise (page 79): Q3 (2) — 5 / _____ = 1

Missing word: 5

Exercise (page 79): Q3 (3) — _____ / 2 = 1

Missing word: 2

Exercise (page 79): Q3 (4) — 4 / _____ = 1

Missing word: 4

Exercise (page 79): Q3 (5) — _____ / 1 = 1

Missing word: 1

Exercise (page 79): Q3 (6) — 7 / 7 = _____

Missing word: 1

Exercise (page 79): Q3 (7) — 9 / _____ = 1

Missing word: 9

Exercise (page 79): Q3 (8) — 8 / 8 = _____

Missing word: 1

Exercise (page 79): Q3 (second 6) — 6 / 6 = _____

Missing word: 1

Exercise (page 79): Q3 (1) is already completed in the book: 3 / 3 = 1. NOTE — the book numbers two different items in Question 3 as '6'; both are solved in order and the repeat is flagged here.

Real-World Word Problems — Division (page 80)

Example (page 80): Raheel had 42 seeds and 7 pots. How many seeds did he plant in each pot? Solution: Raheel had seeds = 42. Raheel had pots = 7. How many seeds put in each pot? 42 / 7 = 6. So, 6 seeds answer.

Exercise (page 80): Word problems

Exercise (page 80): (1) — Sheraz divided Rs. 90 equally among 6 boys. How much money did each boy get?

Step 1: Divide: 90 / 6.

Step 2: 6 x 15 = 90, so the quotient is 15.

Answer: Rs. 15 each

Exercise (page 80): (2) — A tailor has 96 buttons. If he sews 8 buttons on each shirt, how many shirts can he sew?

Step 1: Divide: 96 / 8.

Step 2: 8 x 12 = 96, so the quotient is 12.

Answer: 12 shirts

Exercise (page 80): (3) — A book contains 387 pages. Akram reads 9 pages in a day. How many days are required to read the book?

Step 1: Divide: 387 / 9.

Step 2: 9 x 43 = 387, so the quotient is 43.

Answer: 43 days

Exercise (page 80): (4) — One pencil costs Rs. 6. How many pencils can be bought for Rs. 468?

Step 1: Divide: 468 / 6.

Step 2: 6 x 78 = 468, so the quotient is 78.

Answer: 78 pencils

Exercise (page 80): (5) — There are 7 days in a week. How many weeks are there in 364 days?

Step 1: Divide: 364 / 7.

Step 2: 7 x 52 = 364, so the quotient is 52.

Answer: 52 weeks

Exercise (page 80): (6) — In a school 992 students stood in 8 rows. How many students were there in one row?

Step 1: Divide: 992 / 8.

Step 2: 8 x 124 = 992, so the quotient is 124.

Answer: 124 students in one row

Real-World Word Problems — Mixed Operations

(page 81)

Exercise (page 81): Word problems

Exercise (page 81): (1) — There are 4573 men and 3112 women in a village. What is the total population of the village?

Step 1: Add: 4573 + 3112.

Step 2: Ones: 3 + 2 = 5. Tens: 7 + 1 = 8. Hundreds: 5 + 1 = 6. Thousands: 4 + 3 = 7.

Answer: 7685 people

Exercise (page 81): (2) — Sara bought 2 sarees. The prices of the sarees are Rs. 3250 and Rs. 2550. How much money did she spend in all?

Step 1: Add: 3250 + 2550.

Step 2: Ones: 0 + 0 = 0. Tens: 5 + 5 = 10. Write 0, carry 1.

Step 3: Hundreds: 2 + 5 + 1 (carried) = 8. Thousands: 3 + 2 = 5.

Answer: Rs. 5800

Exercise (page 81): (3) — A nursery planted 3438 red flowers and 2394 yellow flowers. How many flowers were planted in all?

Step 1: Add: 3438 + 2394.

Step 2: Ones: 8 + 4 = 12. Write 2, carry 1.

Step 3: Tens: 3 + 9 + 1 (carried) = 13. Write 3, carry 1.

Step 4: Hundreds: 4 + 3 + 1 (carried) = 8. Thousands: 3 + 2 = 5.

Answer: 5832 flowers

Exercise (page 81): (4) — There are 4865 students in a school. If 2746 are boys, how many girls are there in school?

Step 1: Subtract: 4865 - 2746.

Step 2: Ones: 5 - 6 cannot be done, so regroup 1 ten: 15 - 6 = 9.

Step 3: Tens: 6 - 1 (regrouped) - 4 = 1. Hundreds: 8 - 7 = 1. Thousands: 4 - 2 = 2.

Answer: 2119 girls

Exercise (page 81): (5) — A bakery had 3596 buns. The 2767 buns were sold. How many buns were left?

Step 1: Subtract: 3596 - 2767.

Step 2: Ones: 6 - 7 cannot be done, so regroup 1 ten: 16 - 7 = 9.

Step 3: Tens: 9 - 1 (regrouped) - 6 = 2.

Step 4: Hundreds: 5 - 7 cannot be done, so regroup 1 thousand: 15 - 7 = 8.

Step 5: Thousands: 3 - 1 (regrouped) - 2 = 0.

Answer: 829 buns left

Exercise (page 81): (6) — A milkman carried 2585 bottles of milk. The milkman delivered 1431 bottles. How many bottles were left?

Step 1: Subtract: 2585 - 1431.

Step 2: Ones: 5 - 1 = 4. Tens: 8 - 3 = 5. Hundreds: 5 - 4 = 1. Thousands: 2 - 1 = 1.

Answer: 1154 bottles left

Exercise (page 81): (7) — There are 75 chairs in a hall. How many chairs are there in such 6 halls?

Step 1: Multiply: 75 x 6.

Step 2: Ones: 5 x 6 = 30. Write 0, regroup 3 tens.

Step 3: Tens: 7 x 6 = 42 tens, plus 3 tens = 45 tens. Write 5 in tens and 4 in hundreds.

Answer: 450 chairs

Exercise (page 81): (8) — In a factory 84 school bags are manufactured in one day. How many bags will be manufactured in 8 days?

Step 1: Multiply: 84 x 8.

Step 2: Ones: 4 x 8 = 32. Write 2, regroup 3 tens.

Step 3: Tens: 8 x 8 = 64 tens, plus 3 tens = 67 tens. Write 7 in tens and 6 in hundreds.

Answer: 672 bags

Exercise (page 81): (9) — A shopkeeper sells 125 umbrellas in a day. How many umbrellas will he sell in 7 days?

Step 1: Multiply: 125 x 7.

Step 2: Ones: 5 x 7 = 35. Write 5, regroup 3 tens.

Step 3: Tens: 2 x 7 = 14 tens, plus 3 tens = 17 tens. Write 7, regroup 1 hundred.

Step 4: Hundreds: 1 x 7 = 7 hundreds, plus 1 hundred = 8 hundreds.

Answer: 875 umbrellas

Exercise (page 81): (10) — There are 42 chocolates in a packet. Seema wants to eat an equal amount of chocolate in 6 days. How many chocolates will she eat daily?

Step 1: Divide: 42 / 6.

Step 2: 6 x 7 = 42, so the quotient is 7.

Answer: 7 chocolates daily

Exercise (page 81): (11) — There are 147 books. The 7 books are to be placed on a shelf. How many shelves are there?

Step 1: Divide: 147 / 7.

Step 2: 7 x 21 = 147, so the quotient is 21.

Answer: 21 shelves

Numbers Patterns (page 82)

A number pattern is an arrangement of numbers in sequence and order.

Example (page 82): Patterns increasing and decreasing in tens: increasing: 110, 120, 130, 140, 150, 160, 170, 180, 190, 200; decreasing: 200, 190, 180, 170, 160, 150, 140, 130, 120, 100.

Exercise (page 82): Question 1 — Complete the increasing and decreasing numbers patterns

Exercise (page 82): Q1 (i) — 302, 304, 306, 308, 310, _____, _____, _____, _____, _____

Missing word: 312, 314, 316, 318, 320 (increasing by 2)

Exercise (page 82): Q1 (ii) — 459, 457, 455, 453, 451, _____, _____, _____, _____, _____

Missing word: 449, 447, 445, 443, 441 (decreasing by 2)

Exercise (page 82): Q1 (iii) — 515, 525, 535, 545, 555, _____, _____, _____, _____, _____

Missing word: 565, 575, 585, 595, 605 (increasing by 10)

Exercise (page 82): Q1 (iv) — 750, 740, 730, 720, 710, _____, _____, _____, _____, _____

Missing word: 700, 690, 680, 670, 660 (decreasing by 10)

Exercise (page 82): Q1 (v) — 9050, 9150, 9250, 9350, 9450, _____, _____, _____, _____, _____

Missing word: 9550, 9650, 9750, 9850, 9950 (increasing by 100)

Exercise (page 82): Question 2 — Fill in the gaps in these sequences

Exercise (page 82): Q2 (i) — 110, 120, _____, _____, 150, _____, _____, _____, _____, 200

Missing word: 130, 140, 160, 170, 180, 190 (increasing by 10)

Exercise (page 82): Q2 (ii) — 500, 490, _____, _____, _____, _____, _____, 430, _____

Missing word: 480, 470, 460, 450, 440, 420 (decreasing by 10)

Exercise (page 82): Q2 (iii) — 550, 540, _____, _____, 500, 490, _____, _____, _____, _____

Missing word: 530, 520, 480, 470, 460, 450 (decreasing by 10)

Exercise (page 82): Q2 (iv) — 999, 9899, _____, 9699, _____, 9499, _____, 9299, _____, _____

Missing word: 9799, 9599, 9399, 9199, 9099 (decreasing by 100 from 9899 onward)

Exercise (page 82): Q2 (iv) — the first circle reads '999' at the render, but the rest of the chain (9899, 9699, 9499, 9299) decreases by 100, so the first term should be 9999; the gaps are completed on the -100 step from 9899 onward. Teacher point (page 82): The teacher will write different number patterns using 1, 2, 3 and 4 digit numbers and assist students in completing these patterns — teacher work only.

Numbers Patterns — Finding the Missing Number

(page 83)

To find the missing numbers use the inverse operation. Example (page 83): 20 + w = 100. Solution: Inverse step: 100 - 20. So, w = 80 (missing number). If the missing number is in a subtraction sentence, swap the numbers and subtract. Example (page 83): 45 - x = 32. Solution: Inverse step: 45 - 32. So, x = 13 (missing number). Finding missing number or operation. Example (page 83): Complete the number sentences: a) 19 ___ 14 = 5, b) 20 ___ 5 = 4. Solution: a) 19 - 14 = 5, b) 20 / 5 = 4.

Exercise (page 84): Question 1 — Complete the following number sentences

Exercise (page 84): Q1 (a) — 4 + _____ = 9

Missing word: 5

Exercise (page 84): Q1 (b) — 6 _____ 3 = 18

Missing word: x (6 x 3 = 18)

Exercise (page 84): Q1 (c) — _____ / 4 = 4

Missing word: 16

Exercise (page 84): Q1 (d) — 7 - 5 = _____

Missing word: 2

Exercise (page 84): Q1 (e) — 10 _____ 13 = 23

Missing word: + (10 + 13 = 23)

Exercise (page 84): Q1 (f) — 15 - _____ = 9

Missing word: 6

Exercise (page 84): Q1 (g) — 7 x 4 = _____

Missing word: 28

Exercise (page 84): Q1 (h) — _____ + 18 = 30

Missing word: 12

Exercise (page 84): Q1 (i) — 30 _____ 6 = 5

Missing word: / (30 / 6 = 5)

Exercise (page 84): Q1 (j) — 13 _____ 8 = 5

Missing word: - (13 - 8 = 5)

Exercise (page 84): Q1 (k) — 63 / 7 = _____

Missing word: 9

Exercise (page 84): Q1 (l) — _____ x 8 = 64

Missing word: 8

Exercise (page 84): Q1 (m) — 65 + _____ = 75

Missing word: 10

Exercise (page 84): Q1 (n) — 9 _____ 9 = 81

Missing word: x (9 x 9 = 81)

Unit 2 Summary (page 84)

Summary (page 84): When adding numbers, always add ones to ones, tens to tens, hundreds to hundreds and thousands to thousands. Whenever zero is subtracted from any number, the answer is the number itself. When multiply a number by 0, the answer is always 0. When multiply a number by 1, the answer is always the number itself. Whenever change the order in multiplication the product remains the same. In division, we cannot change the order. When a number is divided by itself, the answer is always 1. Vocabulary: Thousands, estimating, odd sums, differences, multiple, multiplication sentence, divisor, dividend, remainder, product, groups, patterns. This closes Unit 2 (pages 36 to 84).

Unit 3: Fractions (pages 85 to 107)

Students' Learning Outcomes (page 85): Recognise as proper fractions, improper fractions, mixed numbers. Identify equivalent fractions and show families of equivalent fractions. Simplify fractions to the lowest term. Compare and order like fractions using symbols <, > and =. Add and subtract like and unlike fractions (with denominators that are multiples of the same number). Know and recognise that hundredths arise by dividing an object, single digit numbers and quantities into hundred equal parts. Identify that tenths arise by dividing an object, single digit number and quantities into ten equal parts (e.g., 2/10 = 0.2). The unit opener shows a pizza: one half is greater than one quarter, and one quarter is less than one half.

Fractions (page 86)

A fraction is a number that shows parts of a whole. A fraction or common fraction is a number that means parts of an object or parts of a group. A pizza is shown as a Whole, then divided into 2 equal parts, 3 equal parts and 4 equal parts. Numerator and Denominator of a Fraction (page 86): Observe the figure — a hexagon with 3 of its 6 parts shaded, so 3/6 parts of the figure are shaded. Number 3 shows the total shaded parts of a whole. Number 6 shows the total equal parts into which the whole has been divided. Number 3 above the line is known as numerator. Number 6 below the line is known as denominator. Hence, in the fraction, the numerator is 3 and the denominator is 6. The numerator tells how many parts are being used. The denominator tells how many parts are of a whole.

Unit Fraction, Like Fractions, Unlike Fractions (page 87)

Unit Fraction: One part of the whole is called unit fraction. Examples: 1/2, 1/3, 1/5, 1/9, 1/12. Like Fractions: Fractions with the same denominators are called like fractions. Examples: 1/4 and 3/4; 2/6 and 5/6; 5/10 and 3/10. Unlike Fractions: Fractions with different denominators are called unlike fractions. Examples: 1/2 and 1/3; 2/3 and 2/6; 2/5 and 4/8.

Proper and Improper Fractions and Mixed Fractions (pages 88 to 89)

Proper Fraction (page 88): A fraction with the numerator smaller than its denominator is called proper fraction. For example: 1/2, 2/4, 3/5, 2/5. In 1/2, 2/4, 3/5, 2/5, the numerator is smaller than the denominator. Hence these fractions are proper fractions. Improper Fraction (page 88): A fraction with the numerator greater than or equal to its denominator is called improper fraction. Examples: 8/5 and 5/3. Few more examples are: 4/3, 5/4, 6/6. NOTE — the book's list of 'few more examples' also prints 5/7, but 5/7 has a numerator smaller than its denominator, so it is a PROPER fraction, not an improper one; it is left out of the improper list here. Example (page 88): Write proper and improper fractions separately: 3/6, 7/10, 5/4, 9/3, 1/2, 2/6, 8/7, 11/4, 4/5. Proper fractions: 3/6, 7/10, 1/2, 2/6, 4/5. Improper fractions: 5/4, 9/3, 8/7, 11/4.

Mixed Numbers (Fractions) (page 89): A mixed number / fraction is the

combination of a whole number and a proper fraction. For example, 2 3/4. We can show it as: 1 + 1 + 3/4 = 2 + 3/4 = 2 3/4. Mixed number (fraction) 2 3/4 is the sum of whole number 2 and proper fraction 3/4. Key note: We can convert an improper fraction into a mixed number (fraction) and a mixed number into an improper fraction. Example (page 89): Write mixed fractions separately, from: 1 1/2, 2 1/4, 1 1/4, 7/11, 4/5, 2 1/6, 5/13, 5 3/7. Mixed fractions: 1 1/2, 2 1/4, 1 1/4, 2 1/6, 5 3/7. NOTE — the book's printed solution lists only 1 1/4, 2 1/6 and 5 3/7, leaving out 1 1/2 and 2 1/4, which are also mixed fractions; the complete set is given here.

Exercise (page 90): Question 1 — Write the proper and improper fraction in the blanks

Exercise (page 90): Q1 (ii) — 6/9

Missing word: Proper fraction

Exercise (page 90): Q1 (iii) — 7/4

Missing word: Improper fraction

Exercise (page 90): Q1 (iv) — 8/8

Missing word: Improper fraction (numerator equal to denominator)

Exercise (page 90): Q1 (v) — 5/3

Missing word: Improper fraction

Exercise (page 90): Q1 (vi) — 4/6

Missing word: Proper fraction

Exercise (page 90): Q1 (second iv) — 9/5

Missing word: Improper fraction

Exercise (page 90): Q1 (second v) — 3/7

Missing word: Proper fraction

Exercise (page 90): Q1 (second vi) — 9/9

Missing word: Improper fraction (numerator equal to denominator)

Exercise (page 90): Q1 (i) is already completed in the book: 3/6 is a Proper fraction. NOTE — the book numbers two different rows on page 90 as (iv), (v) and (vi); both rows are solved in order and the repeats are flagged here.

Exercise (page 90): Question 2 — Separate the proper and improper fraction in each set

Exercise (page 90): Q2 (i) — 5/3, 2/7, 6/5, 4/2, 2/3, 4/6: Proper fractions

Missing word: 2/7, 2/3, 4/6

Exercise (page 90): Q2 (i) — 5/3, 2/7, 6/5, 4/2, 2/3, 4/6: Improper fractions

Missing word: 5/3, 6/5, 4/2

Exercise (page 90): Q2 (ii) — 7/5, 4/3, 1/2, 8/4, 3/5, 2/4: Proper fractions

Missing word: 1/2, 3/5, 2/4

Exercise (page 90): Q2 (ii) — 7/5, 4/3, 1/2, 8/4, 3/5, 2/4: Improper fractions

Missing word: 7/5, 4/3, 8/4

Exercise (page 90): Q2 (iii) — 9/2, 5/10, 11/4, 13/8, 4/7, 5/8: Proper fractions

Missing word: 5/10, 4/7, 5/8

Exercise (page 90): Q2 (iii) — 9/2, 5/10, 11/4, 13/8, 4/7, 5/8: Improper fractions

Missing word: 9/2, 11/4, 13/8

Exercise (page 90): Q2 (iv) — 2/3, 12/9, 7/3, 13/5, 4/9, 4/5: Proper fractions

Missing word: 2/3, 4/9, 4/5

Exercise (page 90): Q2 (iv) — 2/3, 12/9, 7/3, 13/5, 4/9, 4/5: Improper fractions

Missing word: 12/9, 7/3, 13/5

Exercise (page 90): Question 3 — Separate mixed fraction in each set

Exercise (page 90): Q3 (i) — 3/4, 2 1/2, 5/6, 3 2/5, 3/6, 1 3/5: Mixed fractions

Missing word: 2 1/2, 3 2/5, 1 3/5

Exercise (page 90): Q3 (ii) — 1/3, 3 2/4, 4/5, 2 2/3, 3/6, 4 2/5: Mixed fractions

Missing word: 3 2/4, 2 2/3, 4 2/5

Exercise (page 90): Q3 (iii) — 2/5, 2 3/6, 4/7, 3 4/8, 2/5, 1 4/9: Mixed fractions

Missing word: 2 3/6, 3 4/8, 1 4/9

Exercise (page 90): Q3 (iv) — 4/5, 1 2/5, 3/8, 3 3/7, 4/6, 3 7/8: Mixed fractions

Missing word: 1 2/5, 3 3/7, 3 7/8

Equivalent Fractions (page 91)

Two or more fractions having the same value but different numerators and denominators are called equivalent fractions. Now we will identify the equivalent fractions from the given figures. Umar divided his strip into 2 equal parts. He colored one part or 1/2. His sister Sana divided her strip into 4 parts. She coloured 2 parts or 2/4. Who coloured more? When you compare the two strips, you will see that both parts are equal. We say, the two fractions are equivalent. Mathematically, we write it as: 1/2 = 2/4. Hence, these fractions are known as equivalent fractions. Example (page 91): Use fraction bars to show other fractions that equal 1/2. Solution: 1/2 = 2/4 = 4/8 = 3/6. Equivalent fractions are two or more

fractions that have the same value.

Families of Equivalent Fractions (page 92)

A family of equivalent fractions usually means a set or a group generated by multiplying numerator and denominator by the same number. For example: Family of 1/3: (1 x 2)/(3 x 2) = 2/6; (1 x 3)/(3 x 3) = 3/9; (1 x 4)/(3 x 4) = 4/12. Thus, 1/3, 2/6, 3/9 and 4/12 are all parts of the same family of equivalent fractions of 1/3. Family of 3/5: (3 x 2)/(5 x 2) = 6/10; (3 x 3)/(5 x 3) = 9/15; (3 x 4)/(5 x 4) = 12/20. So, 3/5, 6/10, 9/15 and 12/20 are all parts of another family of equivalent fractions of 3/5.

Teacher note (page 92): equivalent fractions are individual fractions that have the same value, such as 1/2 = 2/4 = 4/8 = 3/6. Families of equivalent fractions: a group of equivalent fractions to one specific original fraction such as 3/5, 6/10, 9/15, 12/20, etc. You can say that: 2/4 is an equivalent fraction of 1/2; 2/4, 3/6 and 4/8 are the family of equivalent fractions.

Exercise (page 93): Question 1 — Write the equivalent fractions of the following

Exercise (page 93): Q1 first set — circles showing 1/2, 2/4 and 3/6

Missing word: 1/2 = 2/4 = 3/6

Exercise (page 93): Q1 second set — circles showing 1/2, 3/6 and 5/10

Missing word: 1/2 = 3/6 = 5/10

Exercise (page 93): Question 2 — Write the fractions and match the equivalent fractions

Exercise (page 93): Q2 — the red bar (1 of 2 parts shaded)

Missing word: 1/2

Exercise (page 93): Q2 — the blue circle (1 of 4 parts shaded)

Missing word: 1/4

Exercise (page 93): Q2 — the green circle (2 of 8 parts shaded)

Missing word: 2/8

Exercise (page 93): Q2 — the blue grid (1 of 4 parts shaded)

Missing word: 1/4

Exercise (page 93): Q2 — the yellow bar 2/6 and the pink grid 4/12 are already given in the book. Matching equivalent fractions: 2/6 = 4/12 (both equal 1/3); 1/4 = 2/8 = 1/4. The fraction 1/2 has no equivalent partner among the other figures in this set.

Exercise (page 93): Question 3 — Write three equivalent fractions for the following

Exercise (page 93): Q3 (i) — 1/2

Missing word: 2/4, 3/6, 4/8

Exercise (page 93): Q3 (ii) — 2/5

Missing word: 4/10, 6/15, 8/20

Exercise (page 93): Q3 (iii) — 3/4

Missing word: 6/8, 9/12, 12/16

Exercise (page 93): Q3 (iv) — 5/8

Missing word: 10/16, 15/24, 20/32

Exercise (page 93): Q3 (v) — 6/11

Missing word: 12/22, 18/33, 24/44

Exercise (page 93): Q3 (vi) — 7/9

Missing word: 14/18, 21/27, 28/36

Exercise (page 94): Question 4 — Write three equivalent fractions by using fraction bar

Exercise (page 94): The fraction bar on the page shows a whole divided into 1/2, 1/3, 1/4, 1/5, 1/6, 1/7, 1/8, 1/9 and 1/10 parts. Equivalents that can be read directly off the bar are marked below; where the bar runs out, the family is continued by the same multiplication rule from page 92.

Exercise (page 94): Q4 (i) — 1/2

Missing word: 2/4, 3/6, 4/8 (all visible on the bar; also 5/10)

Exercise (page 94): Q4 (ii) — 1/4

Missing word: 2/8 (visible on the bar), 3/12, 4/16

Exercise (page 94): Q4 (iii) — 1/3

Missing word: 2/6, 3/9 (visible on the bar), 4/12

Exercise (page 94): Q4 (iv) — 1/5

Missing word: 2/10 (visible on the bar), 3/15, 4/20

Exercise (page 94): Question 5 — Write family of equivalent fractions of

Exercise (page 94): Q5 (i) — 1/4

Missing word: 2/8, 3/12, 4/16

Exercise (page 94): Q5 (ii) — 4/5

Missing word: 8/10, 12/15, 16/20

Exercise (page 94): Q5 (iii) — 2/3

Missing word: 4/6, 6/9, 8/12

Exercise (page 94): Q5 (iv) — 2/7

Missing word: 4/14, 6/21, 8/28

Exercise (page 94): Q5 (v) — 2/9

Missing word: 4/18, 6/27, 8/36

Exercise (page 94): Q5 (vi) — 2/5

Missing word: 4/10, 6/15, 8/20

Simplifying Fractions to its Lowest Form (page 95)

If a fraction's numerator and denominator are divided by the same non-zero number, it reduces the fraction to its lowest form. Example (page 95): Simplify fraction 16/24 to its lowest form. Solution: Step-I: Divide the numerator and denominator of the fraction 16/24 by the common divisor 2: (16 / 2)/(24 / 2) = 8/12. Step-II: Divide the numerator and denominator of the fraction 8/12 by 2 again: (8 / 2)/(12 / 2) = 4/6. Step-III: Divide the numerator and denominator of the fraction 4/6 by 2 again: (4 / 2)/(6 / 2) = 2/3. So, 2/3 is the simplified or the lowest form of the fraction 16/24.

Exercise (page 95): Question 1 — Simplify the following fractions into their lowest form

Exercise (page 95): Q1 (1) — 28/36

Step 1: Divide numerator and denominator by the common divisor 4: (28 / 4)/(36 / 4).

Answer: 7/9

Exercise (page 95): Q1 (2) — 36/27

Step 1: Divide numerator and denominator by the common divisor 9: (36 / 9)/(27 / 9).

Answer: 4/3

Exercise (page 95): Q1 (3) — 40/70

Step 1: Divide numerator and denominator by the common divisor 10: (40 / 10)/(70 / 10).

Answer: 4/7

Exercise (page 95): Q1 (4) — 21/42

Step 1: Divide numerator and denominator by the common divisor 21: (21 / 21)/(42 / 21).

Answer: 1/2

Exercise (page 95): Q1 (5) — 66/99

Step 1: Divide numerator and denominator by the common divisor 33: (66 / 33)/(99 / 33).

Answer: 2/3

Exercise (page 95): Q1 (6) — 84/140

Step 1: Divide numerator and denominator by the common divisor 28: (84 / 28)/(140 / 28).

Answer: 3/5

Exercise (page 95): Q1 (7) — 52/80

Step 1: Divide numerator and denominator by the common divisor 4: (52 / 4)/(80 / 4).

Answer: 13/20

Exercise (page 95): Q1 (8) — 68/84

Step 1: Divide numerator and denominator by the common divisor 4: (68 / 4)/(84 / 4).

Answer: 17/21

Exercise (page 95): Q1 (9) — 36/92

Step 1: Divide numerator and denominator by the common divisor 4: (36 / 4)/(92 / 4).

Answer: 9/23

Exercise (page 95): Q1 (10) — 39/78

Step 1: Divide numerator and denominator by the common divisor 39: (39 / 39)/(78 / 39).

Answer: 1/2

Exercise (page 95): Q1 (11) — 78/120

Step 1: Divide numerator and denominator by the common divisor 6: (78 / 6)/(120 / 6).

Answer: 13/20

Exercise (page 95): Q1 (12) — 90/240

Step 1: Divide numerator and denominator by the common divisor 30: (90 / 30)/(240 / 30).

Answer: 3/8

Comparing Like Fractions (page 96)

Comparing fractions with the same denominators using symbols <, > and =. We can explain comparing fractions with the following example: Umar divides a circle into 4 equal parts. He colours one part of the circle (1/4). Mariam also divides a circle into 4 equal parts. She colours three parts (3/4). Who coloured more? Here (3/4) is greater than 1/4, or 3/4 > 1/4. So, Mariam coloured more parts (3/4) of the circle than Umar. If fractions have the same denominators, then greater the numerator, the greater will be the fraction and smaller the numerator, the smaller will be the fraction. Example (page 96): Write <, > or = to compare fractions of the same denominator: 1) 3/4 □ 1/4. Solution: 3/4 > 1/4. Numerator 3 is greater than 1, i.e., 3 > 1. 2) 2/7 □ 4/7. Solution: 2/7 < 4/7. Numerator 2 is smaller than 4, i.e., 2 < 4. 3) 4/5 □ 4/5. Solution: 4/5 = 4/5. Numerator

of both fractions are equal, i.e., 4 = 4.

Ordering of Fractions (page 97)

Fractions can be arranged in ascending and descending order. Such arrangement is called ordering of fractions. Example (page 97): Arrange the following fractions in ascending order: 2/6, 4/6, 1/6, 3/6. Solution: 1/6 < 2/6 < 3/6 < 4/6. Ascending order: 1/6, 2/6, 3/6, 4/6. Example (page 97): Arrange the following in descending order: 1/5, 3/5, 2/5, 4/5. Solution: 4/5 > 3/5 > 2/5 > 1/5. Descending order: 4/5, 3/5, 2/5, 1/5.

Exercise (page 98): Question 1 — Use the sign <, > or = to compare pairs of fractions

Exercise (page 98): Q1 (ii) — 2/4 and 3/4

Missing word: 2/4 < 3/4

Exercise (page 98): Q1 (iii) — 4/10 and 3/10

Missing word: 4/10 > 3/10

Exercise (page 98): Q1 (iv) — 3/8 and 6/8

Missing word: 3/8 < 6/8

Exercise (page 98): Q1 (i) is already completed in the book: 5/8 > 2/8.

Exercise (page 98): Question 2 — Compare and write <, > or = as suitable sign

Exercise (page 98): Q2 (ii) — 4/5 and 2/5

Missing word: 4/5 > 2/5

Exercise (page 98): Q2 (iii) — 6/8 and 5/8

Missing word: 6/8 > 5/8

Exercise (page 98): Q2 (iv) — 5/9 and 5/9

Missing word: 5/9 = 5/9

Exercise (page 98): Q2 (v) — 3/7 and 4/7

Missing word: 3/7 < 4/7

Exercise (page 98): Q2 (vi) — 3/4 and 3/4

Missing word: 3/4 = 3/4

Exercise (page 98): Q2 (vii) — 6/7 and 4/7

Missing word: 6/7 > 4/7

Exercise (page 98): Q2 (viii) — 3/6 and 5/6

Missing word: 3/6 < 5/6

Exercise (page 98): Q2 (ix) — 7/10 and 2/10

Missing word: 7/10 > 2/10

Exercise (page 98): Q2 (i) is already completed in the book: 1/3 < 2/3.

Exercise (page 99): Question 3 — Arrange the following fractions in ascending order

Exercise (page 99): Q3 (i) — 3/4, 2/4, 1/4, 4/4

Missing word: 1/4, 2/4, 3/4, 4/4

Exercise (page 99): Q3 (ii) — 7/8, 5/8, 3/8, 4/8

Missing word: 3/8, 4/8, 5/8, 7/8

Exercise (page 99): Q3 (iii) — 3/11, 6/11, 5/11, 4/11

Missing word: 3/11, 4/11, 5/11, 6/11

Exercise (page 99): Q3 (iv) — 1/7, 4/7, 2/7, 3/7

Missing word: 1/7, 2/7, 3/7, 4/7

Exercise (page 99): Question 4 — Arrange the following fractions in descending order

Exercise (page 99): Q4 (i) — 3/10, 5/10, 2/10, 4/10

Missing word: 5/10, 4/10, 3/10, 2/10

Exercise (page 99): Q4 (ii) — 3/7, 5/7, 4/7, 6/7

Missing word: 6/7, 5/7, 4/7, 3/7

Exercise (page 99): Q4 (iii) — 3/8, 6/8, 4/8, 5/8

Missing word: 6/8, 5/8, 4/8, 3/8

Exercise (page 99): Q4 (iv) — 3/12, 5/12, 4/12, 6/12

Missing word: 6/12, 5/12, 4/12, 3/12

Addition of Fractions (page 99)

Addition of Like Fractions: Example (page 99): Add 3/8 + 4/8. Solution: 3/8 + 4/8 = 7/8 (add the numerators, keep the denominator). Addition of Unlike Fractions: Example (page 99): Add 2/6 + 3/12. Solution: First make the denominators same: (2 x 2)/(6 x 2) + 3/12 = 4/12 + 3/12. Now add the numerators as follows: (4 + 3)/12 = 7/12.

Exercise (page 100): Question 1 — Add the fractions and colour the shapes

Exercise (page 100): Q1 (ii) — 3/5 + 1/5

Missing word: 4/5 (colour 4 of the 5 parts of the shape)

Exercise (page 100): Q1 (iii) — 3/7 + 2/7

Missing word: 5/7 (colour 5 of the 7 parts of the shape)

Exercise (page 100): Q1 (iv) — 3/9 + 5/9

Missing word: 8/9 (colour 8 of the 9 parts of the shape)

Exercise (page 100): Q1 (v) — 3/6 + 1/6

Missing word: 4/6 (colour 4 of the 6 parts of the shape)

Exercise (page 100): Q1 (vi) — 1/4 + 2/4

Missing word: 3/4 (colour 3 of the 4 parts of the shape)

Exercise (page 100): Q1 (i) is already completed in the book: 3/8 + 4/8 = 7/8, with 7 of the 8 parts coloured.

Exercise (page 100): Question 2 — Solve the following

Exercise (page 100): Q2 (ii) — 1/3 + 2/3

Step 1: The denominators are the same, so add the numerators: 1 + 2 = 3.

Step 2: Keep the denominator 3.

Answer: 3/3 = 1

Exercise (page 100): Q2 (iii) — 4/9 + 4/9

Step 1: Add the numerators: 4 + 4 = 8.

Step 2: Keep the denominator 9.

Answer: 8/9

Exercise (page 100): Q2 (iv) — 5/8 + 7/8

Step 1: Add the numerators: 5 + 7 = 12.

Step 2: Keep the denominator 8.

Answer: 12/8, which is 1 1/2

Exercise (page 100): Q2 (v) — 3/9 + 8/9

Step 1: Add the numerators: 3 + 8 = 11.

Step 2: Keep the denominator 9.

Answer: 11/9, which is 1 2/9

Exercise (page 100): Q2 (vi) — 6/13 + 7/13

Step 1: Add the numerators: 6 + 7 = 13.

Step 2: Keep the denominator 13.

Answer: 13/13 = 1

Exercise (page 100): Q2 (i) is already completed in the book: 2/5 + 1/5 =

3/5.

Exercise (page 100): Question 3 — Solve the following (unlike fractions)

Exercise (page 100): Q3 (i) — 2/3 + 3/6

Step 1: Make the denominators the same: 2/3 = (2 x 2)/(3 x 2) = 4/6.

Step 2: Add the numerators: 4 + 3 = 7. Keep the denominator 6.

Answer: 7/6

Exercise (page 100): Q3 (ii) — 1/2 + 5/8

Step 1: Make the denominators the same: 1/2 = (1 x 4)/(2 x 4) = 4/8.

Step 2: Add the numerators: 4 + 5 = 9. Keep the denominator 8.

Answer: 9/8

Exercise (page 100): Q3 (iii) — 2/5 + 7/10

Step 1: Make the denominators the same: 2/5 = (2 x 2)/(5 x 2) = 4/10.

Step 2: Add the numerators: 4 + 7 = 11. Keep the denominator 10.

Answer: 11/10

Exercise (page 100): Q3 (iv) — 5/12 + 2/6

Step 1: Make the denominators the same: 2/6 = (2 x 2)/(6 x 2) = 4/12.

Step 2: Add the numerators: 5 + 4 = 9. Keep the denominator 12.

Answer: 9/12, which simplifies to 3/4

Exercise (page 100): Q3 (v) — 5/16 + 3/4

Step 1: Make the denominators the same: 3/4 = (3 x 4)/(4 x 4) = 12/16.

Step 2: Add the numerators: 5 + 12 = 17. Keep the denominator 16.

Answer: 17/16

Exercise (page 100): Q3 (vi) — 9/14 + 3/7

Step 1: Make the denominators the same: 3/7 = (3 x 2)/(7 x 2) = 6/14.

Step 2: Add the numerators: 9 + 6 = 15. Keep the denominator 14.

Answer: 15/14

Subtraction of Fractions (page 101)

Subtraction of Like Fractions: Example (page 101): Subtract 7/8 - 3/8. Solution: 7/8 - 3/8 = 4/8 (subtract the numerators, keep the denominator). Subtraction of Unlike Fractions: Example (page 101): Subtract 9/12 - 3/6. Solution: First make the denominators same: 9/12 - (3 x 2)/(6 x 2) = 9/12 - 6/12. Now write the denominator as common and subtract the numerators: (9 - 6)/12 = 3/12.

Exercise (page 101): Question 1 — Subtract the following

Exercise (page 101): Q1 (i) — 2/3 - 1/3

Missing word: 1/3

Exercise (page 101): Q1 (ii) — 3/4 - 1/4

Missing word: 2/4

Exercise (page 101): Q1 (iii) — 5/6 - 3/6

Missing word: 2/6

Exercise (page 101): Q1 (iv) — 8/10 - 7/10

Missing word: 1/10

Exercise (page 101): Q1 (v) — 9/16 - 7/16

Missing word: 2/16

Exercise (page 101): Q1 (vi) — 7/8 - 5/8

Missing word: 2/8

Exercise (page 101): Question 2 — Solve the following (unlike fractions)

Exercise (page 101): Q2 (i) — 5/9 - 1/3

Step 1: Make the denominators the same: 1/3 = (1 x 3)/(3 x 3) = 3/9.

Step 2: Subtract the numerators: 5 - 3 = 2. Keep the denominator 9.

Answer: 2/9

Exercise (page 101): Q2 (ii) — 6/8 - 2/4

Step 1: Make the denominators the same: 2/4 = (2 x 2)/(4 x 2) = 4/8.

Step 2: Subtract the numerators: 6 - 4 = 2. Keep the denominator 8.

Answer: 2/8

Exercise (page 101): Q2 (iii) — 8/10 - 2/5

Step 1: Make the denominators the same: 2/5 = (2 x 2)/(5 x 2) = 4/10.

Step 2: Subtract the numerators: 8 - 4 = 4. Keep the denominator 10.

Answer: 4/10

Exercise (page 101): Q2 (iv) — 7/12 - 3/6

Step 1: Make the denominators the same: 3/6 = (3 x 2)/(6 x 2) = 6/12.

Step 2: Subtract the numerators: 7 - 6 = 1. Keep the denominator 12.

Answer: 1/12

Exercise (page 101): Q2 (v) — 9/15 - 2/5

Step 1: Make the denominators the same: 2/5 = (2 x 3)/(5 x 3) = 6/15.

Step 2: Subtract the numerators: 9 - 6 = 3. Keep the denominator 15.

Answer: 3/15

Exercise (page 101): Q2 (vi) — 11/14 - 4/7

Step 1: Make the denominators the same: 4/7 = (4 x 2)/(7 x 2) = 8/14.

Step 2: Subtract the numerators: 11 - 8 = 3. Keep the denominator 14.

Answer: 3/14

Hundredths (page 102)

Divide one whole into 100 equal parts: 1 part = 1/100 = 1 hundredth. 1 whole = 100 hundredths. We can write it in a place value chart as: Ones 0, Tenths 0, Hundredths 1. We can also show it on a number line as: 0, 1/100, 5/100, 10/100. Example (page 102): What are the hundredths for the shaded parts of a square? 32/100: 32 shaded parts, 100 parts in all = 32 hundredths. We can write it in a place value chart as: Ones 0, Tenths 3, Hundredths 2. On the number line: 30/100, 32/100, 35/100, 40/100. Show in hundredths 1 25/100 = 1 25/100 = 1 whole + 25 hundredths = 100 hundredths + 25 hundredths = 125 hundredths.

Exercise (page 103): Question 1 — Read and write the fractions and hundredths for each shaded parts of the squares

Exercise (page 103): Q1 (ii) — shaded parts of the 10 by 10 square

Missing word: 34/100 = Thirty-four hundredths (3 full rows of 10 plus 4 cells in the fourth row)

Exercise (page 103): Q1 (iii) — shaded parts of the 10 by 10 square

Missing word: 5/100 = Five hundredths (5 cells in the first row)

Exercise (page 103): Q1 (iv) — shaded parts of the 10 by 10 square

Missing word: 18/100 = Eighteen hundredths (1 full row of 10 plus 8 cells in the second row)

Exercise (page 103): Q1 (v) — shaded parts of the 10 by 10 square

Missing word: 97/100 = Ninety-seven hundredths (8 full rows, 9 cells in the ninth row and 8 cells in the tenth row)

Exercise (page 103): Q1 (vi) — shaded parts of the 10 by 10 square

Missing word: 77/100 = Seventy-seven hundredths (7 full rows plus 7

cells in the eighth row)

Exercise (page 103): Q1 (i) is already completed in the book: 9/100 = Nine hundredths. The shaded cells for (ii) to (vi) were counted at 300 dpi, including a zoomed crop for (v).

Exercise (page 103): Question 2 — Write each fraction as hundredths

Exercise (page 103): Q2 (ii) — 43/100

Missing word: Forty-three hundredths

Exercise (page 103): Q2 (iii) — 75/100

Missing word: Seventy-five hundredths

Exercise (page 103): Q2 (i) is already completed in the book: 7/100 = Seven hundredths.

Exercise (page 103): Question 3 — Write each of the following as a fraction

Exercise (page 103): Q3 (ii) — Fifty seven hundredths

Missing word: 57/100

Exercise (page 103): Q3 (iii) — Eighty eight hundredths

Missing word: 88/100

Exercise (page 103): Q3 (iv) — Sixteen hundredths

Missing word: 16/100

Exercise (page 103): Q3 (i) is already completed in the book: Thirty three hundredths = 33/100.

Identifying Tenths (page 104)

We divide a strip into 10 equal parts. 2 shaded parts = 2 tenths = 2/10 = 0.2. 1 whole = 10 tenths. We can write it in a place value chart as: Ones

0, Tenths 2. It is read as 'zero point two'. We call it decimal number. The point is called decimal point. We can also show 0.2 on a number line as: 0, 0.2, 0.5, 1. It is also read as two tenths of a whole. So, the decimal number is 0.2. A decimal number can have one or more digits to the right of the decimal point. Example (page 104): What is the decimal number for the shaded parts of a circle? 3/10: 3 shaded parts, 10 parts in all = 3 tenths = 0.3. In the place value chart: Ones 0, Tenths 3. It is read as three tenths. So, the decimal number is 0.3. Remember: A decimal point separates a whole number from a fractional part of a number.

Dividing single digit numbers and quantities by ten (page 105)

Example (page 105): Show in tenths 2 4/10. Solution: = 2 4/10 = 2 whole + 4 tenths = 20 tenths + 4 tenths = 24 tenths = 2.4. We can write it in a place value chart as: Ones 2, Tenths 4. On the number line: 2, 2.4, 2.5, 3. We read it as 2 and 4 tenths. So, the decimal number is 2.4. It is read as two point four.

Exercise (page 105): Question 1 — Express each of the following fraction as a decimal number

Exercise (page 105): Q1 (ii) — 5/10

Missing word: 0.5

Exercise (page 105): Q1 (iii) — 7/10

Missing word: 0.7

Exercise (page 105): Q1 (iv) — 3 2/10

Missing word: 3.2

Exercise (page 105): Q1 (v) — 5 4/10

Missing word: 5.4

Exercise (page 105): Q1 (vi) — 6 3/10

Missing word: 6.3

Exercise (page 105): Q1 (i) is already completed in the book: 1/10 = 0.1.

Exercise (page 106): Question 2 — Express each decimal number as a fraction

Exercise (page 106): Q2 (ii) — 0.4

Missing word: 4/10

Exercise (page 106): Q2 (iii) — 0.8

Missing word: 8/10

Exercise (page 106): Q2 (iv) — 2.3

Missing word: 2 3/10

Exercise (page 106): Q2 (v) — 4.5

Missing word: 4 5/10

Exercise (page 106): Q2 (vi) — 7.6

Missing word: 7 6/10

Exercise (page 106): Q2 (i) is already completed in the book: 0.6 = 6/10.

Exercise (page 106): Question 2 (second) — Express a fraction and a decimal number for each shaded parts of the given figures

Exercise (page 106): Q2 (ii) — strip of 10 with 3 parts shaded

Missing word: 3/10 = 0.3

Exercise (page 106): Q2 (iii) — strip of 10 with 6 parts shaded

Missing word: 6/10 = 0.6

Exercise (page 106): Q2 (iv) — two strips of 10, first fully shaded and 6 parts of the second shaded

Missing word: 1 6/10 = 1.6

Exercise (page 106): Q2 (v) — two strips of 10, first fully shaded and 9 parts of the second shaded

Missing word: 1 9/10 = 1.9

Exercise (page 106): Q2 (vi) — two strips of 10, first fully shaded and 6 parts of the second shaded

Missing word: 1 6/10 = 1.6

Exercise (page 106): Q2 (i) is already completed in the book: 5/10 = 0.5. NOTE — the page numbers two different questions as '2'; both are solved in order. NOTE — figures (iv) and (vi) show identical shading (one full strip plus 6 parts of the second strip), so both give 1 6/10 = 1.6; the shaded cells were counted at 300 dpi.

Exercise (page 106): Question 3 — Write each as a fraction

Exercise (page 106): Q3 (ii) — Two tenths

Missing word: 2/10 = 0.2

Exercise (page 106): Q3 (iii) — Seven tenths

Missing word: 7/10 = 0.7

Exercise (page 106): Q3 (iv) — Two and five tenths

Missing word: 2 5/10 = 2.5

Exercise (page 106): Q3 (i) is already completed in the book: Five tenths = 5/10 = 0.5.

Unit 3 Summary (page 107)

Summary (page 107): A fraction shows a part of a whole that is divided into equal parts. The numerator tells how many parts are being used. The total number of parts of an object is known as denominator. In a proper

fraction, the numerator is smaller than its denominator. In an improper fraction, the numerator is greater than or equal to the denominator. Fractions that are equal to each other are called equivalent fractions. When we add two fractions with the same denominators, we add only numerators. In two same denominator fractions, we subtract only numerators. Vocabulary: Fraction, numerator, denominator, proper fraction, improper fraction, equivalent fraction, decimal, tenths, hundredths. This closes Unit 3 (pages 85 to 107).

Unit 4: Measurement of Length, Mass, Capacity and Temperature (pages 108 to 134)

Students' Learning Outcomes (page 108): Recognise and use the standard units of length (kilometre, metre, centimetre and millimetre) to measure and record the length of different objects. Add and subtract lengths, given in the same units to solve real-world word problems. Recognise and use the standard units of mass (Kilograms, grams and milligrams) to measure and record the mass of different objects. Add and subtract mass, given in the same units to solve real-world word problems. Recognise and use the standard units of capacity (litre and millilitre) to measure and record the capacity of different objects. Read and write temperature to the nearest appropriate unit i.e., (degrees C) using pictorial representations and relating temperature scale to number line. Compare and order temperature using <, > and = signs. Add and subtract capacities given in the same units to solve real-life word problems.

Standard Units of Length (pages 109 to 110)

We measure the length of objects using standard units of length, which are metre, kilometre, centimetre and millimetre. We measure the length of short objects in millimetres or centimetres and the length of long objects in metres. Long distances are measured in kilometres. Metre (page 109): The metre is the most appropriate unit to measure the length. It is the basic unit of length, and is denoted by 'm'. Key Fact: 1 m = 100 cm. We use metre ruler, metre rod or measuring tape as measuring scale to measure different objects. Few examples to measure the length of objects in metres (page 110): Length of door: 2 m by 1 m. Length of blackboard: 1 m by 2 m. Centimetre (page 110): A centimetre ruler is used to measure the length of short objects, and is denoted as 'cm'. Key Fact: 100 cm = 1 m. Let us observe the lengths of these objects in centimetres: The length of pencil = 10 cm. The length of geometry box = 7 cm.

Millimetre and Kilometre (page 111)

Millimetre (page 111): A millimetre ruler is used to measure the length of very short objects, and it is denoted as 'mm'. Key Fact: 1 cm = 10 mm, so 1 centimetre = 10 millimetres. Lengths of some objects in millimetres (page 111): The length of hair clip = 20 mm. The length of leaf = 50 mm. The length of key = 30 mm. The length of button = 10 mm. Kilometre (page 111): The distance from one place to another is measured in kilometres. In short form we use 'km' for kilometre. We write 1 kilometre as '1 km'. Key Fact: 1 km = 1000 m. Example (page 111): the distance

from Ziarat to Quetta is 125 km.

Exercise (page 112): Question 1 — Write the suitable units of length (m, cm, mm) for measuring the length of the following

Exercise (page 112): Q1 (a) — ring

Missing word: mm (a very small object)

Exercise (page 112): Q1 (b) — book

Missing word: cm

Exercise (page 112): Q1 (c) — candle

Missing word: cm

Exercise (page 112): Q1 (d) — door

Missing word: m

Exercise (page 112): Q1 (e) — hairbrush

Missing word: cm

Exercise (page 112): Q1 (f) — table

Missing word: m

Exercise (page 112): Q1 (j) — Distance between Quetta and Karachi

Missing word: km

Exercise (page 112): Q1 (k) — Distance between Govt. Science College Quetta and Bolan Medical Hospital

Missing word: km

Exercise (page 112): NOTE — Question 1 jumps from item (f) to item (j); items (g), (h) and (i) are not printed in the book, so nothing is invented for them. The unit for (a) is a judgement item: a ring is very small, so millimetres suit it best.

Exercise (page 112): Question 2 — Measure and write the length

of the objects

Exercise (page 112): Q2 (a) — table

Missing word: The length of table is 3 m. The height of table is 1 m.

Exercise (page 112): Q2 (b) — chair

Missing word: The length of chair is 1 m. The height of chair is 2 m.

Exercise (page 112): NOTE — the chair's height is labelled 2 m in the book's figure, so it is answered as printed, although a real chair is shorter.

Exercise (page 113): Question 2 (continued) — Measure and write the length of the objects

Exercise (page 113): Q2 (c) — van

Missing word: The length of van is 6 m. The height of van is 4 m.

Exercise (page 113): Q2 (d) — airplane

Missing word: The length of airplane is 18 m. The height of airplane is 12 m.

Exercise (page 113): Q2 (e) — blackboard

Missing word: The length of blackboard is 4 m. The height of blackboard is 3 m.

Exercise (page 113): Q2 (f) — whiteboard

Missing word: The length of whiteboard is 2 m. The height of whiteboard is 2 m.

Exercise (page 113): Question 3 — Measure and write length of the objects (in centimetres)

Exercise (page 113): Q3 (a) — key

Missing word: Key length is about 4 cm (the key spans from the 0 mark to about the 4 cm mark).

Exercise (page 113): Q3 (a) — flower

Missing word: Flower length is about 6 cm (the flower spans from the 0 mark to about the 6 cm mark).

Exercise (page 113): Q3 (b) — book

Missing word: Book length is about 3 cm (the book spans from the 0 mark to about the 3 cm mark).

Exercise (page 113): Q3 (b) — pencil

Missing word: Pencil length is about 9 cm (the pencil spans from the 0 mark to about the 9 cm mark).

Exercise (page 113): NOTE — these printed objects are read against the ruler to the nearest centimetre; the readings were measured on the 300 dpi page (key 329-731, flower 334-939, book 325-656, pencil 326-1253 against the 0 mark at 332 and 102 pixels per centimetre).

Exercise (page 114): Question 3 (continued) — Measure and write length of the objects (in centimetres)

Exercise (page 114): Q3 (c) — arrow

Missing word: Arrow length is about 8 cm (the arrow spans from the 0 mark to about the 8 cm mark).

Exercise (page 114): Q3 (c) — screwdriver

Missing word: Screwdriver length is about 12 cm (the screwdriver spans from the 0 mark to about the 12 cm mark).

Exercise (page 114): Question 4 — Measure and write the length of the objects in millimetres

Exercise (page 114): Q4 (a) — watch

Missing word: Watch length is 10 mm (the book's guide lines run from the 0 mark to the 10 mm mark).

Exercise (page 114): Q4 (a) — eraser

Missing word: Eraser length is 20 mm (the eraser spans from the 0 mark to the 20 mm mark).

Exercise (page 114): Q4 (b) — leaf

Missing word: Leaf length is 30 mm (the book's guide lines run from the 0 mark to the 30 mm mark).

Exercise (page 114): Q4 (b) — clip

Missing word: Clip length is 40 mm (the clip spans from the 0 mark to the 40 mm mark).

Exercise (page 114): Q4 (c) — toothbrush

Missing word: Toothbrush length is 60 mm (the book's guide lines run from the 0 mark to the 60 mm mark).

Exercise (page 114): Q4 (c) — paintbrush

Missing word: Paintbrush length is 80 mm (the book's guide lines run from the 0 mark to the 80 mm mark).

Exercise (page 115): Question 4 (continued) — Measure and write the length of the objects in millimetres

Exercise (page 115): Q4 (d) — screw

Missing word: Screw length is 20 mm (the book's guide lines run from the 0 mark to the 20 mm mark).

Exercise (page 115): Q4 (d) — nail

Missing word: Nail length is 40 mm (the nail spans from the 0 mark to the 40 mm mark).

Exercise (page 115): Question 5 — Observe and write the distance in kilometres

Exercise (page 115): Q5 (a) — hospital to park

Missing word: The distance from the hospital to the park is 10 km.

Exercise (page 115): Q5 (b) — playland to school

Missing word: The distance from the playland to the school is 8 km.

Exercise (page 115): Q5 (c) — home to bazar

Missing word: The distance from the home to the bazar is 12 km.

Real-World Word Problems: Length (pages 116 to 117)

Real-World Word Problems (page 116): (i) Nadia had 84 cm of red ribbon and 56 cm of blue ribbon. How much ribbon did she have in all?

Step 1: Red ribbon = 84 cm.

Step 2: Blue ribbon = 56 cm.

Step 3: In all means add: 84 + 56.

Step 4: 84 + 56 = 140.

Answer: 140 cm

Real-World Word Problems (page 116): (ii) Arif bought 75 m of red carpet and 68 m of blue carpets. What is the total length of carpet bought by Arif?

Step 1: Red carpet = 75 m.

Step 2: Blue carpet = 68 m.

Step 3: Total means add: 75 + 68.

Step 4: 75 + 68 = 143.

Answer: 143 m

Real-World Word Problems (page 116): (iii) The height of the minarets of Faisal Masjid is 90 m and the height of the minarets of Badshahi Masjid is 70 m. Find the total height of minarets.

Step 1: Faisal Masjid minarets = 90 m.

Step 2: Badshahi Masjid minarets = 70 m.

Step 3: Total means add: 90 + 70.

Step 4: 90 + 70 = 160.

Answer: 160 m

Real-World Word Problems (page 116): (iv) Tanveer drove his car 687 km from Quetta to Karachi and 622 km from Karachi to Gwadar. What is the total distance travelled by him?

Step 1: Quetta to Karachi = 687 km.

Step 2: Karachi to Gwadar = 622 km.

Step 3: Total means add: 687 + 622.

Step 4: 687 + 622 = 1309.

Answer: 1309 km

Real-World Word Problems (page 116): (v) A train covered a distance of 858 km from city A to city B and 763 km from city B to C. Find the total distance covered by the train from A to C.

Step 1: City A to city B = 858 km.

Step 2: City B to city C = 763 km.

Step 3: Total means add: 858 + 763.

Step 4: 858 + 763 = 1621.

Answer: 1621 km

Real-World Word Problems (page 116): (vi) Arshad travelled 755 km by train and 565 km by road. Find the total distance covered by him.

Step 1: Distance by train = 755 km.

Step 2: Distance by road = 565 km.

Step 3: Total means add: 755 + 565.

Step 4: 755 + 565 = 1320.

Answer: 1320 km

Real-World Word Problems (page 116): NOTE — item (v) prints 'fram' for 'from'; the spelling is not changed in the answer.

Real-World Word Problems (page 117): (i) Sana's house is 12 km from her school. Zara's house is 8 km farther away from Sana's house. Find out distance of Zara's house from school.

Step 1: Sana's house from school = 12 km.

Step 2: Zara's house is 8 km farther, so add 8 km.

Step 3: 12 + 8 = 20.

Answer: 20 km

Real-World Word Problems (page 117): (ii) The height of the red building is 52 m and the height of the gray building is 38 m. What is the difference between the heights of the two buildings?

Step 1: Red building = 52 m.

Step 2: Gray building = 38 m.

Step 3: Difference means subtract: 52 − 38.

Step 4: 52 − 38 = 14.

Answer: 14 m

Real-World Word Problems (page 117): (iii) The height of the Faisal Masjid minarets is 90 m and the height of the Badshahi Masjid minarets is 70 m approximately. What is the difference between the height of these two minarets?

Step 1: Faisal Masjid minarets = 90 m.

Step 2: Badshahi Masjid minarets = 70 m.

Step 3: Difference means subtract: 90 − 70.

Step 4: 90 − 70 = 20.

Answer: 20 m

Real-World Word Problems (page 117): (iv) A 200 cm cloth is cut into two pieces. One piece is 155 cm long. What is the length of the other piece?

Step 1: Whole cloth = 200 cm.

Step 2: One piece = 155 cm.

Step 3: Other piece = 200 − 155.

Step 4: 200 − 155 = 45.

Answer: 45 cm

Real-World Word Problems (page 117): (v) The length of two roads are 865 km and 468 km. Find the difference between the lengths of the two roads.

Step 1: First road = 865 km.

Step 2: Second road = 468 km.

Step 3: Difference means subtract: 865 − 468.

Step 4: 865 − 468 = 397.

Answer: 397 km

Real-World Word Problems (page 117): (vi) The distance between Quetta and Islamabad is 896 km. If the distance between Quetta and Zhob is 335 km. What is the distance between Zhob and Islamabad?

Step 1: Quetta to Islamabad = 896 km.

Step 2: Quetta to Zhob = 335 km.

Step 3: Zhob to Islamabad = 896 − 335.

Step 4: 896 − 335 = 561.

Answer: 561 km

Standard Units of Mass (pages 118 to 119)

Standard Units of Mass (page 118): We measure the mass of an object using standard units such as kilograms, grams and milligrams. Kilogram (page 118): We measure the mass of heavy objects in kilograms. It is denoted 'kg'. Some iron masses of kilograms are: 1 kg, 2 kg, 3 kg, 4 kg, 5 kg. Some examples of mass of objects in kilograms (page 118): (i) Weight of Pumpkin is 2 kg. (ii) Weight of Wheat is 10 kg. Gram (page 118): We measure the mass of light objects in grams. In short form, it is denoted as 'g'. Key Fact: 1 kg = 1000 g. Some examples of mass of objects in grams (page 119): (i) Weight of cherries 100 g. (ii) Weight of Coriander seeds 300 g. Milligram (page 119): We measure the mass of extremely light objects in milligrams and it is denoted as 'mg'. Key Fact: 1000 mg = 1 g. Some examples of mass of objects in milligrams (page 119): Mass of fennel seeds 200 mg. Mass of cumin seeds 500 mg.

Exercise (page 119): Question 1 — Write the suitable unit (kg, g,

mg) for measuring the mass of the following

Exercise (page 119): Q1 (a) — Sugar bag

Missing word: kg

Exercise (page 119): Q1 (b) — Cardamom

Missing word: g

Exercise (page 119): Q1 (c) — Necklace

Missing word: g

Exercise (page 120): Question 1 (continued) — Write the suitable unit (kg, g, mg) for measuring the mass of the following

Exercise (page 120): Q1 (d) — Wheat bag

Missing word: kg

Exercise (page 120): Q1 (e) — Black seeds

Missing word: g

Exercise (page 120): Q1 (f) — Paper

Missing word: g

Exercise (page 120): Question 3 — Write the mass in kg or g for the following

Exercise (page 120): Q3 (a) — Mass of Tomatoes (balance weights 200 g, 1 kg, 1 kg)

Step 1: The weights on the pan are 200 g, 1 kg and 1 kg.

Step 2: Change 200 g to kilograms: 200 g = 0.2 kg.

Step 3: Add: 1 + 1 + 0.2 = 2.2.

Answer: 2.2 kg

Exercise (page 120): Q3 (b) — Mass of Watermelon (balance weights 500 g, 1 kg, 2 kg)

Step 1: The weights on the pan are 500 g, 1 kg and 2 kg.

Step 2: Change 500 g to kilograms: 500 g = 0.5 kg.

Step 3: Add: 2 + 1 + 0.5 = 3.5.

Answer: 3.5 kg

Exercise (page 120): Q3 (c) — Mass of Coffee beans (balance weights 1 kg, 5 kg)

Step 1: The weights on the pan are 1 kg and 5 kg.

Step 2: Add: 1 + 5 = 6.

Answer: 6 kg

Exercise (page 120): Q3 (d) — Mass of Fish (balance weight 2 kg)

Step 1: The weight on the pan is 2 kg.

Answer: 2 kg

Exercise (page 120): Q3 (e) — Mass of mangoes (digital scale)

Step 1: The digital scale reads 3 kg.

Answer: 3 kg

Exercise (page 120): Q3 (f) — Mass of vegetable (digital scale)

Step 1: The digital scale reads 1 kg.

Answer: 1 kg

Exercise (page 120): NOTE — the mass exercise is numbered '3' although no Question 2 appears between Question 1 and it; the numbering is kept as printed. The small weight in (a) reads 200 g and in (b) reads 500 g at 300 dpi, so the totals use kilograms with a decimal (2.2 kg and 3.5 kg) as the book's 'kg' answer box requires.

Exercise (page 121): Question 3 (continued) — Write the mass in

g for the following (digital scales)

Exercise (page 121): Q3 (h) — Mass of garlics

Missing word: 600 g

Exercise (page 121): Q3 (i) — Mass of green chillies

Missing word: 600 g

Exercise (page 121): Q3 (j) — Mass of dates

Missing word: 400 g

Exercise (page 121): Q3 (k) — Mass of cherries

Missing word: 400 g

Exercise (page 121): Q3 (l) — Mass of ginger

Missing word: 300 g

Exercise (page 121): Q3 (m) — Mass of blackpaper

Missing word: 200 g

Exercise (page 121): NOTE — Question 3 jumps from item (f) on page 120 to item (h) here; item (g) is not printed in the book, so nothing is invented for it. The scale readings were checked at 300 dpi (green chillies read 600 g, not 800 g as a small rendering suggested).

Exercise (page 121): Question 4 — Write the mass in mg for the following

Exercise (page 121): Q4 (a) — Mass of Turmeric

Missing word: 200 mg

Exercise (page 121): Q4 (b) — Mass of Coriander

Missing word: 300 mg

Exercise (page 122): Question 4 (continued) — Write the mass in mg for the following

Exercise (page 122): Q4 (c) — Mass of Numeg

Missing word: 100 mg

Exercise (page 122): Q4 (d) — Mass of Cumin

Missing word: 100 mg

Real-World Word Problems: Mass (pages 122 to 123)

Real-World Word Problems (page 122): (i) Sana has a bag of rice, with a mass of 57 kg. Raheela has a bag of rice with a mass of 38 kg. What is the total mass of both bags in kilograms?

Step 1: Sana's bag = 57 kg.

Step 2: Raheela's bag = 38 kg.

Step 3: Total means add: 57 + 38.

Step 4: 57 + 38 = 95.

Answer: 95 kg

Real-World Word Problems (page 122): (ii) Ahsan bought 455 kg of flour and 365 kg of rice. Find the total mass.

Step 1: Flour = 455 kg.

Step 2: Rice = 365 kg.

Step 3: Total means add: 455 + 365.

Step 4: 455 + 365 = 820.

Answer: 820 kg

Real-World Word Problems (page 122): (iii) A shopkeeper sold 475 kg of flour and 366 kg of sugar. Find the total mass of sugar

and flour.

Step 1: Flour = 475 kg.

Step 2: Sugar = 366 kg.

Step 3: Total means add: 475 + 366.

Step 4: 475 + 366 = 841.

Answer: 841 kg

Real-World Word Problems (page 122): (iv) Saad buys 648 g black pepper and 453 g of cloves. What is the total mass of both?

Step 1: Black pepper = 648 g.

Step 2: Cloves = 453 g.

Step 3: Total means add: 648 + 453.

Step 4: 648 + 453 = 1101.

Answer: 1101 g

Real-World Word Problems (page 122): (v) Ali bought 385 g of biscuits and 756 g of cake. Find the total mass of both things.

Step 1: Biscuits = 385 g.

Step 2: Cake = 756 g.

Step 3: Total means add: 385 + 756.

Step 4: 385 + 756 = 1141.

Answer: 1141 g

Real-World Word Problems (page 122): (vi) Mass of an empty box is 754 g. The mass of the chocolate in the box is 965 g. Find the total mass.

Step 1: Empty box = 754 g.

Step 2: Chocolate = 965 g.

Step 3: Total means add: 754 + 965.

Step 4: 754 + 965 = 1719.

Answer: 1719 g

Real-World Word Problems (page 123): (i) Usman mass is 84 kg and his brother's is Akram mass is 78 kg. What is the difference between their masses?

Step 1: Usman = 84 kg.

Step 2: Akram = 78 kg.

Step 3: Difference means subtract: 84 − 78.

Step 4: 84 − 78 = 6.

Answer: 6 kg

Real-World Word Problems (page 123): (ii) Ali has two fish in the jar. The mass of one fish is 65 g and the other fish is 58 g. What is difference between the masses of the two fish?

Step 1: First fish = 65 g.

Step 2: Second fish = 58 g.

Step 3: Difference means subtract: 65 − 58.

Step 4: 65 − 58 = 7.

Answer: 7 g

Real-World Word Problems (page 123): (iii) An airline allows 55 kg luggage for its passengers. Aman's luggage weighs 88 kg. How much extra luggage does he have?

Step 1: Allowed luggage = 55 kg.

Step 2: Aman's luggage = 88 kg.

Step 3: Extra means subtract: 88 − 55.

Step 4: 88 − 55 = 33.

Answer: 33 kg

Real-World Word Problems (page 123): (iv) Asthma baked an 845 g cake, and Sara baked 656 g cake. What is the difference between the masses of the two cakes?

Step 1: Asthma's cake = 845 g.

Step 2: Sara's cake = 656 g.

Step 3: Difference means subtract: 845 − 656.

Step 4: 845 − 656 = 189.

Answer: 189 g

Real-World Word Problems (page 123): (v) A contractor has 875 kg of cement at a work place. On the first day 667 kg cement is used. How much cement is left?

Step 1: Cement at first = 875 kg.

Step 2: Cement used = 667 kg.

Step 3: Left means subtract: 875 − 667.

Step 4: 875 − 667 = 208.

Answer: 208 kg

Real-World Word Problems (page 123): (vi) Sadaf bought 975 g of dry fruits. She used 585 g to make a sweet dish. How much dry fruit was left?

Step 1: Dry fruits bought = 975 g.

Step 2: Dry fruits used = 585 g.

Step 3: Left means subtract: 975 − 585.

Step 4: 975 − 585 = 390.

Answer: 390 g

Standard Units of Capacity (pages 124 to 125)

Standard Units of Capacity (page 124): We measure the capacity of different containers using standard units of capacity, such as litre, millilitre. Remember: The amount of liquid a container can hold is called capacity. Litre (page 124): The most appropriate unit to measure the capacity of big containers is the litre. It is denoted as 'l'. We use special types of containers to measure the quantity of liquid. Few examples of quantities of liquid hold in litres (page 124): Capacity of milk bottle = 1 l. Capacity of oil bottle = 3 l. Capacity of oil bottle = 5 l. Millilitre (page 124): We measure the capacity of small containers in millilitre and it is denoted as 'ml'. Key Fact (page 125): 1000 ml = 1 l. We use special types of containers to measure the quantity of liquid (page 125): spoon 50 ml, glass 300 ml, jug 500 ml, jug 200 ml, measure 100 ml. A few examples of liquid quantities measured in millilitres (page 125): Cooking oil 200 ml. Cold Drink 300 ml. Cough Syrup 100 ml.

Exercise (page 125): Question 1 — Write the suitable units (l or ml) for measuring the capacity of the following

Exercise (page 125): Q1 — Water in glass

Missing word: ml

Exercise (page 125): Q1 — Juice packet

Missing word: ml

Exercise (page 125): Q1 — Syringe

Missing word: ml

Exercise (page 125): Q1 — Milk packet

Missing word: l

Exercise (page 125): Q1 — Water in bowl

Missing word: ml

Exercise (page 125): Q1 — Medicine bottle

Missing word: ml

Exercise (page 126): Question 2 — Observe the following and write the capacity in litter

Exercise (page 126): Q2 — yellow bottle (labelled '1 litre')

Missing word: 1 litre

Exercise (page 126): Q2 — milk packet

Missing word: 1 litre

Exercise (page 126): Q2 — cola bottle

Missing word: 2 litres

Exercise (page 126): Q2 — ice cream tub (labelled '2 litre')

Missing word: 2 litres

Exercise (page 126): Q2 — water bottle (labelled '3 litre')

Missing word: 3 litres

Exercise (page 126): Q2 — oil bottle (labelled '3 litre')

Missing word: 3 litres

Exercise (page 126): Q2 — juice bottle (labelled '2 litre')

Missing word: 2 litres

Exercise (page 126): Q2 — oil bottle (labelled '2 litre')

Missing word: 2 litres

Exercise (page 126): NOTE — the milk packet's badge and the cola bottle carry no capacity that can be read even at 300 dpi, so those two are

judgement readings: the milk packet is the book's standard 1-litre packet (the same packet is used on page 125, and the book's milk bottle on page 124 is 1 litre), and the cola bottle is the large family-size bottle, read as 2 litres by size comparison with the labelled bottles. The other six are read exactly from their printed labels.

Exercise (page 126): Question 3 — Observe and write the capacity in millilitre

Exercise (page 126): Q3 — glass of water (labelled '300 ml')

Missing word: 300 ml

Exercise (page 126): Q3 — mug (labelled '250 ml')

Missing word: 250 ml

Exercise (page 126): Q3 — medicine bottle (labelled '50 ml')

Missing word: 50 ml

Exercise (page 126): Q3 — bowl (labelled '100 ml')

Missing word: 100 ml

Exercise (page 126): Question 4 — Write the capacity of water in each container

Exercise (page 126): Q4 — first beaker (liquid surface at the 800 ml mark)

Missing word: 800 ml

Exercise (page 126): Q4 — second beaker (liquid surface at the 200 ml mark)

Missing word: 200 ml

Exercise (page 126): Q4 — third beaker (liquid surface at the 500 ml mark)

Missing word: 500 ml

Real-World Word Problems: Capacity (pages 127 to 128)

Real-World Word Problems (page 127): (i) A jug contains 485 ml of milk. Irum added 347 ml milk to it. Find how much milk is in the jug now?

Step 1: Milk in the jug = 485 ml.

Step 2: Milk added = 347 ml.

Step 3: Now means add: 485 + 347.

Step 4: 485 + 347 = 832.

Answer: 832 ml

Real-World Word Problems (page 127): (ii) Maryam uses 458 ml of cooking oil to cook meat curry and 375 ml of cooking oil to cook rice. How much cooking oil did she use to cook both food items?

Step 1: Oil for meat curry = 458 ml.

Step 2: Oil for rice = 375 ml.

Step 3: Total means add: 458 + 375.

Step 4: 458 + 375 = 833.

Answer: 833 ml

Real-World Word Problems (page 127): (iii) Sara has 675 ml of water in a bottle. She added 665 ml of more water to the bottle. How much water is in the bottle altogether?

Step 1: Water in the bottle = 675 ml.

Step 2: Water added = 665 ml.

Step 3: Altogether means add: 675 + 665.

Step 4: 675 + 665 = 1340.

Answer: 1340 ml

Real-World Word Problems (page 127): (iv) Hamza sold 468 l vegetable oil on Sunday and 389 l on Monday. How much vegetable oil did he sell in two days?

Step 1: Oil sold on Sunday = 468 l.

Step 2: Oil sold on Monday = 389 l.

Step 3: Total means add: 468 + 389.

Step 4: 468 + 389 = 857.

Answer: 857 l

Real-World Word Problems (page 127): (v) A water tank holds 558 l water and another water tank holds 385 l water. Find the total capacity of water in both tanks.

Step 1: First tank = 558 l.

Step 2: Second tank = 385 l.

Step 3: Total means add: 558 + 385.

Step 4: 558 + 385 = 943.

Answer: 943 l

Real-World Word Problems (page 127): (vi) In a restaurant 555 l of milk was used to make tea and 497 l of milk was used to make sweet dishes. Find the total quantity of milk used.

Step 1: Milk for tea = 555 l.

Step 2: Milk for sweet dishes = 497 l.

Step 3: Total means add: 555 + 497.

Step 4: 555 + 497 = 1052.

Answer: 1052 l

Real-World Word Problems (page 128): (i) One juice pack has 356 ml juice, and another juice pack has 475 ml. What is the difference between the capacities of the two juice packs?

Step 1: First juice pack = 356 ml.

Step 2: Second juice pack = 475 ml.

Step 3: Difference means subtract: 475 − 356.

Step 4: 475 − 356 = 119.

Answer: 119 ml

Real-World Word Problems (page 128): (ii) Akhtar has a water bottle that can hold 850 ml of water. He fills it with 677 ml of water. How much more water is needed to fill the bottle?

Step 1: Bottle holds = 850 ml.

Step 2: Water filled = 677 ml.

Step 3: More needed means subtract: 850 − 677.

Step 4: 850 − 677 = 173.

Answer: 173 ml

Real-World Word Problems (page 128): (iii) Samina prepared 677 ml of tea for her guests. At the end 438 ml of tea was left. How much tea did the guests drink?

Step 1: Tea prepared = 677 ml.

Step 2: Tea left = 438 ml.

Step 3: Drunk means subtract: 677 − 438.

Step 4: 677 − 438 = 239.

Answer: 239 ml

Real-World Word Problems (page 128): (iv) Ahmed bought 455 l of paint. He used 378 l paint for his house. How much paint was left?

Step 1: Paint bought = 455 l.

Step 2: Paint used = 378 l.

Step 3: Left means subtract: 455 − 378.

Step 4: 455 − 378 = 77.

Answer: 77 l

Real-World Word Problems (page 128): (v) Sara prepared 655 l squash for party. At the end 459 l squash was left. How much squash did the guests drink?

Step 1: Squash prepared = 655 l.

Step 2: Squash left = 459 l.

Step 3: Drunk means subtract: 655 − 459.

Step 4: 655 − 459 = 196.

Answer: 196 l

Real-World Word Problems (page 128): (vi) A tank holds 565 l of water. 378 l of water is pumped out from it. How much of water is left in the tank now?

Step 1: Water in the tank = 565 l.

Step 2: Water pumped out = 378 l.

Step 3: Left means subtract: 565 − 378.

Step 4: 565 − 378 = 187.

Answer: 187 l

Temperature (pages 129 to 130)

Temperature (page 129): The temperature of something tells us how hot or cold it is. It is the degree of hotness or coldness of an object. A thermometer is used to measure temperature. We measure temperature in degrees celsius (C) or fahrenheit (F). Example (page 129): Write: 40 degrees C. Say / read: 40 degrees celsius. The thermometer rises for hot and falls for cold. Example — observe, read and write the temperature (page 129, already given in the book): Morning 15 degrees C. Afternoon 30 degrees C. Evening 10 degrees C. Night 5 degrees C.

Exercise (page 130): Question 1 — Read and write temperature

Exercise (page 130): Q1 (ii) — ice cube

Missing word: 0 degrees C

Exercise (page 130): Q1 (iii) — children playing in autumn leaves

Missing word: 5 degrees C

Exercise (page 130): Q1 (iv) — summer tree

Missing word: 40 degrees C

Exercise (page 130): Q1 (v) — bare tree in sunshine

Missing word: 20 degrees C

Exercise (page 130): Q1 (vi) — refrigerator

Missing word: 5 degrees C

Exercise (page 130): Q1 (vii) — camp fire

Missing word: 95 degrees C

Exercise (page 130): Q1 (viii) — children playing in the park

Missing word: 30 degrees C

Exercise (page 130): Q1 (i) is already completed in the book: cooking, 20 degrees C. NOTE — the thermometer columns were measured on the 300 dpi page against each thermometer's own 0-100 scale (3.6 pixels per degree, calibrated by the book's own 20 degrees C for item (i)); the measured columns sit about 2 degrees below the round values, so each reading is given to the nearest 5 degrees: (ii) 0, (iii) 5, (iv) 40, (v) 20, (vi) 5, (vii) 95, (viii) 30. Unit 4 continues to page 134; pages 131 to 134 (the rest of the temperature work and the Unit 4 summary, if any) follow next.

Exercise (page 131): Question 2 — Write the temperature in the blanks and show them on number line

Exercise (page 131): Q2 (a) — horizontal thermometer (already given)

Missing word: 50 degrees C, with a dot marked at 50 on the number line.

Exercise (page 131): Q2 (b) — horizontal thermometer

Missing word: 70 degrees C, with a dot marked at 70 on the number line.

Exercise (page 131): Q2 (c) — horizontal thermometer

Missing word: 30 degrees C, with a dot marked at 30 on the number line.

Exercise (page 131): Q2 (d) — horizontal thermometer

Missing word: 15 degrees C, with a dot marked at 15 on the number line.

Exercise (page 131): NOTE — the horizontal bars were measured on the 300 dpi page: the scale runs from the 0 mark to the 100 mark, and the thick red bar ends at 50 for (a, given), 70 for (b), 30 for (c) and 15 for (d). The thin red thread that continues past the thick bar is not the reading.

Comparing Temperature (page 132)

Comparing Temperature — Example (page 132, already given in the book): Read, write and compare the temperature by <, > or = signs. Kite-flying day 30 degrees C < swimming day 40 degrees C. NOTE — the example's right thermometer is drawn with its column near 47 degrees, although the book labels it 40 degrees C; the comparison 30 < 40 is kept exactly as printed.

Exercise (page 132): Question 1 — Read, write and compare the temperature by using <, > or = signs

Exercise (page 132): Q1 row 1 — autumn day and snowman day

Missing word: Autumn day 20 degrees C > snowman day 0 degrees C.

Exercise (page 132): Q1 row 2 — skiing day and spring day

Missing word: Skiing day 0 degrees C < spring day 40 degrees C.

Exercise (page 132): Q1 row 3 — snowy house night and snowy mountain day

Missing word: Snowy house night 10 degrees C = snowy mountain day 10 degrees C.

Exercise (page 132): NOTE — the six exercise columns were measured on the 300 dpi page against each thermometer's own 0-100 scale (3.09 pixels per degree), then corrected by the same about-2-degree offset found on page 130 (where the book's own 20 degrees C for item (i) measured 18.1) and rounded to the nearest 5 degrees: autumn 20, snowman 0, skiing 0, spring 40, snowy house 10, snowy mountain 10. The two row-3 columns are the same height (within 1 pixel), so their comparison is '='.

Exercise (page 133): Question 2 — Compare and write the suitable signs <, > or =

Exercise (page 133): Q2 — 30 degrees C and 50 degrees C

Missing word: 30 degrees C < 50 degrees C

Exercise (page 133): Q2 — 40 degrees C and 10 degrees C

Missing word: 40 degrees C > 10 degrees C

Exercise (page 133): Q2 — 20 degrees C and 20 degrees C

Missing word: 20 degrees C = 20 degrees C

Exercise (page 133): Question 3 — Tick the greatest and encircle the least temperature in each part

Exercise (page 133): Q3 (i) — 40 degrees C, 60 degrees C, 30 degrees C

Missing word: Greatest: 60 degrees C (tick). Least: 30 degrees C (encircled).

Exercise (page 133): Q3 (ii) — 70 degrees C, 50 degrees C, 80 degrees C

Missing word: Greatest: 80 degrees C (tick). Least: 50 degrees C (encircled).

Exercise (page 133): Q3 (iii) — 90 degrees C, 20 degrees C, 60 degrees C

Missing word: Greatest: 90 degrees C (tick). Least: 20 degrees C (encircled).

Ordering Temperature (page 133)

Ordering Temperature — Example (page 133, already given in the book): Write the temperature in decreasing order: 10 degrees C, 15 degrees C, 50 degrees C, 30 degrees C, 40 degrees C, 25 degrees C. Solution: Temperature in decreasing order: 50 degrees C > 40 degrees C > 30 degrees C > 25 degrees C > 15 degrees C > 10 degrees C. Example (page 133, already given in the book): Write the temperature in increasing order: 70 degrees C, 60 degrees C, 90 degrees C, 75 degrees C, 40 degrees C, 50 degrees C. Solution: Temperature in increasing order:

40 degrees C < 50 degrees C < 60 degrees C < 70 degrees C < 75 degrees C < 90 degrees C.

Exercise (page 133): Question 1 — Write the temperature in decreasing order

Exercise (page 133): Q1 (i) — 80 degrees C, 50 degrees C, 10 degrees C, 40 degrees C, 20 degrees C, 30 degrees C

Missing word: 80 degrees C > 50 degrees C > 40 degrees C > 30 degrees C > 20 degrees C > 10 degrees C

Exercise (page 134): Question 1 (continued) — Write the temperature in decreasing order

Exercise (page 134): Q1 (ii) — 50 degrees C, 20 degrees C, 90 degrees C, 70 degrees C, 10 degrees C, 40 degrees C

Missing word: 90 degrees C > 70 degrees C > 50 degrees C > 40 degrees C > 20 degrees C > 10 degrees C

Exercise (page 134): Q1 (iii) — 90 degrees C, 40 degrees C, 70 degrees C, 50 degrees C, 60 degrees C, 30 degrees C

Missing word: 90 degrees C > 70 degrees C > 60 degrees C > 50 degrees C > 40 degrees C > 30 degrees C

Exercise (page 134): Question 2 — Write the temperature in increasing order

Exercise (page 134): Q2 (i) — 70 degrees C, 30 degrees C, 60 degrees C, 20 degrees C, 90 degrees C, 40 degrees C

Missing word: 20 degrees C < 30 degrees C < 40 degrees C < 60 degrees C < 70 degrees C < 90 degrees C

Exercise (page 134): Q2 (ii) — 80 degrees C, 30 degrees C, 60 degrees C, 0 degrees C, 40 degrees C, 50 degrees C

Missing word: 0 degrees C < 30 degrees C < 40 degrees C < 50 degrees C < 60 degrees C < 80 degrees C

Exercise (page 134): Q2 (iii) — 100 degrees C, 70 degrees C, 50 degrees C, 90 degrees C, 10 degrees C, 30 degrees C

Missing word: 10 degrees C < 30 degrees C < 50 degrees C < 70 degrees C < 90 degrees C < 100 degrees C

Unit 4 Summary (page 134)

Summary (page 134): There are 100 centimetres in 1 metre and 10 millimetres in 1 centimetre. There are 1000 metres in 1 kilometre. There are 1000 grams in 1 kilogram and 1000 milligrams in 1 gram. Milligrams is used to weigh the mass of extremely light objects. The amount of liquid that a container can hold is called capacity. Temperature is the degree of hotness and coldness. Thermometer is used to measure temperature. Vocabulary: Length, millimetre, metre, kilometre, mass, gram, capacity, litre, temperature degrees. This closes Unit 4 (pages 108 to 134).

Unit 5 — Time (page 135)

Unit 5 Time — Students' Learning Outcomes (page 135): Read and write time in hours and minutes from analogue and digital clocks. Recognise and use a.m. and p.m. Add and subtract measures of time given in the same units to solve real-life word problems. Read and write days and dates from the Solar Calendar. The opener page shows four analogue clocks as pictures (no exercise).

Analogue and Digital clock (page 136)

Analogue and Digital clock (page 136): An analogue clock looks like this. The hour hand is between 7 and 8. The minute hand is at 20. The time is seven twenty. We write: 07:20. We read: 20 minutes past 7. A digital clock looks like this. Digital clock show the time using numbers only. In a digital clock, the time can be read as accurate as to 1 minute. The time is three fifty. We write: 03:50. We read: 50 minutes past 3. Key Fact: 60 minutes = 1 hour.

Exercise (page 136): Question 1 — In the analogue clock read and write time in hours and minutes

Exercise (page 136): Q1 clock 1 (already given)

Missing word: 3:10 — 10 minutes past 3.

Exercise (page 136): Q1 clock 2

Missing word: 5:15 — 15 minutes past 5.

Exercise (page 136): Q1 clock 3

Missing word: 11:35 — 35 minutes past 11.

Exercise (page 136): Q1 clock 4

Missing word: 5:05 — 5 minutes past 5.

Exercise (page 136): NOTE — the clock hands were measured on the 300 dpi page (purple hand = hour, teal hand = minute). The artist draws the hour hand near the hour numeral rather than at its exact proportional position, so each clock was read by matching the minute hand exactly and taking the hour whose true position is nearest the drawn hour hand.

Exercise (page 137): Question 1 (continued) — In the analogue clock read and write time in hours and minutes

Exercise (page 137): Q1 clock 5

Missing word: 1:30 — 30 minutes past 1.

Exercise (page 137): Q1 clock 6

Missing word: 7:45 — 45 minutes past 7.

Exercise (page 137): Q1 clock 7

Missing word: 8:55 — 55 minutes past 8.

Exercise (page 137): Q1 clock 8

Missing word: 6:15 — 15 minutes past 6.

Exercise (page 137): Question 2 — In the digital clock read and write the time

Exercise (page 137): Q2 clock 1 (already given)

Missing word: 05:20 — 20 minutes past 5.

Exercise (page 137): Q2 clock 2

Missing word: 10:25 — 25 minutes past 10.

Exercise (page 137): Q2 clock 3

Missing word: 12:35 — 35 minutes past 12.

Exercise (page 137): Q2 clock 4

Missing word: 03:15 — 15 minutes past 3.

Exercise (page 137): Q2 clock 5

Missing word: 07:10 — 10 minutes past 7.

Exercise (page 137): Q2 clock 6

Missing word: 11:05 — 5 minutes past 11.

Exercise (page 138): Question 3 — Match analogue clock to the digital clock

Exercise (page 138): Q3 analogue clock 1 (hour near 5, minute at 2)

Missing word: 5:10 — matches digital 5:10.

Exercise (page 138): Q3 analogue clock 2 (hour near 4, minute at 3)

Missing word: 4:15 — matches digital 4:15.

Exercise (page 138): Q3 analogue clock 3 (hour between 1 and 2, minute at 9)

Missing word: 1:45 — matches digital 1:45.

Exercise (page 138): Q3 analogue clock 4 (hour between 10 and 11, minute at 10)

Missing word: 10:50 — NO matching digital is printed: the digital set offers 11:40, which does not match the drawn hands (minute 50, hour near 11). FLAGGED: the book's digital set does not contain 10:50, so this clock is left unmatched; ask the teacher.

Exercise (page 138): Q3 analogue clock 5 (hour between 5 and 6, minute at 11)

Missing word: 5:55 — NO matching digital is printed: the digital set offers 6:55, which does not match the drawn hands (minute 55, hour in the 5th hour). FLAGGED: the book's digital set does not contain 5:55, so this clock is left unmatched; ask the teacher.

Exercise (page 138): Q3 analogue clock 6 (hour at 8, minute at 6)

Missing word: 8:30 — matches digital 8:30.

Exercise (page 138): Q3 analogue clock 7 (hour between 2 and 3, minute at 7)

Missing word: 2:35 — matches digital 2:35.

Exercise (page 138): Q3 analogue clock 8 (hour between 4 and 5, minute at 10)

Missing word: 4:50 — matches digital 4:50.

Exercise (page 138): Question 4 — Encircle the correct time

Exercise (page 138): Q4 clock 1 (hour at 10, minute at 4; options 11:10, 10:20, 9:50)

Missing word: 10:20 (encircled).

Exercise (page 138): Q4 clock 2 (hour at 9, minute at 4; options 5:00, 4:50, 9:20)

Missing word: 9:20 (encircled).

Exercise (page 138): Q4 clock 3 (hour at 12, minute at 2; options 10:15, 7:30, 12:10)

Missing word: 12:10 (encircled).

Exercise (page 138): Q4 clock 4 (hour between 3 and 4, minute at 6; options 8:15, 4:30, 6:45)

Missing word: 4:30 (encircled). NOTE — the hour hand is drawn just short of the 4:30 position, but 4:30 is the only option that fits an hour near 4 with 30 minutes.

Exercise (page 138): Q4 clock 5 (hour between 2 and 3, minute at 10; options 12:35, 2:50, 8:15)

Missing word: 2:50 (encircled).

Exercise (page 138): Q4 clock 6 (hour at 9, minute at 1; options 9:05, 10:00, 7:35)

Missing word: 9:05 (encircled).

Recognizing and using a.m. and p.m. (page 139)

Recognizing and using a.m. and p.m. (page 139): As we know there are 24 hours in a day. We divide a day into two equal parts: The first 12-hours period of the day is known as the before noon. The next 12 hours period of the day is known as the afternoon. The time, from 12 o'clock midnight to 12 o'clock noon is called a.m. The time from 12

o'clock noon to 12 o'clock midnight is called p.m. Example: Nasir has breakfast at 7 o'clock in the morning. We can also write the time as 7:00 am. Example: Nasir goes to sleep at 9 o'clock at night. We can also write the time as 9:00 pm. Key Note: Do not write a.m. and p.m. with 12:00 midnight and 12:00 noon. 1 day = 24 hour.

Using a.m. and p.m. (page 140, already given in the book): We use a.m. and p.m to tell time from 12 hour clock. The time is five o'clock in the morning — 5:00 a.m. The time is eight o'clock in the morning — 8:00 a.m. The time is twelve fifteen in the afternoon — 12:15 p.m. The time is five o'clock in the afternoon — 5:00 p.m. The time is seven thirty in the afternoon — 7:30 p.m. NOTE — the book's words for the last row read 'nine thirty', but its clock shows 7:30 and its time box reads 7:30 p.m.; 7:30 p.m. is used and the 'nine thirty' wording is a book slip.

Exercise (page 141): Question 1 — Write the time in a.m and p.m in the boxes

Exercise (page 141): Q1 — six o'clock in the morning

Missing word: 6:00 a.m.

Exercise (page 141): Q1 — eight o'clock at night

Missing word: 8:00 p.m.

Exercise (page 141): Q1 — five thirty in the evening

Missing word: 5:30 p.m.

Exercise (page 141): Q1 — nine forty in the night

Missing word: 9:40 p.m.

Exercise (page 141): Question 2 — Fill in the boxes with a.m or p.m

Exercise (page 141): Q2 — sunrise picture, clock 6:00

Missing word: It is 6:00 a.m.

Exercise (page 141): Q2 — homework picture, clock 4:15

Missing word: It is 4:15 p.m.

Exercise (page 141): Q2 — playing in the park picture, clock 10:30

Missing word: It is 10:30 a.m.

Exercise (page 141): Q2 — moonlit night picture, clock 9:00

Missing word: It is 9:00 p.m.

Exercise (page 142): Question 3 — Write the time in a.m. and p.m.

Exercise (page 142): Q3 (i) — Before noon (already given)

Missing word: 10:00 a.m.

Exercise (page 142): Q3 (ii) — After noon, clock 3:00

Missing word: 3:00 p.m.

Exercise (page 142): Q3 (iii) — Morning, clock 8:00

Missing word: 8:00 a.m.

Exercise (page 142): Q3 (iv) — Evening, clock 5:00

Missing word: 5:00 p.m.

Exercise (page 142): Q3 (v) — Mid night, clock 12:00

Missing word: 12:00 midnight (no a.m. or p.m. is written with 12:00 midnight, as per the Key Note on page 139).

Exercise (page 142): Q3 (vi) — Before mid night, clock 11:00

Missing word: 11:00 p.m.

Exercise (page 142): Question 4 — Read and write the time shown in clocks

Exercise (page 142): Q4 (i) — We get up at (already given)

Missing word: 6:00 a.m.

Exercise (page 142): Q4 (ii) — We go to school at, clock 7:30

Missing word: 7:30 a.m.

Exercise (page 142): Q4 (iii) — We play games at, clock 5:00

Missing word: 5:00 p.m.

Exercise (page 142): Q4 (iv) — We recite Holy Quran at, clock 6:00

Missing word: 6:00 p.m. NOTE — the book gives no picture clue for this row; p.m. is used because the row follows the 5:00 p.m. games row in the day's order (morning 6:00 is already used by the 'get up' row).

Addition of Units of Time in Hours (page 143)

Addition of Units of Time in Hours — Examples (page 143, already given in the book): (i) Add 5 hours and 7 hours = 12 hours. (ii) Add 18 hours and 16 hours = 34 hours.

Exercise (page 143): Question 1 — Solve the following

Exercise (page 143): Q1 (1) — 25 hours + 18 hours

Step 1: 25 hours + 18 hours

Step 2: 5 + 8 = 13, write 3, carry 1

Step 3: 2 + 1 + 1 = 4

Answer: 43 hours

Exercise (page 143): Q1 (2) — 33 hours + 26 hours

Step 1: 33 hours + 26 hours

Step 2: 3 + 6 = 9

Step 3: 3 + 2 = 5

Answer: 59 hours

Exercise (page 143): Q1 (3) — 45 hours + 38 hours

Step 1: 45 hours + 38 hours

Step 2: 5 + 8 = 13, write 3, carry 1

Step 3: 4 + 3 + 1 = 8

Answer: 83 hours

Exercise (page 143): Q1 (4) — 78 hours + 63 hours

Step 1: 78 hours + 63 hours

Step 2: 8 + 3 = 11, write 1, carry 1

Step 3: 7 + 6 + 1 = 14

Answer: 141 hours

Exercise (page 143): Q1 (5) — Add 64 hours and 38 hours

Step 1: 64 hours + 38 hours

Step 2: 4 + 8 = 12, write 2, carry 1

Step 3: 6 + 3 + 1 = 10

Answer: 102 hours

Exercise (page 143): Q1 (6) — Add 78 hours and 54 hours

Step 1: 78 hours + 54 hours

Step 2: 8 + 4 = 12, write 2, carry 1

Step 3: 7 + 5 + 1 = 13

Answer: 132 hours

Real Life Word Problems — Addition of Time (page 144)

Real Life Word Problems — Example (page 144, already given in the

book): Rahim traveled for 16 hours by train and 7 hours by bus to reach his grandfather's village. Find the total time he spent in traveling.

Solution: Travelled by train = 16 hours, Travelled by bus = + 7 hours, Total time spent = 23 hours. Hence, Rahim spent 23 hours in traveling.

Exercise (page 144): Real Life Word Problems

Exercise (page 144): Q1 — Umer drove 14 hours from Karachi to Quetta and then he drove 13 hours more to reach Rahim Yar Khan. Find total time he spent in traveling

Step 1: Time Karachi to Quetta = 14 hours

Step 2: More time to Rahim Yar Khan = 13 hours

Step 3: Total = 14 + 13 = 27 hours

Answer: Umer spent 27 hours in traveling.

Exercise (page 144): Q2 — Awais read a history book in 46 hours and another story book in 17 hours. How much time did he spend reading the books altogether?

Step 1: History book = 46 hours

Step 2: Story book = 17 hours

Step 3: Total = 46 + 17 = 63 hours

Answer: Awais spent 63 hours reading the books.

Exercise (page 144): Q3 — In March, Aslam played hockey for 32 hours and football for 16 hours. How much total time he spent playing in the month of March?

Step 1: Hockey = 32 hours

Step 2: Football = 16 hours

Step 3: Total = 32 + 16 = 48 hours

Answer: Aslam spent 48 hours playing in March.

Exercise (page 144): Q4 — A plumber worked 28 hours the during the first three days of the week and worked 32 hours the second three days of the week. How many hours did he work in a week?

Step 1: First three days = 28 hours

Step 2: Second three days = 32 hours

Step 3: Total = 28 + 32 = 60 hours

Answer: The plumber worked 60 hours in a week.

Subtraction of Units of Time in Hours (page 145)

Subtraction of Units of Time in Hours — Examples (page 145, already given in the book): (i) Subtract 5 hours from 14 hours = 9 hours. (ii) Subtract 27 hours from 53 hours = 26 hours.

Exercise (page 145): Question 1 — Solve the following

Exercise (page 145): Q1 (1) — 27 hours − 12 hours

Step 1: 27 hours − 12 hours

Step 2: 7 − 2 = 5

Step 3: 2 − 1 = 1

Answer: 15 hours

Exercise (page 145): Q1 (2) — 65 hours − 23 hours

Step 1: 65 hours − 23 hours

Step 2: 5 − 3 = 2

Step 3: 6 − 2 = 4

Answer: 42 hours

Exercise (page 145): Q1 (3) — 82 hours − 63 hours

Step 1: 82 hours − 63 hours

Step 2: 2 − 3 cannot be done, borrow 1 ten: 12 − 3 = 9

Step 3: 7 − 6 = 1

Answer: 19 hours

Exercise (page 145): Q1 (4) — 170 hours − 139 hours

Step 1: 170 hours − 139 hours

Step 2: 0 − 9 cannot be done, borrow 1 ten: 10 − 9 = 1

Step 3: 6 − 3 = 3

Step 4: 1 − 1 = 0

Answer: 31 hours

Exercise (page 145): Q1 (5) — Subtract 18 hours from 42 hours

Step 1: 42 hours − 18 hours

Step 2: 2 − 8 cannot be done, borrow 1 ten: 12 − 8 = 4

Step 3: 3 − 1 = 2

Answer: 24 hours

Exercise (page 145): Q1 (6) — Subtract 76 hours from 93 hours

Step 1: 93 hours − 76 hours

Step 2: 3 − 6 cannot be done, borrow 1 ten: 13 − 6 = 7

Step 3: 8 − 7 = 1

Answer: 17 hours

Real Life Word Problems — Subtraction of Time

(page 146)

Real Life Word Problems — Example (page 146, already given in the

book): Aslam travelled for 22 hours and Nasir travelled for 15 hours by train. How many more hours did Aslam travel? Solution: Aslam travelled = 22 hours, Nasir travelled = − 15 hours, Difference = 7 hours. Hence, Aslam travelled 7 more hours than Nasir.

Exercise (page 146): Real Life Word Problems

Exercise (page 146): Q1 — Student A completed his homework in 48 minutes, while student B did completed the same homework in 39 minutes. How much more time did student A take?

Step 1: Student A = 48 minutes

Step 2: Student B = 39 minutes

Step 3: Difference = 48 − 39 = 9 minutes

Answer: Student A took 9 minutes more.

Exercise (page 146): Q2 — A carpenter makes a furniture set in 70 hours, while another carpenter makes a similar furniture set in 54 hours. How much more time does the first carpenter take?

Step 1: First carpenter = 70 hours

Step 2: Second carpenter = 54 hours

Step 3: Difference = 70 − 54 = 16 hours

Answer: The first carpenter takes 16 hours more.

Exercise (page 146): Q3 — Ahsan worked in a factory for 315 hours in one month while Arshad worked for 248 hours. How many more hours did Ahsan work in the same moth than Arshad?

Step 1: Ahsan = 315 hours

Step 2: Arshad = 248 hours

Step 3: Difference = 315 − 248 = 67 hours

Answer: Ahsan worked 67 hours more than Arshad.

Exercise (page 146): Q4 — A rice factory ran for 465 hours in July. It also ran 388 hours in August. How many more hours did the factory run in July than in August?

Step 1: July = 465 hours

Step 2: August = 388 hours

Step 3: Difference = 465 − 388 = 77 hours

Answer: The factory ran 77 hours more in July than in August.

Days and Dates from Solar Calendar (page 147)

Days and Dates from Solar Calendar (page 147): We know that a year has 12 months. There are 365 days in the solar year. There are 366 days in a leap year. A year that is exactly divisible by 4 is called a leap year. For example, 2020, 2024 are leap years. A calendar shows the days of the week and the months in a year. The numbers on the calendar represent the dates. The book prints the full 2026 calendar (January to December) on this page.

Days and Dates (page 148)

Days and Dates (page 148): Dates from a calendar can be written as: Day/Month/Year, Day/Month/Year, Day-Month-Year. Example (already given in the book): Write the date of Pakistan day. We can write: 23rd March 1940, 23/03/1940, 23 - 03 - 1940. Teacher Point: Teacher should hang the current calendar in the classroom and use during teaching learning process. Teacher may use calendar time to time during educational activities and events.

Exercise (page 148): Question 1 — Now look at the calendar and

answer the questions

Exercise (page 148): Q1 (i) — On which day does the January start?

Missing word: Thursday. NOTE — the book has 'Wednesday' pre-filled in this box, but the book's own 2026 calendar (page 147) places 1 January under Thursday (first row: 1, 2, 3 under Thursday, Friday, Saturday); Thursday is the calendar-correct answer and the pre-filled 'Wednesday' is a book slip.

Exercise (page 148): Q1 (ii) — How many days are in February 2026? (already given)

Missing word: 28 days.

Exercise (page 148): Q1 (iii) — How many Sunday's are in March?

Missing word: 5 Sundays (1, 8, 15, 22 and 29 March 2026 fall on Sunday in the book's calendar).

Exercise (page 148): Q1 (iv) — On which day does the June end?

Missing word: Tuesday (30 June 2026 falls under Tuesday in the book's calendar: last row 28, 29, 30 under Sunday, Monday, Tuesday).

Exercise (page 148): Q1 (v) — How many days are in July?

Missing word: 31 days.

Exercise (page 148): Q1 (vi) — How many days are in December?

Missing word: 31 days.

Exercise (page 149): Question 2 — Encircle the following important days in calendar and write dates in the blanks

Exercise (page 149): Q2 (i) — What is the date on which we celebrate Independence Day? (already given)

Missing word: 14 August 2026.

Exercise (page 149): Q2 (ii) — On which date do we celebrate Pakistan

Day?

Missing word: 23 March 2026.

Exercise (page 149): Q2 (iii) — On which date do we celebrate Quaid-e-Azam Day?

Missing word: 25 December 2026.

Exercise (page 149): Q2 (iv) — What is the date on which we celebrate Iqbal Day?

Missing word: 9 November 2026.

Exercise (page 149): Q2 (v) — What is the date of Labour day?

Missing word: 1 May 2026.

Exercise (page 149): Q2 (vi) — What is your birthday date?

Missing word: LEFT BLANK — this is the pupil's own birthday date, which the book does not give and which cannot be filled from the book.

Exercise (page 149): Question 3 — Write the dates in short form given below

Exercise (page 149): Q3 (i) — 1st January, 2026 (already given)

Missing word: 01-01-2026.

Exercise (page 149): Q3 (ii) — 3rd March, 2026

Missing word: 03-03-2026.

Exercise (page 149): Q3 (iii) — 25th September, 2026

Missing word: 25-09-2026.

Exercise (page 149): Q3 (iv) — 06th December, 2026

Missing word: 06-12-2026.

Exercise (page 149): Q3 (v) — 30th July, 2026

Missing word: 30-07-2026.

Exercise (page 149): Q3 (vi) — 14th August, 2026

Missing word: 14-08-2026.

Exercise (page 150): Question 4 — Write the following dates and the days from the calendar?

Exercise (page 150): Q4 (i) — 24-01-2026 (already given in the book, but corrected)

Missing word: 24 January, 2026 — Saturday. NOTE — the book has 'Friday' pre-filled in this box, but the book's own 2026 calendar (page 147) places 24 January under Saturday (January row: 18, 19, 20, 21, 22, 23, 24 under Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday). The pre-filled 'Friday' matches the 2025 calendar, not the printed 2026 calendar, so Saturday is used.

Exercise (page 150): Q4 (ii) — 23-03-2026

Missing word: 23 March, 2026 — Monday.

Exercise (page 150): Q4 (iii) — 14-08-2026

Missing word: 14 August, 2026 — Friday.

Exercise (page 150): Q4 (iv) — 06-09-2026

Missing word: 6 September, 2026 — Sunday.

Exercise (page 150): Q4 (v) — 31-12-2026

Missing word: 31 December, 2026 — Thursday.

Unit 5 Summary (page 150)

Summary (page 150): An analog clock has a clock face with a dial of number from 1 to 12. On a digital clock, the first two digits show us the hours, the last two digits show the minutes. The time from 12 o'clock noon to 12 o'clock midnight is called p.m. The time from 12 o'clock midnight to 12 o'clock noon is called a.m. There are 365 days in a Solar

year. A calendar shows the days of the week and the months of the year. Vocabulary: Analog clock, digital clock, hour hand, minute hand, noon, mid night, before noon, after noon. This closes Unit 5 (pages 135 to 150).

Unit 6 — Geometry (page 151)

Unit 6 Geometry — Students' Learning Outcomes (page 151): Recognise and identify the units of measurement of area and perimeter and find the perimeter and area of 2-D figures (Squares and rectangles) and irregular figures (figures covers either 1/2 or 1 square unit only) on a square grid. Differentiate and classify polygons with respect to their attributes (pentagon, hexagon, octagon and decagon). Identify and differentiate between prisms and pyramids with respect to their attributes. Describe the movement of objects (i.e., slide and rotation). Recognise point, line, ray and line segment; and draw and measure line segments. Identify the centre, radius and diameter of a circle. Recognise and identify quarter turns and identify quarter turns as right angles (and vice versa). Identify half and 3-quarter turns (clockwise and anti-clockwise) as two and three right angles respectively. Identify reflective symmetry in 2-D shapes.

Perimeter and Area (page 152)

Perimeter (page 152): The sum of lengths of all sides of a closed figure is called its perimeter. The perimeter is measured in units such as centimetres, metres. They are denoted as cm, m. Area (page 152): Area is the amount of space occupied by the surface of an object. It is measured in square units such as square centimetres, square metre. In short, they are denoted by cm squared and m squared.

Perimeter of Squares and Rectangles — Examples (page 152, already given in the book): 1. To find the perimeter of a square, we add the lengths of all its sides: Perimeter of a square = 4 cm + 4 cm + 4 cm + 4 cm = 16 cm. 2. To find the perimeter of a rectangle, we add the lengths of all its sides: Perimeter of a rectangle = 3 cm + 4 cm + 3 cm + 4 cm = 14 cm.

Perimeter of irregular figures — Example 3 (page 153, already given in the book): To find the perimeter of a irregular figure, we add the lengths of all its sides. Perimeter of an irregular figure = 2 m + 2 m + 1 m + 2 m + 4 m + 2 m + 1 m + 2 m = 16 m.

Area of Squares and Rectangle on a square grid — Examples (page 153, already given in the book): 1. To find the area of the square in cm squared: Number of squares = 9. Therefore, the area of square is 9 cm squared. 2. To find the area of the rectangle in cm squared: Number of squares = 20. Therefore, the area of the rectangle is 20 cm squared. Area of an irregular figure (page 153, already given in the book): To find the area of the irregular figure in m squared: Number of squares = 13. Therefore, the area of the irregular figure is 13 m squared.

Exercise (page 154): Question 1 — Find the perimeter of the given figures in cm

Exercise (page 154): Q1 (i) — rectangle 4 cm by 3 cm on the grid

Step 1: Sides: 4 cm, 3 cm, 4 cm, 3 cm

Step 2: Perimeter = 4 + 3 + 4 + 3 = 14 cm

Answer: 14 cm

Exercise (page 154): Q1 (ii) — rectangle 7 cm by 4 cm on the grid

Step 1: Sides: 7 cm, 4 cm, 7 cm, 4 cm

Step 2: Perimeter = 7 + 4 + 7 + 4 = 22 cm

Answer: 22 cm

Exercise (page 154): Question 2 — Find the perimeter of the irregular figure in cm

Exercise (page 154): Q2 (i) — U-shaped figure, 5 cm wide and 4 cm tall with a 1 cm by 2 cm notch

Step 1: Trace the outline: top of left block 2 cm, down the notch 2 cm, notch bottom 1 cm, up the notch 2 cm, top of right block 2 cm, right side 4 cm, bottom 5 cm, left side 4 cm

Step 2: Perimeter = 2 + 2 + 1 + 2 + 2 + 4 + 5 + 4 = 22 cm

Answer: 22 cm

Exercise (page 154): Q2 (ii) — plus-shaped figure, vertical bar 5 cm tall and horizontal bar 5 cm wide, each 1 cm thick

Step 1: Trace the outline: top 1 cm, down 2 cm, across 2 cm, down 1 cm, back 2 cm, down 2 cm, bottom 1 cm, up 2 cm, back 2 cm, up 1 cm, across 2 cm, up 2 cm

Step 2: Perimeter = 1 + 2 + 2 + 1 + 2 + 2 + 1 + 2 + 2 + 1 + 2 + 2 = 20 cm

Answer: 20 cm

Exercise (page 154): Question 3 — Find the area of the given figures in cm squared

Exercise (page 154): Q3 (i) — rectangle 5 by 3 on the grid (cells numbered 1 to 15)

Missing word: Number of squares = 15. Therefore, the area is 15 cm squared.

Exercise (page 154): Q3 (ii) — square 4 by 4 on the grid (cells numbered 1 to 16)

Missing word: Number of squares = 16. Therefore, the area is 16 cm squared.

Exercise (page 154): Question 4 — Find the area of the irregular figures in m squared

Exercise (page 154): Q4 (i) — irregular figure (cells numbered 1 to 9)

Missing word: Number of squares = 9. Therefore, the area is 9 m squared.

Exercise (page 154): Q4 (ii) — irregular figure (cells numbered 1 to 8)

Missing word: Number of squares = 8. Therefore, the area is 8 m squared.

Exercise (page 155): Question 5 — Find the perimeter of the given figures in cm

Exercise (page 155): Q5 (i) — square 5 cm by 5 cm on the grid

Step 1: Sides: 5 cm, 5 cm, 5 cm, 5 cm

Step 2: Perimeter = 5 + 5 + 5 + 5 = 20 cm

Answer: 20 cm

Exercise (page 155): Q5 (ii) — rectangle 6 cm by 3 cm on the grid

Step 1: Sides: 6 cm, 3 cm, 6 cm, 3 cm

Step 2: Perimeter = 6 + 3 + 6 + 3 = 18 cm

Answer: 18 cm

Exercise (page 155): Question 6 — Find perimeter of irregular figures in m

Exercise (page 155): Q6 (i) — stepped figure, top block 2 m wide, full block 4 m wide, 3 m tall

Step 1: Trace the outline: top 2 m, down 1 m, right 2 m, down 2 m, bottom 4 m, left side 3 m

Step 2: Perimeter = 2 + 1 + 2 + 2 + 4 + 3 = 14 m

Answer: 14 m

Exercise (page 155): Q6 (ii) — figure with a 3 m top block, a 5 m middle row and a 3 m bottom block

Step 1: Trace the outline: top 3 m, right side 3 m, bottom 3 m, up 1 m, left 2 m, up 1 m, right 2 m, up 1 m

Step 2: Perimeter = 3 + 3 + 3 + 1 + 2 + 1 + 2 + 1 = 16 m

Answer: 16 m

Exercise (page 155): Question 7 — Find the area of the given figures in cm squared

Exercise (page 155): Q7 (i) — square 3 by 3 on the grid

Missing word: Number of squares = 9. Therefore, the area is 9 cm squared.

Exercise (page 155): Q7 (ii) — rectangle 10 by 7 on the grid

Missing word: Number of squares = 70. Therefore, the area is 70 cm squared.

Exercise (page 155): Question 8 — Find the area of the given figures in m squared

Exercise (page 155): Q8 (i) — figure with a 3-square top block, a 5-square middle row and a 3-square bottom block

Missing word: Number of squares = 3 + 5 + 3 = 11. Therefore, the area is 11 m squared.

Exercise (page 155): Q8 (ii) — staircase figure, 3 squares in each of the top two rows and 5 squares in each of the bottom two rows

Missing word: Number of squares = 3 + 3 + 5 + 5 = 16. Therefore, the area is 16 m squared.

Classifying Polygons with respect to their attributes (page 156)

Classifying Polygons with respect to their attributes (page 156): Polygons are two-dimensional shapes made up of straight sides. A polygon is a closed plane figure bounded by three or more line segments. Triangles, all kinds of quadrilaterals, five sided figures, seven sided figures etc., are all examples of polygons. Example (page 156, already given in the book): Polygons: a pentagon, a triangle, and a star-shaped polygon. Not Polygons: an open figure and an oval (curved, not made of straight sides). Names of some polygons (page 156): Triangle — 3 sides. Quadrilateral — 4 sides. Pentagon — 5 sides. Hexagon — 6 sides. Octagon — 8 sides. Decagon — 10 sides.

Polygons (page 157)

Polygons (page 157): Regular polygon: A regular polygon has sides of equal length and equal angles of equal measure. Irregular polygon: An irregular polygon have sides and angles that are not all equal. Regular Polygons (pictures): a pentagon, a triangle, an octagon. Irregular Polygons (pictures): an arrow-shaped polygon, a quadrilateral with unequal sides, a scalene triangle. The name of a polygon depends upon the number of sides it has. Table: Five — Pentagon. Six — Hexagon. Eight — Octagon. Ten — Decagon.

Differentiation based on attributes (pages 157-158): Pentagon: It is a five sided figure, like the shape of given house. A regular pentagon has all 5

sides and angles of equal measure. Each angle measures 108 degrees. Hexagon: It is a six sided figure like a honey comb cell. Regular hexagon's has six sides and angles are of equal measure. Each angle measures 120 degrees. Octagon: It is an eight sided figure like a stop sign. Regular octagon's all sides and all angles are of equal measure. Each angle measures 135 degrees. Decagon: It is a ten-sided figure often used in tilling patterns. Regular decagon's all 10 sides and all angles are of equal measure. Each angle measures 144 degrees.

Exercise (page 159): Question 1 — Look at the figures. Which figures are polygons? and which are not?

Exercise (page 159): Q1 — figures A, B, C, D, E

Missing word: Polygons: A (octagon, closed, straight sides), C (triangle), E (square). Not Polygons: B (open figure, not closed), D (oval, curved sides, not straight). NOTE — the book's boxes print more blank slots than there are figures; only the figures above fill them.

Exercise (page 159): Question 2 — Identify shapes as regular or irregular polygons

Exercise (page 159): Q2 (a) — equilateral triangle

Missing word: Regular polygon (all sides and angles equal).

Exercise (page 159): Q2 (b) — quadrilateral with unequal sides

Missing word: Irregular polygon (sides and angles are not all equal).

Exercise (page 159): Q2 (c) — regular pentagon

Missing word: Regular polygon (all sides and angles equal).

Exercise (page 159): Q2 (d) — arrow-shaped polygon

Missing word: Irregular polygon (sides and angles are not all equal).

Exercise (page 159): Question 3 — Write the name of each polygon

Exercise (page 159): Q3 (a) — pink figure with 6 sides

Missing word: Hexagon.

Exercise (page 159): Q3 (b) — green figure with 7 sides

Missing word: Heptagon. NOTE — the book's table on page 157 lists 5, 6, 8 and 10 sides only and does not name the 7-sided polygon, but the figure counts 7 sides at 300 dpi, and the standard name for a 7-sided polygon is heptagon.

Exercise (page 159): Q3 (c) — purple figure with 10 sides

Missing word: Decagon.

Exercise (page 160): Question 4 — Which polygon has six sides?

Exercise (page 160): Question 4

Which polygon has six sides?

- Pentagon
- ✓ Hexagon
- Octagon
- Decagon

Exercise (page 160): Question 5 — Pentagon each angle is _____ degrees

Exercise (page 160): Question 5

Pentagon each angle is _____ degrees

- 118 degrees
- ✓ 108 degrees
- 120 degrees
- 128 degrees

Exercise (page 160): Question 6 — Fill in the blanks

Exercise (page 160): Q6 (1) — A _____ has 8 sides

Missing word: An Octagon has 8 sides.

Exercise (page 160): Q6 (2) — A polygon with 5 sides is called a _____

Missing word: A polygon with 5 sides is called a Pentagon.

Exercise (page 160): Q6 (3) — A decagon has _____ sides

Missing word: A decagon has 10 sides.

Exercise (page 160): Question 7 — Match the column A with column B

Exercise (page 160): Q7 — Match column A with column B

Pentagon — 5 sides (already matched in the book)

Hexagon — 6 sides

Octagon — 8 sides

Decagon — 10 sides

Differentiating Prisms and Pyramids (page 161)

Differentiating Prisms and Pyramids — Prism (page 161): A prism is a 3-D solid with two identical and parallel polygonal bases connected by rectangular or parallelogram faces. OR: A prism is a polyhedron whose top and bottom faces are identical polygons, with their corresponding parallel sides. The shape of the base or end face, describes the prism. (a) Triangular prism: A triangular prism has two equal triangular bases and three faces that are parallelogram. Examples of triangular prisms include a tent and triangular chocholate bar (like toblerone). (b) Rectangular Prism: A rectangular prism has two rectangular bases eight corners and no curved faces. A example of rectangular prism is a box. Pyramid (page 161): A pyramid is 3D solid with one polygonal base and triangular faces that meet at a single point called the vertex (or apex). OR: A three

dimensional shape with a polygon-shaped base and triangular faces meeting at one vertex is called a pyramid.

Pyramids continued (page 162): (a) Triangular pyramid: A triangular pyramid has a triangular base and three triangular faces that meet at one vertex. (b) Square pyramid: A square based pyramid has a square base and four triangular faces that meet at one vertex. An example is the Egyptian pyramids. Similarities: Prism and pyramid both have polygonal bases and are types of polyhedrons (solids with flat faces). Differences: Prism is defined by having two parallel bases, while pyramid have a single base and converge to an apex.

Exercise (page 162): Question 1 — Complete by mark the prisms with a tick and pyramid with a cross

Exercise (page 162): Q1 (i) — rectangular prism (already ticked in the book)

Missing word: Prism — tick.

Exercise (page 162): Q1 (ii) — triangular prism (tent shape)

Missing word: Prism — tick.

Exercise (page 162): Q1 (iii) — square pyramid

Missing word: Pyramid — cross.

Exercise (page 162): Q1 (iv) — triangular pyramid

Missing word: Pyramid — cross.

Exercise (page 162): NOTE — the table prints columns (v) and (vi) but no figures (v) or (vi) are printed anywhere in the book (page 163 begins with Question 2); columns (v) and (vi) cannot be marked and are left blank — nothing invented for them.

Exercise (page 163): Question 2 — Which of the following is a prism?

Exercise (page 163): Question 2

Which of the following is a prism?

A shape with one base and triangular faces.

A shape with one base and rectangular faces.

A shape with only one vertex and one base.

✓ A shape with circular bases.

Exercise (page 163): NOTE on Question 2 — options (a), (b) and (c) all describe one-base shapes, which are pyramids, not prisms; only option (d) describes a two-base shape, so (d) is the intended answer. Strictly, the book defines a prism with polygonal bases (page 161), so a shape with circular bases is a cylinder rather than a prism in the strict sense; the question is kept with (d) as the intended answer and this caveat.

Exercise (page 163): Question 3 — Match the shape to its type

Exercise (page 163): Q3 — Match the shape to its type

Square base pyramid — Square pyramid picture (second in column B)

Rectangular prism — Rectangular prism picture (first in column B)

Triangular prism — Triangular prism picture (third in column B)

Triangular pyramid — Triangular pyramid picture (fourth in column B)

Movement of Objects: Slide and Rotation (page 164)

Movement of Objects: Slide and Rotation (page 164): The movement of objects can be described in two ways: slide (translation) and rotation. Slide (Translation): When an object moves in a straight line from one position to another without rotating or changing its direction, it is called

slide. Slide also called Translation. Example: Sliding a book straight across a table without tilting or turning it. Every part of the book moves in the same way, so it appear the same throughout the movement. Another example is a slide for children in parks and play grounds/areas.

Characteristics: The shape and orientation of the object remain the same. Every point on the object moves the same distance in the same direction.

Rotation (Turn) (page 165): When an object turns around a fixed point or axis, either within itself or outside of it, is called rotation. For example, rotating swings for children in parks and play areas demonstrate rotation.

Other examples: If a clock's hands rotates around the centre, each hand rotates in circular path. Similarly, if you rotate a spinner on its spin it turns around the fixed point but stays in one spot. Characteristics: The object changes its orientation as its moves. The points on the object follow a circular path around the centre of rotation, and distance from the center to any point on the object remains the same. Both movements can occur individually or simultaneously, depending on the motion of the object. For example, a rotating wheel combines sliding (as it moves forward) and rotation (as it spins).

Key differences (page 166): In sliding (translation) an object moves straight along a path, while in rotation, it spins around a centre or its axis.

Example (page 166, already given in the book): Show slide and rotation (turn). Slide: the triangle keeps its orientation and moves straight. Rotation: the triangle turns and changes its orientation.

Exercise (page 166): Question 1 — Tick the slide and cross the rotation

Exercise (page 166): Q1 (i) left pair — L-shape moves right keeping its orientation (straight dashed arrows)

Missing word: Slide — tick.

Exercise (page 166): Q1 (i) right pair — L-shape turns and changes its orientation (curved arrow)

Missing word: Rotation — cross.

Exercise (page 167): Question 1 (continued) — Tick the slide and cross the rotation

Exercise (page 167): Q1 (ii) left pair — shape moves right keeping its orientation (straight dashed arrows)

Missing word: Slide — tick.

Exercise (page 167): Q1 (ii) right pair — shape turns 90 degrees and changes its orientation (curved arrow)

Missing word: Rotation — cross.

Exercise (page 167): Q1 (iii) left pair — semicircle turns and changes its orientation (curved arrow)

Missing word: Rotation — cross.

Exercise (page 167): Q1 (iii) right pair — semicircle moves right keeping its orientation (straight arrows)

Missing word: Slide — tick.

Exercise (page 167): Q1 (iv) left pair — arrow turns from pointing right to pointing down (curved arrow)

Missing word: Rotation — cross.

Exercise (page 167): Q1 (iv) right pair — arrow moves right keeping its orientation (straight dashed arrows)

Missing word: Slide — tick.

Exercise (page 167): Q1 (v) left pair — trapezoid moves right keeping its orientation (straight dashed arrows)

Missing word: Slide — tick.

Exercise (page 167): Q1 (v) right pair — trapezoid turns and changes its orientation (curved arrow)

Missing word: Rotation — cross.

Exercise (page 167): Q1 (vi) left pair — shape turns and changes its orientation (curved arrow)

Missing word: Rotation — cross.

Exercise (page 167): Q1 (vi) right pair — shape moves right keeping its orientation (straight dashed arrows)

Missing word: Slide — tick.

Point, Line, Ray and Line Segment (page 168)

Point, Line, Ray and Line Segment (page 168): Point: A dot fixed on a paper or any surface to show the location of a place is an example of a point (Fig.1). The capital letters A, B, C, D etc. are used for naming the point (Fig.2). We read these points as point A, point B, point C and point D. Line: A line extends in two opposite directions without end. In other words, A line is a long, straight path made up of points that goes on forever in two opposite directions. For example, Fig.3. It is written as: line AB and read as: line AB. Ray: It has one end point and goes forever in only one direction. OR: If we extend a line segment from one end in a particular direction and put an arrow as shown in Fig.4. It is written as ray AB and read as ray AB.

Line Segment (page 169): Look at the (Fig.1), which is the shortest distance between point A and B. This shortest distance is shown here by a line segment AB. Let us mark two points A and B on a paper (fig.2). Join two point A and B, then get a line segment AB (fig.3). It is written as: line segment AB and read as: line segment AB. The point A and B are called

the end points of line segment. Remember — Lines and Lines Segments: A line segment join two points on a line. In the world of maths, a line itself goes on forever in two opposite direction, but in everyday life, most people describe a line segment as a line too.

Measuring a line segment (pages 169-170): Step 1: The ruler is placed in such a way along the line segment AB that the 0 mark of the ruler should at be point A. Step 2: The point B corresponds to the number 4 on the ruler. Step 3: The line segment of AB is 4 cm long (fig.a). Measuring PQ: Step 1: The ruler is placed in such a way along the line segment PQ that the '0' mark of the ruler should be at the point P. Step 2: The point Q corresponds to the number 7 on the ruler. Step 3: The line segment of PQ is 7 cm long (fig.b). Drawing a line segment (page 170): Let us draw a line segment whose length is 5 cm. Steps to be followed: 1. Mark a point L and put the ruler so that its 0 mark coincides with L. (The remaining drawing steps continue on page 171.)

Drawing a line segment continued (page 171): 2. Mark another point M corresponding to the number 5 mark on the ruler. 3. Put your pencil at L and move it along the edge of the ruler till you reach the point M. Remove the ruler. The line segment LM is 5 cm long.

Exercise (page 171): Question 1 — Identify and write the name of the given figure

Exercise (page 171): Q1 (i) — A and B joined by a segment with two end dots and no arrows

Missing word: Line Segment AB.

Exercise (page 171): Q1 (ii) — C and D joined by a line with arrows at both ends

Missing word: Line CD.

Exercise (page 171): Q1 (iii) — arrow at L, dot at M

Missing word: Ray ML (the ray starts at the end point M and goes through L, where the arrow is).

Exercise (page 171): Q1 (iv) — dot at Y, arrow at Z

Missing word: Ray YZ.

Exercise (page 171): Q1 (v) — a single dot labelled G

Missing word: Point G.

Exercise (page 171): Q1 (vi) — W and X joined by a segment with two end dots and no arrows

Missing word: Line Segment WX.

Exercise (page 171): Question 2 — Fill in the blanks (table)

Exercise (page 171): Q2 — Ray row (symbol AB with ray arrow, read as Ray AB)

Missing word: Figure: a ray starting at point A with an arrow at the B end.

Exercise (page 171): Q2 — Line row (symbol NM with double arrow, read as Line NM)

Missing word: Figure: a line through points N and M with arrows at both ends.

Exercise (page 171): Q2 — Point row (figure: dot B, read as Point B)

Missing word: Symbolically Written: (blank — a point is named by a capital letter only; it has no line symbol).

Exercise (page 172): Question 3 — Measure and write the length of the line segments L, M, N and O by ruler

Exercise (page 172): Q3 — segment LM (3 cm is printed under it in the book)

Missing word: 3 cm.

Exercise (page 172): Q3 — segment MN

Missing word: 3 cm (measured against the printed centimetre scale: from the 4 cm mark to the 7 cm mark).

Exercise (page 172): Q3 — segment NO

Missing word: 3 cm (measured against the printed centimetre scale: from the 8 cm mark to the 11 cm mark).

Exercise (page 172): Q3 — segment starting at O

Missing word: 3 cm (measured against the printed centimetre scale: from the 12 cm mark to the 15 cm mark).

Exercise (page 172): Question 4 — Measure and write the length of the line segments E, F and G by ruler

Exercise (page 172): Q4 — segment EF (4 cm is printed under it in the book)

Missing word: 4 cm.

Exercise (page 172): Q4 — segment FG

Missing word: 4 cm (measured against the printed centimetre scale: from the 5 cm mark to the 9 cm mark).

Exercise (page 172): Q4 — segment starting at G

Missing word: 5 cm (measured against the printed centimetre scale: from the 10 cm mark to the 15 cm mark).

Exercise (page 172): Question 5 — Measure the length of the following line segments

Exercise (page 172): Q5 (a) — segment EF

Missing word: 6 cm (measured on the 300 dpi page against the 86.8 pixels-per-centimetre scale of the printed rulers on the same page).

Exercise (page 172): Q5 (b) — segment LM

Missing word: 7 cm (measured on the 300 dpi page against the same

scale).

Exercise (page 172): Q5 (c) — segment CD

Missing word: 6 cm (measured on the 300 dpi page against the same scale).

Exercise (page 172): Q5 (d) — segment PQ

Missing word: 6 cm (measured on the 300 dpi page against the same scale).

Exercise (page 172): Question 6 — Draw line segments of the following lengths and name them

Exercise (page 172): Q6 (i) — 4 cm

Missing word: Draw a line segment 4 cm long and name it (for example, AB).

Exercise (page 172): Q6 (ii) — 8 cm

Missing word: Draw a line segment 8 cm long and name it.

Exercise (page 172): Q6 (iii) — 7 cm

Missing word: Draw a line segment 7 cm long and name it.

Exercise (page 172): Q6 second (iii) — 6 cm

Missing word: Draw a line segment 6 cm long and name it. NOTE — the book numbers this item '(iii)' a second time and skips (v); the items are kept in printed order.

Exercise (page 172): Q6 (iv) — 5 cm

Missing word: Draw a line segment 5 cm long and name it.

Exercise (page 172): Q6 (vi) — 10 cm

Missing word: Draw a line segment 10 cm long and name it.

Exercise (page 172): Teacher Point: Teacher will ask students to measure different objects from classroom and their school bag's.

Centre, Radius and Diameter of a Circle (page 173)

Centre, Radius and Diameter of a Circle (page 173): Circle: A circle is a round shape enclosed by a curved line. It has no corners (vertices) and no sides. Centre of the circle: The centre of a circle is a 'point O' inside the circle that is always at the same distance from all points on the boundary of the circle. Radius of the circle: The distance from centre 'O' to the point B is called the radius of the circle. Symbolically, it is written as OB or AO. Diameter of the circle: The line segment AB is known as diameter. The diameter must pass through the centre of the circle. We can express symbolically diameter as AB. Key Note: Diameter of a circle is twice of its radius i.e. Diameter = 2 times radius. Plural of radius is radii.

Exercise (page 174): Question 1 — Write the name of the diameters of the given circles

Exercise (page 174): Q1 circle 1 — segment AB through the centre

Missing word: Diameter AB.

Exercise (page 174): Q1 circle 2 — segment PQ through the centre

Missing word: Diameter PQ.

Exercise (page 174): Q1 circle 3 — segment CD through the centre

Missing word: Diameter CD.

Exercise (page 174): Question 2 — Write the name of the radius of the given circles

Exercise (page 174): Q2 circle 1 — segment from L on the circle to centre O

Missing word: Radius LO.

Exercise (page 174): Q2 circle 2 — segment from centre A to D on the circle

Missing word: Radius AD.

Exercise (page 174): Q2 circle 3 — segment from centre Y to Z on the circle

Missing word: Radius YZ.

Exercise (page 174): Question 3 — Write the name of the diameter, radius and centre of the following circles

Exercise (page 174): Q3 circle 1 (already given)

Missing word: Diameter CD. Radii OD. Centre O.

Exercise (page 174): Q3 circle 2 — diameter PR through centre Q

Missing word: Diameter PR. Radii QP and QR. Centre Q.

Exercise (page 174): Q3 circle 3 — diameter XZ through centre Y

Missing word: Diameter XZ. Radii YX and YZ. Centre Y.

Quarter Turn and Quarter Turn as Right Angle (page 175)

Quarter Turn and Quarter Turn as Right Angle (page 175): Identifying Quarter Turn: A quarter turn occurs when an object rotates by 90 degrees in either direction i.e., clockwise or anti-clockwise. This amount of turn represents one-fourth of a full rotation (360 degrees). It is called quarter turn. Quarter Turn as Right Angle: A quarter turn creates a right angle (90 degrees). A right angle is formed when two lines or sides meet to make an 'L' shape. Therefore, if an object makes a quarter turn, it rotates by a right angle. Conversely, if you see a right angle, you can think of it as a

quarter turn from the original position. Examples — Clock Hands: If the minute hand in a clock moves from 12 to 3, it makes a quarter turn (90 degrees) clockwise forming a right angle.

Turning in Place (page 176): Imagine standing still and then turning 90 degrees to face your side. This is called a quarter turn, which creates a right angle with your original direction. So, a quarter turn is a 90 degree rotation equivalent to one right angle. Identifying a quarter turn as a right angle (and vice versa) helps us visualise and measure turns more easily.

Exercise (page 176): Question 1 — Cross the shapes that have quarter turn

Exercise (page 176): Q1 box 1 — M turned on its side (already crossed in the book)

Missing word: Quarter turn — cross (M rotated 90 degrees clockwise).

Exercise (page 176): Q1 box 2 — F turned into a backwards F

Missing word: NOT a quarter turn — no cross (the shape is a mirror image, a flip, not a rotation).

Exercise (page 176): Q1 box 3 — pencil turned upside down

Missing word: NOT a quarter turn — no cross (the pencil is flipped end to end, a half turn of 180 degrees, not 90 degrees).

Exercise (page 176): Q1 box 4 — flag shape with the notch moved from the right edge to the top edge

Missing word: Quarter turn — cross (the shape is rotated 90 degrees anti-clockwise).

Exercise (page 176): Question 1 (second, same number in the book) — Draw the shapes which they have shown a quarter turn in a clockwise direction

Exercise (page 176): Q1 (second) box 1 — vertical cylinder (already

drawn in the book)

Missing word: Drawn lying horizontal (the cylinder turned 90 degrees clockwise).

Exercise (page 176): Q1 (second) box 2 — arrow pointing right

Missing word: Draw the arrow pointing down (turned 90 degrees clockwise).

Exercise (page 176): Q1 (second) box 3 — letter L

Missing word: Draw the L lying on its side with its foot pointing down (turned 90 degrees clockwise).

Exercise (page 176): Q1 (second) box 4 — right triangle with vertical left side and flat bottom

Missing word: Draw the triangle turned 90 degrees clockwise, with a flat top and a vertical right side. NOTE — the book numbers both exercises on this page as '1'; they are kept as printed.

Exercise (page 177): Question 3 — Draw the shapes which they have shown a quarter turn in an anti-clockwise direction

Exercise (page 177): Q3 box 1 — horizontal rectangle (already drawn in the book)

Missing word: Drawn standing vertical (turned 90 degrees anti-clockwise).

Exercise (page 177): Q3 box 2 — letter P

Missing word: Draw the P lying on its side with its stem along the bottom and its bowl opening upward (turned 90 degrees anti-clockwise).

Exercise (page 177): Q3 box 3 — arrow pointing down

Missing word: Draw the arrow pointing right (turned 90 degrees anti-clockwise).

Exercise (page 177): Q3 box 4 — letter R

Missing word: Draw the R lying on its side with its stem along the bottom, its bowl opening upward and its leg pointing to the upper left (turned 90 degrees anti-clockwise).

Exercise (page 177): Question 4 — Tick clockwise quarter turns

Exercise (page 177): Q4 diagram 1 — arrow curving from the top to the left

Missing word: NOT ticked (anti-clockwise quarter turn).

Exercise (page 177): Q4 diagram 2 — arrow curving from the top to the right

Missing word: Ticked (clockwise quarter turn).

Exercise (page 177): Q4 diagram 3 — arrow sweeping from the top around the right to the bottom

Missing word: NOT ticked (clockwise half turn of 180 degrees, not a quarter turn).

Exercise (page 177): Question 5 — Tick anti-clockwise quarter turns

Exercise (page 177): Q5 diagram 1 — arrow curving from the top to the right

Missing word: NOT ticked (clockwise quarter turn).

Exercise (page 177): Q5 diagram 2 — arrow sweeping from the right to the bottom

Missing word: NOT ticked (clockwise quarter turn).

Exercise (page 177): Q5 diagram 3 — arrow curving from the top to the left

Missing word: Ticked (anti-clockwise quarter turn).

Exercise (page 177): Question 6 — Quarter turn is equivalent to:

Exercise (page 177): Question 6

Quarter turn is equivalent to:

- ✓ One right angle
- Two right angles
- three right angles
- Four right angles

Half and 3 Quarter Turns as Two and Three Right Angles (page 178)

Half and 3 Quarter Turns as Two and Three Right Angles (page 178): Identifying Half Turn: A half turn is a rotation of 180 degrees, which is equivalent to two right angles (90 degrees each). Clockwise half turn: If a person or object starts facing forward, a clockwise half turn will make them face the opposite direction. Anti-clockwise half turn: An anti-clockwise half turn also results in facing the opposite direction. Example: If you turn a book from facing up to facing down, that's a half turn. You have changed its position by 2 right angles (180 degrees). Identifying three quarter turn: A three quarter turn is a rotation of 270 degrees, which is equivalent to three right angles.

Three-quarter turns (page 179): Clockwise three-quarter turn: Starting from the initial position, a clockwise three quarter turn will result in the object or person facing to the left of the original position. Anti clockwise three-quarter turn: Similarly, an anti-clockwise three quarter turn also result in facing the opposite direction. Example: If the minute hand on a clock moves from 12 to 6 in clockwise direction, it makes a two-quarter turn, which is equal 180 degrees. Hence, half turn (180 degrees) is equal

to 2 right angles (90 degrees each). Similarly, if the minute hands on clock moves from 12 to 9 in the clockwise direction, it makes three-quarter turns, which equal to 270 degrees. Hence, three quarter turn (270 degrees) is equal to 3 right angles (90 degrees each). NOTE — facing after a three-quarter turn: clockwise 270 degrees from forward faces left, and anti-clockwise 270 degrees from forward faces right; only a half turn (180 degrees) faces exactly opposite, so the book's 'opposite direction' wording for the anti-clockwise case is loose and the exact facings above are used.

Exercise (page 180): Question 1 — Half turn is equivalent to:

Exercise (page 180): Question 1

Half turn is equivalent to:

- 90 degrees
- ✓ 180 degrees
- 270 degrees
- 360 degrees

Exercise (page 180): Question 2 — Three quarter is equivalent to:

Exercise (page 180): Question 2

Three quarter is equivalent to:

- One right angle
- Two right angles
- ✓ Three right angles
- Four right angles.

Exercise (page 180): Question 3 — How many degree angle make a 3-quarter turn

Exercise (page 180): Question 3

How many degree angle make a 3-quarter turn.

- 90 degrees

180 degrees

✓ 270 degrees

360 degrees

Exercise (page 180): Question 4 — If a clock minute hand moves from 12 to 9, how many right angles does it pass?

Exercise (page 180): Question 4

If a clock minute hand moves from 12 to 9, how many right angles does it pass?

1

2

✓ 3

4

Reflective Symmetry in 2-D Shapes (page 181)

Reflective Symmetry in 2-D Shapes (page 181): Reflective symmetry in 2-D shapes occurs when a shape can be divided into two identical halves. Along a line of symmetry, each half is a mirror image of the other. In figure, MN is the line of symmetry as it divides the shape exactly in half. NOTE — the book also states that RS is not the line of symmetry because it does not divide the shape exactly in half when folded along the line, but the figure is a rectangle and its vertical centre line RS does divide it into two identical halves; moreover page 182 states that a rectangle has symmetry along the vertical and horizontal lines through its centre. The book's RS statement is therefore incorrect for the rectangle shown, and both MN and RS are lines of symmetry of that rectangle. 1. Circle: Any line that passes through the centre of a circle creates two identical halves. Infinite Lines of Symmetry. 2. Square: A square has symmetry

along both its diagonals and the vertical line through the center, and the horizontal line through the centre. 4 lines of symmetry.

Reflective Symmetry continued (page 182): 3. Rectangle: A rectangle has symmetry along the vertical and horizontal lines through its centre, but not along the diagonals. 2 lines of symmetry. 4. Triangle: Each line of symmetry passes through a vertex and the midpoint of the opposite side. 3 lines of symmetry. Example (page 182, already given in the book): Look at a few letters of the English alphabets to identify the different lines of symmetry. A — Vertical line of symmetry. B — Horizontal line of symmetry. O — Both axes of symmetry.

Exercise (page 183): Question 1 — Look at the shapes and identify the correct lines of symmetry and trace them

Exercise (page 183): Q1 (a) — circle with a horizontal line through the centre and a diagonal line off the centre (already given)

Missing word: Horizontal line of symmetry (the diagonal line does not pass through the centre, so it is not a line of symmetry).

Exercise (page 183): Q1 (b) — triangle with a horizontal line across the upper part and a vertical line through the apex

Missing word: Vertical line of symmetry (the horizontal line does not divide the triangle into mirror halves).

Exercise (page 183): Q1 (c) — square with a vertical line through the centre and a horizontal line below the centre

Missing word: Vertical line of symmetry (the horizontal line is drawn off-centre, so it is not a line of symmetry).

Exercise (page 183): Q1 (d) — regular hexagon with a horizontal line and a vertical line through the centre

Missing word: Both lines of symmetry (the horizontal line and the vertical

line each divide the hexagon into mirror halves).

Exercise (page 183): Q1 (e) — rectangle with a horizontal line and a vertical line through the centre

Missing word: Both lines of symmetry (the horizontal line and the vertical line through the centre).

Exercise (page 183): Q1 (f) — regular pentagon with a horizontal line across the lower part and a vertical line through the apex

Missing word: Vertical line of symmetry (a horizontal line does not divide a regular pentagon into mirror halves).

Exercise (page 183): Question 2 — Which of the following letters are symmetrical. Draw lines of symmetry

Exercise (page 183): Q2 — letters S, T, O, B, M

Missing word: Symmetrical: T (vertical line), O (vertical and horizontal lines), B (horizontal line), M (vertical line). Not symmetrical: S (no line of symmetry).

Exercise (page 183): Question 2 (second, same number in the book) — Draw the lines of symmetry for 2-D figures

Exercise (page 183): Q2 (second) (a) — diamond (square turned)

Missing word: 4 lines of symmetry (vertical, horizontal, and both diagonals).

Exercise (page 183): Q2 (second) (b) — triangle

Missing word: 3 lines of symmetry (each through a vertex and the midpoint of the opposite side).

Exercise (page 183): Q2 (second) (c) — circle

Missing word: Infinite lines of symmetry (any line through the centre).

Exercise (page 183): Q2 (second) (d) — rectangle

Missing word: 2 lines of symmetry (vertical and horizontal through the

centre).

Exercise (page 183): Q2 (second) (e) — regular hexagon

Missing word: 6 lines of symmetry (through opposite vertices and through midpoints of opposite sides).

Exercise (page 183): Q2 (second) (f) — arrow pointing up

Missing word: 1 line of symmetry (vertical through the centre). NOTE — the book numbers both exercises on this page as '2'; they are kept as printed.

Unit 6 Summary (page 184)

Summary (page 184): The length of a closed figure is called perimeter. Classification of polygon, with respect to their attributes. Prisms are defined by having two parallel bases, while pyramids have a single base and converge to an apex. Polygon is a closed plane figure bounded by three or more than three line segments. A regular polygons has equal sides and equal angles of measure. Ray has one end point and goes on forever in one direction. A line segment is a part of a line that is bounded by two different endpoints. The radius of any circle is half of its diameter. The line of symmetry divides a figure into two equal halves. A prism has two identical parallel bases connected by flat side. Area is the space occupied by any covered surface of any object. Vocabulary: Perimeter, line segment, polygon, prism, point, pyramid, ray, diameter, circle, radius, centre. This closes Unit 6 (pages 151 to 184).

Unit 7: Data Handling and Probability (page 185)

Unit 7: Data Handling and Probability (page 185): Students' Learning Outcomes: Represent data; read and interpret data using Carroll Diagram (including real-world problems). Describe the likelihood that everyday events will occur, using mathematical language (i.e., impossible, possible, less likely, more likely, equally likely, unlikely and certain).

Carroll Diagram (page 186)

Carroll Diagram (page 186): A Carroll diagram is a method used to organize or sort objects in a chart or table according to their characteristics, figures, numbers and shapes we can sort different things in Carroll diagram. Representing Data in Carroll Diagram and Interpreting it. Example: Arif has some circular and square cards in yellow and other colours. Sort these cards, represent them in Carroll diagram and interpret data. The cards are (counted from the picture): yellow circles 4, green circle 1, blue circle 1, yellow square 1, blue square 1, green square 1, brown square 1 — 10 cards in all: 6 circles and 4 squares, 5 yellow and 5 of other colours. Here we are to: Sort the cards based on shape and colour. Represent data in a Carroll diagram. Interpret data. Represent these circular and square cards in Carroll diagram according to their shapes and colours (the diagram is on page 187).

Carroll Diagram, the same data (page 187): Picture table — Yellow colour: Circles 4, Squares 2; Not Yellow colour: Circles 2 (green and brown pictured), Squares 3 (green, blue and brown pictured). Numbers table as printed — Yellow colour: Circles 4, Squares 2; Not Yellow colour: Circles 2, Squares 1. Interpret Data (as printed): 1. 4 circular cards and 2 square cards are yellow. 2. 2 circular cards and 3 square cards are not yellow. 3. 4 circular cards are yellow and 2 circle cards are not yellow. 4. 2 square

cards are yellow and 3 square cards are not yellow. NOTE — the book's page 187 tables do not match its own page 186 card set, and the numbers table contradicts the picture table and the interpretation: the page 186 set has only 1 yellow square (not 2) and 3 squares that are not yellow (not 1 as the numbers table prints), and its non-yellow circles are green and blue (not green and brown as the picture table draws). The correct sort of the page 186 set is: Yellow Circles 4, Yellow Squares 1, Not Yellow Circles 2, Not Yellow Squares 3 — 10 cards in all. The book's printed values are kept above, and this corrected sort is the answer to use.

Example (page 188, already given in the book): Sort out and put the numbers in correct position in Carroll diagram. Numbers: 2, 4, 10, 12, 16, 19, 21, 24, 26, 28, 31, 36. Greater than 20 and Divisible by 4: 24, 28, 36. Less than 20 and Divisible by 4: 4, 12, 16. Greater than 20 and Not divisible by 4: 21, 26, 31. Less than 20 and Not divisible by 4: 2, 10, 19. In above Carroll diagram, we can observe that: 1. The numbers divisible by 4 less than 20 are: 4, 12, 16. 2. The numbers divisible by 4 greater than 20 are: 24, 28, 36. 3. The numbers not divisible by 4 less than 20 are: 2, 10, 19. 4. The numbers not divisible by 4 greater than 20 are: 21, 26, 31 (the book prints '29' here, but 29 is not in the number list and the table gives 31; 31 is used). 5. The number divisible by 4 are: 4, 12, 16, 24, 28, 36.

Exercise (page 188): Question 1 — Complete the Carroll diagram using the given numbers: 2, 4, 5, 7, 11, 12, 15, 16, 18

Exercise (page 188): Q1 — Carroll diagram, Even and Odd against Less than 10 and More than 10

Missing word: Even and Less than 10: 2, 4. Even and More than 10: 12, 16, 18. Odd and Less than 10: 5, 7. Odd and More than 10: 11, 15.

Exercise (page 189): Question 2 — Put the numbers in correct position in the Carroll diagram: 3, 7, 15, 21, 26, 28

Exercise (page 189): Q2 — Carroll diagram, Multiple of 3 and Not a Multiple of 3 against Greater than 20 and Less than 20

Missing word: Multiple of 3 and Greater than 20: 21. Multiple of 3 and Less than 20: 3, 15. Not a Multiple of 3 and Greater than 20: 26, 28. Not a Multiple of 3 and Less than 20: 7.

Exercise (page 189): Question 3 — Look at the Carroll diagram and answer the questions given below (In 4 time table: Less than 15: 4, 8, 12; More than 15: 16, 20, 24. Not in 4 time table: Less than 15: 3, 5, 9; More than 15: 17, 19, 21)

Exercise (page 189): Q3 (1) — Which numbers are less than 15 and in 4 time table?

Missing word: 4, 8, 12.

Exercise (page 189): Q3 (2) — Which numbers are more than 15 but not in the 4 time table?

Missing word: 17, 19, 21.

Exercise (page 189): Q3 (3) — Which is the smallest number less than 15?

Missing word: 3 (the smallest number in the whole diagram).

Exercise (page 189): Q3 (4) — Which number are less than 15 but not in the 4 time table?

Missing word: 3, 5, 9.

Exercise (page 189): Q3 (5) — Which is the greatest number that is not in the 4 time table?

Missing word: 21.

Exercise (page 189): Q3 (6) — Which numbers are in the 4 time table that

are more than 15?

Missing word: 16, 20, 24.

Exercise (page 190): Question 4 — Look at the Carroll diagram and answer the questions given below (Even: In 5 time table: 10, 20, 40, 60; Not in 5 time table: 14, 26, 38. Odd: In 5 time table: 5, 15, 35, 55; Not in 5 time table: 13, 37, 49)

Exercise (page 190): Q4 (1) — Write even numbers in the 5 times table.

Missing word: 10, 20, 40, 60.

Exercise (page 190): Q4 (2) — Write odd numbers not in the 5 times table.

Missing word: 13, 37, 49.

Exercise (page 190): Q4 (3) — Write odd numbers in the 5 times table.

Missing word: 5, 15, 35, 55.

Exercise (page 190): Q4 (4) — Write even numbers not in the 5 times table.

Missing word: 14, 26, 38.

Exercise (page 190): Q4 (5) — Write even and odd numbers in the 5 time table.

Missing word: 10, 20, 40, 60, 5, 15, 35, 55 (all eight numbers in the 'In 5 time table' column).

Exercise (page 190): Q4 (6) — Which numbers in the 5 times table are even and odd?

Missing word: 5, 10, 15, 20, 35, 40, 55, 60 (every number in the 'In 5 time table' column; the even ones are 10, 20, 40, 60 and the odd ones are 5, 15, 35, 55).

Exercise (page 190): Question 5 — Fatima has blue and yellow color balloons, with stars and without stars. Which shown in a

Carroll diagram

Exercise (page 190): Q5 — Fatima's balloons Carroll diagram (counted from the picture)

Missing word: Blue with Stars: 4. Blue without Stars: 2. Yellow with Stars: 2. Yellow without Stars: 2. 10 balloons in all.

Exercise (page 191): Fatima's balloons — Look at the Carroll diagram and answer the questions

Exercise (page 191): Balloons (1) — How many blue balloons with stars?

Missing word: 4.

Exercise (page 191): Balloons (2) — How many blue balloons without stars?

Missing word: 2.

Exercise (page 191): Balloons (3) — How many yellow balloons with stars?

Missing word: 2.

Exercise (page 191): Balloons (4) — How many yellow balloons without stars?

Missing word: 2.

Exercise (page 191): Balloons (5) — How many blue balloons with and without stars?

Missing word: 6 (4 with stars + 2 without stars).

Exercise (page 191): Question 6 — Arshad has some ice creams and donuts with chocolate and without chocolate. Which shown in a Carroll diagram

Exercise (page 191): Q6 — Arshad's ice creams and donuts Carroll diagram (counted from the picture)

Missing word: Chocolate Ice Creams: 3. Chocolate Donuts: 2. Not

Chocolate Ice Creams: 2. Not Chocolate Donuts: 2. 9 items in all.

Exercise (page 191): Arshad's diagram — Look at the Carroll diagram and answer the questions

Exercise (page 191): Arshad (1) — How many chocolate ice creams?

Missing word: 3.

Exercise (page 191): Arshad (2) — How many chocolate donuts?

Missing word: 2.

Exercise (page 191): Arshad (3) — How many ice creams without chocolate?

Missing word: 2.

Exercise (page 191): Arshad (4) — How many donuts without chocolate?

Missing word: 2.

Exercise (page 191): Arshad (5) — How many ice creams with and without chocolate?

Missing word: 5 (3 with chocolate + 2 without chocolate).

Probability (page 192)

Probability (page 192): Probability is the chance or likelihood that an event will occur. Probability of an event can be described using words such as certain, possible, impossible, less likely, more likely, unlikely, equally likely. Certain: An event is certain if it always happen. Example: It is certain that the sun sets every evening. There are only red balls in a pot. It is certain to pick a red ball. Impossible: An event is impossible, if it will never happen. Example: It is impossible to have 13 months in a year. There are only red balls in a pot. It is impossible to pick a yellow ball. Possible: An event that can happen but does not always. Example: It is

possible that I will have cereal for breakfast tomorrow. KEY NOTE: Picking a red ball from a bag is an example of an event. An event is something that happens. The probability of an event is the chance of likelihood of it happening.

Probability continued (page 193): There are six red balls and one blue ball. Then it is possible to pick red ball. Less likely: When comparing two events, one has a less chance of occurring. Example: It is less likely to see a tiger than a cat in your street. There are fewer red balls than blue balls in the pot. It is less likely to pick a red balls. More Likely: When comparing two events one has a greater chance of occurring. Example: There are more blue balls than red balls in a pot. It is more likely to pick a blue balls. Unlikely: An event that has a low chance of happening. Example: It is unlikely that it will snow in the desert. There are more yellow balls and only one red ball. It is unlikely to pick a red ball.

Probability continued (page 194): Equally Likely: Two events have the same chance of happening. Example: When you toss a coin, there is chance to get heads or tails. There are same number of yellow and blue balls, there is equal chance to pick yellow and blue balls. NOTE — the book prints 'yellow and red balls' in this last sentence, but the jar pictured holds yellow and blue balls and the sentence itself says 'yellow and blue balls'; 'blue' is used here.

Exercise (page 194): Question 1 — Write possible, impossible, certain, unlikely, more likely and less likely, equally, likely (ball counts verified at 300 dpi)

Exercise (page 194): Q1 (1) — You will pick a red ball from a jar with 3 yellow and 3 blue balls (already given)

Missing word: Impossible (there is no red ball in the jar).

Exercise (page 194): Q1 (2) — You will pick a yellow ball from a jar with 6

yellow balls only

Missing word: Certain (every ball in the jar is yellow).

Exercise (page 194): Q1 (3) — You will pick a blue ball from a jar with 2 red and 5 blue balls

Missing word: More likely (5 blue balls against 2 red balls).

Exercise (page 194): Q1 (4) — You will pick a red ball from a jar with 5 red and 2 yellow balls

Missing word: More likely (5 red balls against 2 yellow balls).

Exercise (page 194): Q1 (5) — You will pick a yellow ball from a jar with 3 yellow and 3 red balls

Missing word: Equally likely (3 yellow and 3 red balls, the same chance).

Exercise (page 194): Q1 (6) — You will pick a blue ball from a jar with 1 blue and 6 red balls

Missing word: Less likely (only 1 blue ball against 6 red balls; a 1-in-7 chance is also low enough to be called unlikely, but 'less likely' is the comparative word for the two colours in the jar).

Exercise (page 195): Question 1 continued — Write possible, impossible, certain, unlikely, more likely and less likely, equally, likely

Exercise (page 195): Q1 (7) — You will pick a red ball from a jar with 2 red and 5 blue balls

Missing word: Less likely (only 2 red balls against 5 blue balls).

Exercise (page 195): Q1 (8) — You will pick a green ball from a jar with 5 green and 2 blue balls

Missing word: More likely (5 green balls against 2 blue balls).

Exercise (page 195): Question 2 — Choose the correct answer

Exercise (page 195): Question 2 (i)

You leave your water bottle in the sun. It will feel warm.

- Unlikely
- More likely
- Less likely
- ✓ Certain

Exercise (page 195): Question 2 (ii)

You roll a number cube (1-6). Getting an even number (2,4,6)

- More likely
- Certain
- ✓ equally likely
- Impossible

Exercise (page 195): Question 2 (iii)

It might rain today the sky is cloudy.

- Impossible
- Certain
- ✓ Possible
- More likely

Unit 7 Summary (page 195)

Summary (page 195): A Carroll diagram is a method to organize or sort objects according to their characteristics. Figures, numbers, shapes and different things can be sorted in a Carroll Diagram. Probability is the chance or likely hood that an event will happen. An event is certain if it always happen. An event is possible that can happen but does not always happen. Vocabulary: Carroll diagram, Figures, Shapes, Impossible, Possible, Certain, Probability. This closes Unit 7 (pages 185 to 195).

Unit 8: Glossary — Reference Pages Summary

(page 196)

Reference Pages Summary — Glossary (page 196, no exercises in the book; the terms are summarised, not reprinted as solved items): Addition: Add two numbers, two fractions and measurements. Angle: The space between two lines that meet at a point. Axis: The vertical or horizontal line in a graph. Axes: The plural of axis. Clock wise: Moving around in a circle in the same direction as a clock hand moves. Centimeter: Small units used to measure size: one hundredth of a metre. Digital Clock: Clock which shows time in the form of digital. Division: It is repeated subtraction of numbers. Direction: The way that something is moving. Equal: Evenly balanced and of the same proportion. Horizontal: Means in a sideways direction. Litre: It is standard unit for measuring capacity of an object. Multiplication: It is repeated addition of numbers. Order: The way things are organized. Polygon: It is a two dimensional geometric figure that has many sides. Perimeter: The total length of all sides of a shape. Pictograph: A graph in which data is represented through pictures. Roman Number: Numbers invented and used by ancient Romans. Ray: A ray is a part of a line that has one endpoint. Radius: Distance between center and boundary of a circle. Subtraction: Taking away two or more items to make a new item. Solar year: The time taken by one revolution of earth around sun is called solar year. Tally: A way of counting in sets of fives. Temperature: A measure of how hot or cold something is. Turn: Move around in a circular motion. Vertical: Upright, in an upwards direction. Width: Measure how wide an object is.